

Topological Energy Condensate Theory (TECT): 35

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I. CANONICAL LOCKED-BASIS READING OF THE LONGITUDINAL ACTUAL PDE COEFFICIENT

II. 1. DIRECT FORMULA FOR $u_{\parallel}$

Let the shell-relevant reduced J -matrix at the selected intervalley harmonic be

$$\mathcal{U}_j(0) = (\mathcal{U}_{j,ab}(0))_{a,b=1}^2, \quad \mathcal{U}_{j,ab}(0) = \langle -, a | \hat{J}_j(G_*; 0) | +, b \rangle. \quad (1)$$

Then the shell-normal longitudinal singlet coefficient is

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 \hat{n}^j \mathcal{U}_j(0)). \quad (2)$$

In the diagonal Pauli basis $\Sigma_3 = \text{diag}(1, -1)$,

$$u_{\parallel} = \frac{1}{2} \left[(\hat{n}^j \mathcal{U}_j(0))_{11} - (\hat{n}^j \mathcal{U}_j(0))_{22} \right]. \quad (3)$$

So $u_{\parallel}$ is literally the shell-normal diagonal asymmetry of the reduced J -block.

III. 2. DIRECT FORMULA FOR $m_{\parallel}$

In the canonical locked frame,

$$m^a(P_0) = m_{\parallel} \delta^{a3}. \quad (4)$$

Therefore

$$m_{\parallel} = m^3(P_0). \quad (5)$$

No further reduction is needed.

IV. 3. THE ACTUAL LONGITUDINAL SCALAR

Hence the unresolved longitudinal scalar is

$$L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2} \text{Tr}(\Sigma_3 \hat{n}^j \mathcal{U}_j(0)). \quad (6)$$

Equivalently,

$$L_* = \frac{m_{\parallel}}{2} \left[(\hat{n}^j \mathcal{U}_j(0))_{11} - (\hat{n}^j \mathcal{U}_j(0))_{22} \right]. \quad (7)$$

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V. 4. DIRECT LONGITUDINAL PDE COEFFICIENT

On the current canonical branch,

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X L_* I_{\theta'^3}. \quad (8)$$

Comparing with the reference reduced sheet

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3,$$

one reads, on this branch,

$$\lambda_c = \alpha_X L_*. \quad (9)$$

Thus the actual longitudinal reduced coefficient is

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X m_{\parallel} u_{\parallel} I_{\theta'^3}. \quad (10)$$

Using the explicit overlap

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad (11)$$

we get

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X m_{\parallel} u_{\parallel} \theta_0^3 \kappa^3. \quad (12)$$

VI. 5. WHAT IS NOW LITERALLY LEFT FOR THE LONGITUDINAL SLOT

The longitudinal slot is therefore no longer structurally open.

The only remaining finite microscopic inputs are

$$m_{\parallel} = m^3(P_0), \quad u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 \hat{n}^j \mathcal{U}_j(0)). \quad (13)$$

Once these two numbers are computed from the actual microscopic branch/operator data, the longitudinal actual PDE coefficient is fully populated.

VII. DIRECT READING OF THE ODD TRANSVERSE COEFFICIENT ON THE ACTIVE ROUTE

The active explicit odd extra-carrier slot is

$$\mathcal{I}_O = X_i^{(O)} \cos \theta(\xi) J_j D_j \psi.$$

Its induced tangent transport block is

$$V_{EO}^{\mu} = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^{\mu}{}_a \Sigma_a + \tilde{\Lambda}_O J^{\mu}{}_a \Sigma_a \right].$$

Hence the odd projector reads

$$\tilde{\Lambda}_O = \frac{\mathfrak{P}_O[V_{EO}]}{\theta'(\xi) \cos \theta(\xi)}.$$

At one-mediator level,

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1.$$

In canonical shell normalization $C_O = 1$,

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} J_1.$$

Therefore the reduced odd transverse coefficient is

$$\beta = I_c \tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1 I_c.$$

If one uses the transport-side canonical explicit normalization that absorbs J_1 ,

$$\beta = -\frac{\beta_X \gamma_O}{M_X^2} I_c.$$

Under the current reference branch choice

$$\gamma_E = \gamma_O =: \gamma_\perp \neq 0,$$

the reference low-order sheet becomes

$$\alpha = \beta = -\frac{\beta_X \gamma_\perp}{M_X^2} I_c.$$

$$\left[i\partial_t + \frac{3\sqrt{3}\pi}{1024} \alpha_X m_\parallel (u_{11} - u_{22}) \theta_0^3 \kappa^3 \sigma_3(-i\partial_\parallel) + \frac{\beta_X \gamma_\perp}{M_X^2} I_c (\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) + \frac{\beta_X \gamma_\perp}{M_X^2} I_c (-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0$$

VIII. FINITE 2×2 LONGITUDINAL EXTRACTION FOR THE ACTUAL REDUCED PDE

IX. 1. VALLEY-DIAGONAL PROJECTED MATRICES

Fix the explicit carrier basis

$$|+, a\rangle := |G_*, \chi_a\rangle, \quad |-, a\rangle := |-G_*, \chi_a\rangle, \quad a = 1, 2.$$

The two valley-diagonal projected blocks at $p = 0$ are

$$(E_+(0))_{ab} = \langle +, a | H_{\text{shell}}^{(N)}(0) | +, b \rangle, \quad (E_-(0))_{ab} = \langle -, a | H_{\text{shell}}^{(N)}(0) | -, b \rangle. \quad (14)$$

Equivalently, write them in general Hermitian form

$$E_+(0) = \begin{pmatrix} a_+ & c_+ \\ \bar{c}_+ & b_+ \end{pmatrix}, \quad E_-(0) = \begin{pmatrix} a_- & c_- \\ \bar{c}_- & b_- \end{pmatrix}, \quad (15)$$

with

$$a_\pm, b_\pm \in \mathbb{R}, \quad c_\pm \in \mathbb{C}.$$

X. 2. LOW EIGENVALUES AND NORMALIZED LOW EIGENVECTORS

The lower eigenvalues are

$$\varepsilon_{\pm}^{(\text{low})} = \frac{a_{\pm} + b_{\pm}}{2} - \sqrt{\left(\frac{a_{\pm} - b_{\pm}}{2}\right)^2 + |c_{\pm}|^2}. \quad (16)$$

Define the standard ratio parameter

$$r_{\pm} = -\frac{c_{\pm}}{a_{\pm} - \varepsilon_{\pm}^{(\text{low})}}. \quad (17)$$

Then the normalized low eigenvectors can be written as

$$\eta_{\pm} = \frac{1}{\sqrt{1 + |r_{\pm}|^2}} \begin{pmatrix} 1 \\ r_{\pm} \end{pmatrix}. \quad (18)$$

This is the exact finite 2×2 solution of the valley-side low-mode extraction.

XI. 3. EXACT CLOSED FORMULA FOR Δ_{00}

The intervalley Bloch asymmetry is

$$\Delta_{00} = \eta_+^\dagger \Sigma_3 \eta_-. \quad (19)$$

Substituting (18), one gets the exact closed form

$$\Delta_{00} = \frac{1 - \bar{r}_+ r_-}{\sqrt{(1 + |r_+|^2)(1 + |r_-|^2)}}. \quad (20)$$

Hence the exact longitudinal cancellation condition is

$$\Delta_{00} = 0 \iff \bar{r}_+ r_- = 1. \quad (21)$$

So the first longitudinal finite audit is reduced completely to the microscopic matrix entries

$$a_{\pm}, \quad b_{\pm}, \quad c_{\pm}.$$

XII. 4. REDUCED SHELL-NORMAL INTERVALLEY J -MATRIX

Define the shell-normal intervalley J -matrix

$$U_{\parallel} = \hat{n}^j \mathcal{U}_j(0) = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}, \quad (\mathcal{U}_j(0))_{ab} = \langle -, a | \hat{J}_j(G_*; 0) | +, b \rangle. \quad (22)$$

The longitudinal bias is

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}) = \frac{1}{2}(u_{11} - u_{22}). \quad (23)$$

Therefore the actual unresolved longitudinal scalar is

$$L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2}(u_{11} - u_{22}). \quad (24)$$

XIII. 5. FINITE HARMONIC FORMULA FOR $u_{\parallel}$

In the shell-harmonic basis,

$$u_a^{\pm}(x) = \sum_{G \in \mathcal{S}_N} c_{a,G}^{(\pm)} e^{iG \cdot x}, \quad J_j(x) = \sum_Q \hat{J}_j(Q; 0) e^{iQ \cdot x}.$$

Then

$$(\mathcal{U}_j(0))_{ab} = V_{\text{cell}} \sum_{G, G' \in \mathcal{S}_N} \overline{c_{a,G}^{(+)}} \hat{J}_j(G_* + G - G'; 0) c_{b,G'}^{(-)}. \quad (25)$$

Hence

$$u_{\parallel} = \frac{V_{\text{cell}}}{2} \sum_{G, G' \in \mathcal{S}_N} \left(\overline{c_{1,G}^{(+)}} c_{1,G'}^{(-)} - \overline{c_{2,G}^{(+)}} c_{2,G'}^{(-)} \right) \hat{n}^j \hat{J}_j(G_* + G - G'; 0). \quad (26)$$

Equivalently,

$$L_* = \frac{m_{\parallel} V_{\text{cell}}}{2} \sum_{G, G' \in \mathcal{S}_N} \left(\overline{c_{1,G}^{(+)}} c_{1,G'}^{(-)} - \overline{c_{2,G}^{(+)}} c_{2,G'}^{(-)} \right) \hat{n}^j \hat{J}_j(G_* + G - G'; 0). \quad (27)$$

XIV. 6. LONGITUDINAL COEFFICIENT OF THE ACTUAL REDUCED PDE

On the current canonical branch,

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X L_* I_{\theta'^3}. \quad (28)$$

Using the explicit overlap

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad (29)$$

we obtain

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3 = -\frac{3\sqrt{3}\pi}{1024} \alpha_X m_{\parallel} (u_{11} - u_{22}) \theta_0^3 \kappa^3. \quad (30)$$

XV. 7. ACTUAL REDUCED PDE COEFFICIENT SHEET

The active reference-route reduced PDE coefficient sheet is therefore

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{1024} \alpha_X m_{\parallel} (u_{11} - u_{22}) \theta_0^3 \kappa^3, \quad (31)$$

$$\alpha = \beta = -\frac{\beta_X \gamma_{\perp}}{M_X^2} I_c. \quad (32)$$

Thus the current reference reduced PDE is

$$\left[i\partial_t + \frac{3\sqrt{3}\pi}{1024} \alpha_X m_{\parallel} (u_{11} - u_{22}) \theta_0^3 \kappa^3 \sigma_3 (-i\partial_{\parallel}) + \frac{\beta_X \gamma_{\perp}}{M_X^2} I_c (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) + \frac{\beta_X \gamma_{\perp}}{M_X^2} I_c (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0. \quad (33)$$

The first sign is positive because the PDE contains $-\lambda_{\parallel}$, while $\lambda_{\parallel}$ itself is negative in (30).

XVI. 8. WHAT REMAINS LITERALLY TO EVALUATE

No structural ambiguity remains in the longitudinal slot.
The remaining microscopic inputs are exactly

$$\boxed{a_{\pm}, \quad b_{\pm}, \quad c_{\pm}, \quad u_{11}, \quad u_{22}, \quad m_{\parallel}}. \quad (34)$$

Equivalently, at the operator level, the true remaining frontier is to evaluate

$$\boxed{c_{a,G}^{(\pm)}, \quad \widehat{J}_j(Q;0), \quad P_0, \quad m^3(P_0)}$$

for the final refined locked microscopic operator.

XVII. FINITE 2×2 LONGITUDINAL INSERTION SHEET

XVIII. 1. VALLEY-DIAGONAL PROJECTED MATRICES

Write

$$E_+(0) = \begin{pmatrix} a_+ & c_+ \\ \bar{c}_+ & b_+ \end{pmatrix}, \quad E_-(0) = \begin{pmatrix} a_- & c_- \\ \bar{c}_- & b_- \end{pmatrix},$$

with

$$a_{\pm}, b_{\pm} \in \mathbb{R}, \quad c_{\pm} \in \mathbb{C}.$$

Their low eigenvalues are

$$\varepsilon_{\pm}^{(\text{low})} = \frac{a_{\pm} + b_{\pm}}{2} - \sqrt{\left(\frac{a_{\pm} - b_{\pm}}{2}\right)^2 + |c_{\pm}|^2}.$$

Define

$$r_{\pm} = -\frac{c_{\pm}}{a_{\pm} - \varepsilon_{\pm}^{(\text{low})}} = -\frac{c_{\pm}}{\frac{a_{\pm} - b_{\pm}}{2} + \sqrt{\left(\frac{a_{\pm} - b_{\pm}}{2}\right)^2 + |c_{\pm}|^2}}.$$

Then the normalized low eigenvectors can be chosen as

$$\eta_{\pm} = \frac{1}{\sqrt{1 + |r_{\pm}|^2}} \begin{pmatrix} 1 \\ r_{\pm} \end{pmatrix}.$$

XIX. 2. EXACT FINITE FORMULA FOR Δ_{00}

$$\Delta_{00} = \eta_+^{\dagger} \Sigma_3 \eta_-, \quad \Sigma_3 = \text{diag}(1, -1).$$

Hence

$$\boxed{\Delta_{00} = \frac{1 - \bar{r}_+ r_-}{\sqrt{(1 + |r_+|^2)(1 + |r_-|^2)}}.}$$

Therefore

$$\boxed{\Delta_{00} = 0 \iff \bar{r}_+ r_- = 1.}$$

XX. 3. SHELL-NORMAL REDUCED J -MATRIX

Define

$$U_{\parallel} := \hat{n}^j \mathcal{U}_j(0) = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}, \quad (\mathcal{U}_j(0))_{ab} = \langle -, a | \hat{J}_j(G_*; 0) | +, b \rangle.$$

Then the longitudinal J -singlet bias is

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}) = \frac{1}{2}(u_{11} - u_{22}).$$

Hence

$$L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2}(u_{11} - u_{22}).$$

XXI. 4. ACTUAL LONGITUDINAL COEFFICIENT

On the current canonical branch,

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X L_* I_{\theta'^3}.$$

Using

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3,$$

we obtain

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{1024} \alpha_X m_{\parallel} (u_{11} - u_{22}) \theta_0^3 \kappa^3.$$

XXII. 5. CURRENT ACTUAL REDUCED PDE COEFFICIENT SHEET

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{1024} \alpha_X m_{\parallel} (u_{11} - u_{22}) \theta_0^3 \kappa^3, \quad \alpha = \beta = -\frac{\beta_X \gamma_{\perp}}{M_X^2} I_c.$$

XXIII. 6. LITERAL REMAINING MICROSCOPIC DATA

$$a_{\pm}, \quad b_{\pm}, \quad c_{\pm}, \quad u_{11}, \quad u_{22}, \quad m_{\parallel}.$$

Equivalently:

$$E_{\pm}(0), \quad U_{\parallel} = \hat{n}^j \mathcal{U}_j(0), \quad m_{\parallel} = m^3(P_0).$$

XXIV. FINAL FINITE INSERTION SHEET FOR THE ACTUAL REDUCED PDE

XXV. 1. VALLEY-SIDE 2×2 PROJECTED BLOCKS

Write

$$E_+(0) = \begin{pmatrix} a_+ & c_+ \\ \overline{c_+} & b_+ \end{pmatrix}, \quad E_-(0) = \begin{pmatrix} a_- & c_- \\ \overline{c_-} & b_- \end{pmatrix},$$

with

$$a_{\pm}, b_{\pm} \in \mathbb{R}, \quad c_{\pm} \in \mathbb{C}.$$

Their lower eigenvalues are

$$\varepsilon_{\pm}^{(\text{low})} = \frac{a_{\pm} + b_{\pm}}{2} - \sqrt{\left(\frac{a_{\pm} - b_{\pm}}{2}\right)^2 + |c_{\pm}|^2}.$$

Define

$$r_{\pm} = -\frac{c_{\pm}}{a_{\pm} - \varepsilon_{\pm}^{(\text{low})}}, \quad \eta_{\pm} = \frac{1}{\sqrt{1 + |r_{\pm}|^2}} \begin{pmatrix} 1 \\ r_{\pm} \end{pmatrix}.$$

Then the exact intervalley asymmetry is

$$\Delta_{00} = \eta_+^\dagger \Sigma_3 \eta_- = \frac{1 - \overline{r_+} r_-}{\sqrt{(1 + |r_+|^2)(1 + |r_-|^2)}}.$$

XXVI. 2. SHELL-NORMAL REDUCED J -MATRIX

Define

$$U_{\parallel} := \hat{n}^j \mathcal{U}_j(0) = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}, \quad (\mathcal{U}_j(0))_{ab} = \langle -, a | \hat{\mathcal{J}}_j(G_*; 0) | +, b \rangle.$$

Then

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}) = \frac{1}{2}(u_{11} - u_{22}).$$

Equivalently, in the shell-harmonic basis,

$$(\mathcal{U}_j(0))_{ab} = V_{\text{cell}} \sum_{G, G' \in \mathcal{S}_N} \overline{c_{a,G}^{(+)}} \hat{\mathcal{J}}_j(G_* + G - G'; 0) c_{b,G'}^{(-)}.$$

Hence

$$u_{\parallel} = \frac{V_{\text{cell}}}{2} \sum_{G, G' \in \mathcal{S}_N} \left(\overline{c_{1,G}^{(+)}} c_{1,G'}^{(-)} - \overline{c_{2,G}^{(+)}} c_{2,G'}^{(-)} \right) \hat{n}^j \hat{\mathcal{J}}_j(G_* + G - G'; 0).$$

XXVII. 3. LONGITUDINAL SCALAR AND COEFFICIENT

Let

$$m_{\parallel} = m^3(P_0).$$

Then

$$L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2} (u_{11} - u_{22}).$$

Using

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3,$$

the actual longitudinal reduced coefficient is

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X L_* I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{1024} \alpha_X m_{\parallel} (u_{11} - u_{22}) \theta_0^3 \kappa^3.$$

XXVIII. 4. TRANSVERSE COEFFICIENTS ON THE ACTIVE REFERENCE ROUTE

The overlap dictionary is

$$I_c = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \cos \theta(\xi).$$

On the active explicit odd extra-carrier route,

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = I_c \tilde{\Lambda}_E, \quad \beta = I_c \tilde{\Lambda}_O.$$

At one-mediator level in canonical explicit normalization,

$$\tilde{\Lambda}_E^{(1)} = -\frac{\beta_X \gamma_E}{M_X^2}, \quad \tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2}.$$

Under the current reference choice

$$\gamma_E = \gamma_O = \gamma_{\perp},$$

this becomes

$$\alpha = \beta = -\frac{\beta_X \gamma_{\perp}}{M_X^2} I_c.$$

XXIX. 5. FINAL CURRENT REFERENCE REDUCED PDE

Therefore the current actual/reference reduced PDE is

$$\left[i\partial_t + \frac{3\sqrt{3}\pi}{1024} \alpha_X m_{\parallel} (u_{11} - u_{22}) \theta_0^3 \kappa^3 \sigma_3 (-i\partial_{\parallel}) + \frac{\beta_X \gamma_{\perp}}{M_X^2} I_c (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) + \frac{\beta_X \gamma_{\perp}}{M_X^2} I_c (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0.$$

XXX. 6. LITERAL REMAINING MICROSCOPIC INPUTS

So the only remaining microscopic data to insert are

$$a_{\pm}, \quad b_{\pm}, \quad c_{\pm}, \quad u_{11}, \quad u_{22}, \quad m_{\parallel}, \quad I_c.$$

Equivalently:

$$E_{\pm}(0), \quad U_{\parallel}, \quad m^3(P_0), \quad I_c.$$

Once these are evaluated from the actual microscopic branch/operator data, the coefficient point

$$(\lambda_{\parallel}, \alpha, \beta)$$

is fully populated.

$$\left[i\partial_t + \frac{3\sqrt{3}\pi}{512} \alpha_X \theta_0^3 \kappa^3 \sigma_3(-i\partial_{\parallel}) + \frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa (\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) + \frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa (-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0$$

$$\left[i\partial_t + \frac{3\sqrt{3}\pi}{512} \alpha_X \theta_0^3 \kappa^3 \sigma_3(-i\partial_{\parallel}) + \frac{\sqrt{6}\pi}{16} \beta_X \theta_0 \kappa (\sigma_1(-i\partial_{q_1}) + \sigma_2(-i\partial_{q_2})) \right] \chi = 0$$

XXXI. FINAL FULL-MICROSCOPIC EXTRACTION ROUTE

XXXII. 1. SHELL-PROJECTED FINITE OPERATOR

Choose the folded Bragg point

$$k_* = \frac{G_*}{2}, \quad k = k_* + p, \quad |p| \ll |G_*|.$$

Let S_* be the finite near-resonant harmonic set around k_* , and define the shell projector

$$\Pi_{S_*} : \mathcal{H}_{\text{Bloch}}(k) \rightarrow \mathcal{H}_{S_*}(k).$$

Then the first finite-dimensional operator belonging to the actual full-PDE derivation is

$$H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*}.$$

This is the true computational bottleneck. :contentReference[oaicite:2]index=2

XXXIII. 2. LOW VALLEY PROJECTION

Assume $H_{\text{shell}}(0)$ contains an isolated two-dimensional low-energy sector $\mathcal{V}_*$. Let

$$P_* : \mathcal{H}_{S_*}(0) \rightarrow \mathcal{V}_*$$

be the spectral projector onto that valley subspace.

Define the first derivatives

$$K_i := \partial_{p_i} H_{\text{shell}}(p)|_{p=0}.$$

Then the projected linear valley operator is

$$H_{\text{val}}^{(1)}(p) = p_i P_* K_i P_*.$$

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XXXIV. 3. UNIFIED COEFFICIENT READOUT

Inside the valley space, define

$$M_{\mu i} = \frac{1}{2} \text{Tr}_{\mathcal{H}_*}(\sigma_\mu P_* K_i P_*), \quad \mu = 1, 2, 3.$$

In the shell-adapted frame $(\hat{n}, e_1, e_2)$,

$$\lambda_{\parallel} = \hat{n}_i M_{3i},$$

$$\alpha = \frac{1}{2}(e_{1i} M_{1i} + e_{2i} M_{2i}), \quad \beta = \frac{1}{2}(e_{2i} M_{1i} - e_{1i} M_{2i}).$$

Equivalently,

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 P_* K_{\parallel} P_*), \quad K_{\parallel} = \hat{n}_i K_i.$$

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XXXV. 4. MINIMAL AXIAL 4×4 SHELL ANSATZ

On the folded-Bragg 12-mode shell pair, the minimal axial ansatz is

$$K_i = A \hat{n}_i \tau^3 \otimes \mathbf{1}_2 + B(e_{1i} \tau^1 \otimes s_1 + e_{2i} \tau^1 \otimes s_2) + C(-e_{2i} \tau^1 \otimes s_1 + e_{1i} \tau^1 \otimes s_2).$$

Hence

$$K_{\parallel} = A \tau^3 \otimes \mathbf{1}_2,$$

$$K_1 = B \tau^1 \otimes s_1 + C \tau^1 \otimes s_2, \quad K_2 = -C \tau^1 \otimes s_1 + B \tau^1 \otimes s_2.$$

Projecting to the selected positive-valley two-band block,

$$P_+(\tau^3 \otimes \mathbf{1}_2)P_+ \mapsto \sigma^3, \quad P_+(\tau^1 \otimes s_1)P_+ \mapsto \sigma^1, \quad P_+(\tau^1 \otimes s_2)P_+ \mapsto \sigma^2,$$

so that

$$h_+(p) = A p_{\parallel} \sigma^3 + B(p_1 \sigma^1 + p_2 \sigma^2) + C(-p_2 \sigma^1 + p_1 \sigma^2) + O(|p|^2).$$

Therefore

$$\lambda_{\parallel} = A, \quad \alpha = B, \quad \beta = C.$$

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XXXVI. 5. WHAT MUST BE COMPUTED CONCRETELY NOW

The folded-Bragg notes localize the remaining finite audit to the primitive-cell Fourier data

$$\widehat{U}_{\text{eff}}(G_*), \quad \widehat{T}_{\text{eff}}^1(G_*), \quad \widehat{T}_{\text{eff}}^2(G_*), \quad v_{\parallel}.$$

The constant intervalley Bragg block and enriched transverse blocks are controlled by

$$M_0 = \widehat{U}_{\text{eff}}(G_*) \mathbf{1}_2, \quad W^\alpha = \widehat{T}_{\text{eff}}^\alpha(G_*) s^\alpha.$$

The local Weyl data are then extracted from

$$K_i = \partial_{k_i} L_{\text{act}}(k) \Big|_{k=k_*}, \quad P_* K_i P_*.$$

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XXXVII. 6. HONEST NEXT STEP

So the next nonredundant computation is not another symbolic reduction.
It is exactly this finite operator build:

$$H_{\text{eff}}(k_* + p) \rightarrow H_{\text{shell}}(p) \rightarrow P_* \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (A, B, C)$$

with

$$(\lambda_{\parallel}, \alpha, \beta) = (A, B, C).$$

XXXVIII. EXPLICIT 4×4 FOLDED-BRAGG SHELL OPERATOR

XXXIX. 1. SHELL BASIS

Work in the folded-Bragg shell basis

$$\mathcal{H}_* = \text{span}\{\Phi_{+,1}, \Phi_{+,2}, \Phi_{-,1}, \Phi_{-,2}\},$$

with

$$\Phi_{+,a}(x) = e^{+iG_* \cdot x/2} u_a, \quad \Phi_{-,a}(x) = e^{-iG_* \cdot x/2} u_a, \quad a = 1, 2.$$

In this basis, the shell operator is a 4×4 block matrix of the form

$$H_{\text{shell}}(p) = \begin{pmatrix} E_+^{\text{eff}}(p) & B^{\text{eff}}(p) \\ B^{\text{eff}}(p)^\dagger & E_-^{\text{eff}}(p) \end{pmatrix},$$

where each block is 2×2 .

XL. 2. FIRST-ORDER EXPANSION NEAR $p = 0$

Expand the valley-diagonal blocks as

$$E_{\pm}^{\text{eff}}(p) = E_* \mathbf{1}_2 \pm v_{\parallel} p_{\parallel} \mathbf{1}_2 + O(|p|^2),$$

where $v_{\parallel}$ is extracted from the $p_{\parallel}$ -derivative of the valley-diagonal difference. The intervalley block expands as

$$B^{\text{eff}}(p) = M_0 + p_1 W^1 + p_2 W^2 + O(|p|^2),$$

with

$$M_0 = B^{\text{eff}}(E_*, 0), \quad W^i = \partial_{p_i} B^{\text{eff}}(E_*, p)|_{p=0}.$$

Therefore the explicit first-order shell operator is

$$H_{\text{shell}}(p) = E_* \mathbf{1}_4 + v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1}_2 + \tau^1 \otimes M_0 + p_1 \tau^1 \otimes W^1 + p_2 \tau^1 \otimes W^2 + O(|p|^2).$$

XLI. 3. TRANSVERSE SHELL MATRICES

The transverse shell matrices are

$$W^1 = \hat{T}_{\text{eff}}^1(G_*) s^1, \quad W^2 = \hat{T}_{\text{eff}}^2(G_*) s^2.$$

So the first-order operator becomes

$$H_{\text{shell}}(p) = E_* \mathbf{1}_4 + v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1}_2 + \tau^1 \otimes M_0 + p_1 \hat{T}_{\text{eff}}^1(G_*) \tau^1 \otimes s^1 + p_2 \hat{T}_{\text{eff}}^2(G_*) \tau^1 \otimes s^2 + O(|p|^2).$$

XLII. 4. MATRIX FORM

Writing

$$M_0 = \begin{pmatrix} m_{11} & m_{12} \\ \overline{m}_{12} & m_{22} \end{pmatrix}, \quad s^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad s^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix},$$

the shell operator is explicitly

$$H_{\text{shell}}(p) = \begin{pmatrix} E_* + v_{\parallel} p_{\parallel} & 0 & m_{11} & m_{12} + \widehat{T}_{\text{eff}}^1 p_1 - i \widehat{T}_{\text{eff}}^2 p_2 \\ 0 & E_* + v_{\parallel} p_{\parallel} & \overline{m}_{12} + \widehat{T}_{\text{eff}}^1 p_1 + i \widehat{T}_{\text{eff}}^2 p_2 & m_{22} \\ m_{11} & \overline{m}_{12} + \widehat{T}_{\text{eff}}^1 p_1 - i \widehat{T}_{\text{eff}}^2 p_2 & E_* - v_{\parallel} p_{\parallel} & 0 \\ m_{12} + \widehat{T}_{\text{eff}}^1 p_1 + i \widehat{T}_{\text{eff}}^2 p_2 & m_{22} & 0 & E_* - v_{\parallel} p_{\parallel} \end{pmatrix} + O(|p|^2).$$

Here $\widehat{T}_{\text{eff}}^{\alpha} = \widehat{T}_{\text{eff}}^{\alpha}(G_*)$.

XLIII. 5. PRINCIPAL SYMBOL AFTER THE SCALAR BRAGG SLOT IS REMOVED

On the graded bypass / carrier block, the constant scalar Bragg block is discarded, so the principal first-order shell symbol is

$$A_1(p) = v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1}_2 + p_1 \widehat{T}_{\text{eff}}^1(G_*) \tau^1 \otimes s^1 + p_2 \widehat{T}_{\text{eff}}^2(G_*) \tau^1 \otimes s^2.$$

If the transverse amplitudes are equal after calibration,

$$\widehat{T}_{\text{eff}}^1(G_*) = \widehat{T}_{\text{eff}}^2(G_*) =: v_{\perp},$$

then

$$A_1(p) = v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1}_2 + v_{\perp} (p_1 \tau^1 \otimes s^1 + p_2 \tau^1 \otimes s^2).$$

XLIV. 6. FINAL PROJECTED READOUT

Let P_* be the spectral projector onto the isolated two-dimensional low valley block. Define

$$K_i := \partial_{p_i} H_{\text{shell}}(p) \big|_{p=0}.$$

Then

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma^3 P_* K_{\parallel} P_*), \quad K_{\parallel} = \hat{n}_i K_i,$$

$$\alpha = \frac{1}{4} \sum_{a=1}^2 \text{Tr}(\sigma^a P_* (e_{ai} K_i) P_*),$$

$$\beta = \frac{1}{4} \left[\text{Tr}(\sigma^2 P_* (e_{1i} K_i) P_*) - \text{Tr}(\sigma^1 P_* (e_{2i} K_i) P_*) \right].$$

Equivalently, in the minimal axial ansatz

$$A_1(p) = A p_{\parallel} \sigma^3 + B(p_1 \sigma^1 + p_2 \sigma^2) + C(-p_2 \sigma^1 + p_1 \sigma^2),$$

one simply has

$$(\lambda_{\parallel}, \alpha, \beta) = (A, B, C).$$

XIV. DIRECT READOUT OF $v_{\parallel}$ AND $\hat{T}_{\text{eff}}^{1,2}(G_*)$

A. 1. Build the complete effective Bloch operator

Start from the complete microscopic Bloch Hessian

$$\mathcal{H}_{\text{Bloch}}^{\text{comp}}(k),$$

take the Schur complement against the remote sector, and obtain

$$H_{\text{eff}}^{\text{comp}}(k) = H_{\text{low,low}}(k) - H_{\text{low,rem}}(k) H_{\text{rem,rem}}(k)^{-1} H_{\text{rem,low}}(k).$$

Then project to the folded-Bragg shell subspace S_* :

$$H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}^{\text{comp}}(k_* + p) \Pi_{S_*}.$$

B. 2. Explicit 4×4 first-order shell form

In the folded basis

$$\{\Phi_{+,1}, \Phi_{+,2}, \Phi_{-,1}, \Phi_{-,2}\},$$

write

$$H_{\text{shell}}(p) = E_* \mathbf{1}_4 + v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1}_2 + \tau^1 \otimes M_0 + p_1 \tau^1 \otimes W^1 + p_2 \tau^1 \otimes W^2 + O(|p|^2).$$

The constant intervalley block is

$$M_0 = B^{\text{eff}}(E_*, 0),$$

and the first transverse derivatives are

$$W^i = \partial_{p_i} B^{\text{eff}}(E_*, p)|_{p=0}.$$

At the shell-fixed current route,

$$W^1 = \hat{T}_{\text{eff}}^1(G_*) s^1, \quad W^2 = \hat{T}_{\text{eff}}^2(G_*) s^2.$$

So the principal symbol is

$$A_1(p) = v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1}_2 + p_1 \hat{T}_{\text{eff}}^1(G_*) \tau^1 \otimes s^1 + p_2 \hat{T}_{\text{eff}}^2(G_*) \tau^1 \otimes s^2.$$

C. 3. Trace formulas for the three remaining inputs

Define

$$K_{\parallel} := \partial_{p_{\parallel}} H_{\text{shell}}(p)|_0, \quad K_1 := \partial_{p_1} H_{\text{shell}}(p)|_0, \quad K_2 := \partial_{p_2} H_{\text{shell}}(p)|_0.$$

Then the raw shell coefficients are read by

$$v_{\parallel} = \frac{1}{4} \text{Tr}[(\tau^3 \otimes \mathbf{1}_2) K_{\parallel}],$$

$$\hat{T}_{\text{eff}}^1(G_*) = \frac{1}{4} \text{Tr}[(\tau^1 \otimes s^1) K_1],$$

$$\widehat{T}_{\text{eff}}^2(G_*) = \frac{1}{4} \text{Tr}[(\tau^1 \otimes s^2) K_2].$$

Equivalently, before taking derivatives one may read them from the block decomposition:

$$v_{\parallel} = \frac{1}{4} \text{Tr}[(\tau^3 \otimes \mathbf{1}_2) \partial_{p_{\parallel}} H_{\text{shell}}(p)|_0],$$

$$\widehat{T}_{\text{eff}}^1(G_*) = \frac{1}{2} \text{Tr}_{2 \times 2} [s^1 \partial_{p_1} B^{\text{eff}}(E_*, p)|_0],$$

$$\widehat{T}_{\text{eff}}^2(G_*) = \frac{1}{2} \text{Tr}_{2 \times 2} [s^2 \partial_{p_2} B^{\text{eff}}(E_*, p)|_0].$$

D. 4. Final reduced coefficients from the projected two-band block

After projecting to the isolated low valley block P_* ,

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma^3 P_* K_{\parallel} P_*),$$

$$\alpha = \frac{1}{2} \left[\frac{1}{2} \text{Tr}(\sigma^1 P_* K_1 P_*) + \frac{1}{2} \text{Tr}(\sigma^2 P_* K_2 P_*) \right],$$

$$\beta = \frac{1}{2} \left[\frac{1}{2} \text{Tr}(\sigma^2 P_* K_1 P_*) - \frac{1}{2} \text{Tr}(\sigma^1 P_* K_2 P_*) \right].$$

In the minimal axial ansatz this reduces simply to

$$(\lambda_{\parallel}, \alpha, \beta) = (v_{\parallel}, \widehat{T}_{\text{eff}}^1(G_*), \widehat{T}_{\text{eff}}^2(G_*))$$

up to the final calibration/projection normalization of the chosen P_* -basis.

E. 5. What must be computed literally now

So the literal remaining computation is only:

$$(i) \text{ build } H_{\text{eff}}^{\text{comp}}(k), \quad (ii) \text{ restrict to } H_{\text{shell}}(p), \quad (iii) \text{ differentiate at } p = 0, \quad (iv) \text{ evaluate the three traces above.}$$

No further structural reduction is needed.

XLVI. IMMEDIATE FULL-MICROSCOPIC ROUTE

$$\mathcal{H}_{\text{Bloch}}^{\text{comp}}(k) \rightarrow H_{\text{eff}}^{\text{comp}}(k) \rightarrow H_{\text{shell}}(p) \rightarrow P_* K_i P_* \rightarrow (\lambda_c, \Lambda_E, \Lambda_O) \rightarrow (\lambda_{\parallel}, \alpha, \beta).$$

$$\mathcal{L}_{\text{mix}}^{\text{comp}} = \alpha_X X_i^a J_i^a + \beta_X X_i^a K_i^a + \gamma_L \mathcal{I}_L + \gamma_E \mathcal{I}_E + \gamma_O \mathcal{I}_O + \text{h.c.}$$

After eliminating X_i^a ,

$$\mathcal{L}_{\text{eff}}^{(X)} = -\frac{1}{2M_X^2} \mathcal{J}_i^{a,\text{tot}} \mathcal{J}_i^{a,\text{tot}}.$$

On the shell-adapted seed family,

$$g_c = \nu_{\parallel} \Gamma c_{\parallel}, \quad g_E = \nu_{\perp} \Gamma c_I, \quad g_O = \nu_{\perp} \Gamma c_J, \quad \Gamma = \frac{\lambda g_X m}{M_X^2}.$$

Therefore the full microscopic closure is reduced to actual evaluation of

$$\gamma_L, \gamma_E, \gamma_O \quad \text{equivalently} \quad (\lambda_c, \Lambda_E, \Lambda_O).$$

XLVII. LOWEST-ORDER COMPLETE MICROSCOPIC SLOT REALIZATION

XLVIII. 1. THREE EXACT LOWEST-ORDER CARRIER SLOTS

On the complete minimal extra mixing sector, the three lowest-order invariant candidates are

$$\mathcal{I}_L^{(2)} = X_i^a n^i \text{Tr} \left[T^a (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} \right], \quad (35)$$

$$\mathcal{I}_E^{(1,1)} = X_i^a \mathcal{E}^{ia}, \quad (36)$$

$$\mathcal{I}_O^{(1,1)} = X_i^a \mathcal{O}^{ia}. \quad (37)$$

These generate, respectively,

$$V_{(L)}^{\mu}(\xi) \propto \theta'(\xi)^3 n^{\mu} \sigma_3, \quad V_{(E)}^{\mu}(\xi) \propto \theta'(\xi) E^{\mu}{}_a \sigma_a, \quad V_{(O)}^{\mu}(\xi) \propto \theta'(\xi) J^{\mu}{}_a \sigma_a.$$

Hence the complete lowest-order extra mixing sector is

$$\mathcal{L}_{\text{mix}}^{\text{extra,LO}} = \gamma_L \mathcal{I}_L^{(2)} + \gamma_E \mathcal{I}_E^{(1,1)} + \gamma_O \mathcal{I}_O^{(1,1)} + \text{h.c.} \quad (38)$$

XLIX. 2. EXACT HEAVY-FIELD ELIMINATION

Introduce the total X -source

$$\mathcal{J}_i^{a,\text{tot}} = \alpha_X J_i^a[\Psi] + \beta_X K_i^a[P, \psi] + \gamma_L \mathfrak{L}_i^a + \gamma_E \mathfrak{E}_i^a + \gamma_O \mathfrak{D}_i^a, \quad (39)$$

where

$$\mathfrak{L}_i^a, \quad \mathfrak{E}_i^a, \quad \mathfrak{D}_i^a$$

are the local operator densities corresponding to $\mathcal{I}_L^{(2)}, \mathcal{I}_E^{(1,1)}, \mathcal{I}_O^{(1,1)}$.

Then the X -dependent piece is

$$\mathcal{L}_X^{\text{tot}} = \frac{1}{2} M_X^2 X_i^a X_i^a + X_i^a \mathcal{J}_i^{a,\text{tot}}. \quad (40)$$

Integrating out X_i^a gives exactly

$$\mathcal{L}_{\text{eff}}^{(X)} = -\frac{1}{2M_X^2} \mathcal{J}_i^{a,\text{tot}} \mathcal{J}_i^{a,\text{tot}}. \quad (41)$$

L. 3. SHELL-ADAPTED SEED FAMILY AND INDUCED THREE-DIRECTION COEFFICIENTS

At the explicit Class II quadratic template level, the reduced X -sector is

$$L_X^{(2)} = \frac{M_X^2}{2} X_i^a X_i^a + \lambda m^a X_i^a \partial_i \phi_0 + g_X X_i^a C_{ij}^{ab} (D_j \psi) \xi^b + \text{h.c.} \quad (42)$$

Integrating out X_i^a yields

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}. \quad (43)$$

Now choose the shell-adapted seed family

$$m^a = m \delta^{a3}, \quad C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j, \quad (44)$$

$$C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}. \quad (45)$$

Define

$$\Gamma := \frac{\lambda g_X m}{M_X^2}. \quad (46)$$

Then the three shell-reduced axial coefficients are

$$\delta A = -\Gamma c_{\parallel}, \quad \delta B = -\Gamma c_I, \quad \delta C = -\Gamma c_J. \quad (47)$$

Using the exact shell/defect dictionary

$$A_{\text{eff}} \leftrightarrow g_c, \quad B_{\text{eff}} \leftrightarrow g_E, \quad C_{\text{eff}} \leftrightarrow g_O,$$

we obtain

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J. \quad (48)$$

LI. 4. DIRECT REDUCED-COEFFICIENT DICTIONARY

The exact local defect/projector dictionary is

$$\lambda_{\parallel} = \frac{1}{2} g_c I_{\theta'^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}. \quad (49)$$

Equivalently, in the carrier notation already fixed in the working notes,

$$g_c \leftrightarrow \lambda_c, \quad g_E \leftrightarrow \Lambda_E, \quad g_O \leftrightarrow \Lambda_O.$$

So the complete lowest-order closure condition is simply

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0), \quad (50)$$

or equivalently

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0). \quad (51)$$

LII. WITNESS FUNCTIONALS FOR THE FINAL CLASS II COEFFICIENT READOUT

LIII. 1. AXIAL TENSOR FAMILY

In the shell-adapted frame $(\hat{n}, e_1, e_2)$, the shell-relevant axial tensor family is

$$\boxed{N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}.} \quad (52)$$

LIV. 2. EXACT WITNESS FUNCTIONALS

The exact witness functionals are

$$\boxed{W_{\parallel}[N] := \hat{n}_i \hat{n}_j N_{ij}^3,} \quad (53)$$

$$\boxed{W_I[N] := \frac{1}{2} \left(e_{1j} \hat{n}_i N_{ij}^1 + e_{2j} \hat{n}_i N_{ij}^2 \right),} \quad (54)$$

$$\boxed{W_J[N] := \frac{1}{2} \left(e_{1j} \hat{n}_i N_{ij}^2 - e_{2j} \hat{n}_i N_{ij}^1 \right).} \quad (55)$$

For any tensor in (52), these extract exactly

$$\boxed{W_{\parallel}[N] = A, \quad W_I[N] = B, \quad W_J[N] = C.} \quad (56)$$

So the final Class II coefficient problem is reduced to constructing the microscopic tensor N_{ij}^a and applying these three linear functionals.

IV. 3. MINIMAL CLASS II SEED FAMILY

Take the canonical shell-adapted Class II seed family

$$\boxed{m^a(P_0) = m \delta^{a3},} \quad (57)$$

$$\boxed{C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j,} \quad (58)$$

$$\boxed{C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}.} \quad (59)$$

Inside the exact concrete quadratic Class II reduction,

$$\boxed{L_X^{(2)} = \frac{M_X^2}{2} X_i^a X_i^a + \lambda m^a X_i^a \partial_i \phi_0 + g_X X_i^a C_{ij}^{ab} (D_j \psi) \xi^b + \text{h.c.}} \quad (60)$$

and integrating out X_i^a gives

$$\boxed{\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}.} \quad (61)$$

Define

$$\boxed{\Gamma := \frac{\lambda g_X m}{M_X^2}.} \quad (62)$$

Then the induced axial coefficients are

$$\boxed{\delta A = -\Gamma c_{\parallel}, \quad \delta B = -\Gamma c_I, \quad \delta C = -\Gamma c_J.} \quad (63)$$

LVI. 4. DIRECT READOUT OF THE THREE SEED COEFFICIENTS

Applying (56) to δN , we get

$$\boxed{W_{\parallel}[\delta N] = -\Gamma c_{\parallel}}, \quad (64)$$

$$\boxed{W_I[\delta N] = -\Gamma c_I}, \quad (65)$$

$$\boxed{W_J[\delta N] = -\Gamma c_J}. \quad (66)$$

Hence the full microscopic Class II coefficient problem is exactly the finite extraction problem for

$$c_{\parallel}, \quad c_I, \quad c_J.$$

LVII. 5. REDUCED PDE COEFFICIENT DICTIONARY

Using the local defect/projector transport dictionary,

$$A_{\text{eff}} \leftrightarrow g_c, \quad B_{\text{eff}} \leftrightarrow g_E, \quad C_{\text{eff}} \leftrightarrow g_O,$$

and therefore

$$\boxed{g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J}. \quad (67)$$

In the concrete Class II transport normalization,

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16} g_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} g_O \theta_0 \kappa}. \quad (68)$$

Therefore the exact nondegeneracy condition is

$$\boxed{g_c \neq 0, \quad (g_E, g_O) \neq (0, 0)}, \quad (69)$$

equivalently

$$\boxed{c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0)}. \quad (70)$$

LVIII. 6. LITERAL REMAINING MICROSCOPIC COMPUTATION

Compute the actual microscopic projected tensor

$$N_{ij}^{a(\text{mic})}$$

from the final Bloch harmonic matrices $H_{\Psi X}(k)$ and $H_{XX}(k)$, then evaluate

$$\boxed{W_{\parallel}[N^{(\text{mic})}], \quad W_I[N^{(\text{mic})}], \quad W_J[N^{(\text{mic})}]}.$$

This yields

$$\boxed{c_{\parallel}^{(\text{mic})}, \quad c_I^{(\text{mic})}, \quad c_J^{(\text{mic})}},$$

and hence

$$\boxed{g_c, \quad g_E, \quad g_O},$$

and finally

$$\boxed{\lambda_{\parallel}, \quad \alpha, \quad \beta}.$$

LIX. EXACT PROJECTION FORMULA FROM $K_i^{(\text{II})}$ TO $N_{ij}^{a(\text{mic})}$

A. 1. Exact microscopic source

The exact Schur-complement derivative formula is

$$K_i^{(\text{II})} = -\Pi_{S_*} \left[\partial_{k_i} \left(H_{\Psi X}(k) H_{XX}(k)^{-1} H_{X\Psi}(k) \right) \right]_{k=k_*} \Pi_{S_*}.$$

At strict heavy-mass leading order,

$$K_i^{(\text{II})} = -\frac{1}{M_X^2} \Pi_{S_*} \left[(\partial_{k_i} \mathcal{C}) \mathcal{C}^\dagger + \mathcal{C} (\partial_{k_i} \mathcal{C}^\dagger) \right]_{k=k_*} \Pi_{S_*}, \quad \mathcal{C}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k).$$

B. 2. Shell-enriched tensor definition

After shell reduction, the induced first-order Class II operator is defined by

$$\delta L_{\text{enr}}^{(\text{II})} = N_{ij}^{a(\text{II})} (\partial_i \phi_0) (-i \partial_j) \otimes s^a.$$

So $N_{ij}^{a(\text{mic})}$ is not a new ansatz. It is the unique shell-enriched tensor obtained by matching the projected first-order operator generated by $K_i^{(\text{II})}$ to the canonical shell-enriched form above. [:contentReference\[oaicite:2\]index=2](#)
[:contentReference\[oaicite:3\]index=3](#)

C. 3. Exact shell seed decomposition

The shell witness theorem fixes the only three allowed seed channels:

$$N_{ij}^{a(\text{II})} = c_{\parallel} B_{\parallel ij}^a + c_I B_{I ij}^a + c_J B_{J ij}^a,$$

with

$$B_{\parallel ij}^a = \hat{n}_i \hat{n}_j \delta^{a3},$$

$$B_{I ij}^a = \hat{n}_i e_{1j} \delta^{a1} + \hat{n}_i e_{2j} \delta^{a2},$$

$$B_{J ij}^a = \hat{n}_i e_{2j} \delta^{a1} - \hat{n}_i e_{1j} \delta^{a2}.$$

So the whole microscopic Class II coefficient problem is reduced to the three scalars

$$c_{\parallel}, \quad c_I, \quad c_J.$$

[:contentReference\[oaicite:4\]index=4](#) [:contentReference\[oaicite:5\]index=5](#)

D. 4. Exact witness functionals

Define the three witness functionals

$$W_{\parallel}[N] := \hat{n}_i \hat{n}_j N_{ij}^3,$$

$$W_I[N] := \frac{1}{2} \left(e_{1j} \hat{n}_i N_{ij}^1 + e_{2j} \hat{n}_i N_{ij}^2 \right),$$

$$W_J[N] := \frac{1}{2} \left(e_{1j} \hat{n}_i N_{ij}^2 - e_{2j} \hat{n}_i N_{ij}^1 \right).$$

For any tensor in the seed family,

$$W_{\parallel}[N] = c_{\parallel}, \quad W_I[N] = c_I, \quad W_J[N] = c_J.$$

Therefore the exact projection formula is simply

$$c_{\parallel} = W_{\parallel}[N^{(\text{mic})}], \quad c_I = W_I[N^{(\text{mic})}], \quad c_J = W_J[N^{(\text{mic})}].$$

This is the wanted exact projection map from $K_i^{(\text{II})}$ to the three microscopic Class II coefficients. :contentReference[oaicite:6]index=6

E. 5. Matching to the reduced transport coefficients

The reduced shell coefficients are then

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J.$$

On the shell-level reduced Dirac symbol,

$$\lambda_{\parallel} = \kappa_* |G_*| c_{\parallel}, \quad \alpha = \kappa_* |G_*| c_I, \quad \beta = \kappa_* |G_*| c_J,$$

or equivalently, in the local defect/projector dictionary,

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}.$$

So once $c_{\parallel}, c_I, c_J$ are computed from the actual Bloch harmonic matrices, the actual reduced PDE coefficients are fixed. :contentReference[oaicite:7]index=7 :contentReference[oaicite:8]index=8 :contentReference[oaicite:9]index=9

F. 6. Literal remaining computation

The honest last open step is now exactly:

$$\text{compute } \mathcal{J}(k), \mathcal{K}(k), H_{XX}(k) \text{ on the final working branch,}$$

$$\text{build } K_i^{(\text{II})}, \text{ match it to } N_{ij}^{a(\text{mic})}, \text{ and apply } W_{\parallel}, W_I, W_J.$$

Nothing structural remains beyond this finite coefficient extraction. :contentReference[oaicite:10]index=10 :contentReference[oaicite:11]index=11

LX. FINAL SUMMARY

The exact microscopic Class II projection route is

$$\begin{aligned}
& \boxed{K_i^{(\text{II})} = -\Pi_{S_*} [\partial_{k_i} (H_{\Psi X} H_{XX}^{-1} H_{X\Psi})]_{k=k_*} \Pi_{S_*}} \\
& \implies \boxed{\delta L_{\text{enr}}^{(\text{II})} = N_{ij}^{a(\text{II})} (\partial_i \phi_0) (-i \partial_j) \otimes s^a} \\
& \implies \boxed{N_{ij}^{a(\text{II})} = c_{\parallel} B_{\parallel ij}^a + c_I B_{I ij}^a + c_J B_{J ij}^a} \\
& \implies \boxed{c_{\parallel} = W_{\parallel}[N^{(\text{mic})}], \quad c_I = W_I[N^{(\text{mic})}], \quad c_J = W_J[N^{(\text{mic})}]} \\
& \implies \boxed{(\lambda_{\parallel}, \alpha, \beta) = (\kappa_* |G_*| c_{\parallel}, \kappa_* |G_*| c_I, \kappa_* |G_*| c_J)}
\end{aligned}$$

or equivalently

$$\boxed{\lambda_{\parallel} = \frac{g_c}{2} I_{\theta^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}.}$$

So the remaining microscopic frontier is now exactly the finite extraction of

$$\boxed{c_{\parallel}, \quad c_I, \quad c_J}$$

from the actual Bloch harmonic matrices.

LXI. IMMEDIATE EXPLICIT CALCULATIONS ON THE MIXED ODD-ACTIVE TARGET BRANCH

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3}$$

$$\boxed{\alpha^2 + \beta^2 = 2 \left(\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa \right)^2 = \frac{3\pi^2}{128} \beta_X^2 \theta_0^2 \kappa^2}$$

Hence the 2×2 Dirac/Weyl determinant factor is

$$\boxed{\det M = \lambda_{\parallel} (\alpha^2 + \beta^2) = -\frac{9\sqrt{3}\pi^3}{65536} \alpha_X \beta_X^2 L_* \theta_0^5 \kappa^5.}$$

In rotated transverse coordinates

$$q_1 = \frac{p_1 - p_2}{\sqrt{2}}, \quad q_2 = \frac{p_1 + p_2}{\sqrt{2}},$$

the transverse velocity is

$$\boxed{v_{\perp} = \sqrt{2} |\alpha| = \frac{\sqrt{6}\pi}{16} |\beta_X \theta_0 \kappa|.}$$

Therefore the branch-level dispersion is

$$E(p)^2 = \left(\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3 \right)^2 p_{\parallel}^2 + \frac{3\pi^2}{128} \beta_X^2 \theta_0^2 \kappa^2 (q_1^2 + q_2^2).$$

Equivalently, since $q_1^2 + q_2^2 = p_1^2 + p_2^2$,

$$E(p)^2 = \left(\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3 \right)^2 p_{\parallel}^2 + \frac{3\pi^2}{128} \beta_X^2 \theta_0^2 \kappa^2 (p_1^2 + p_2^2).$$

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_{0;OO}^{(3)}, \quad \alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa E_*, \quad \beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa O_*.$$

$$\left[i\partial_t + \frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_{0;OO}^{(3)} \sigma_3(-i\partial_{\parallel}) + \frac{\sqrt{3}\pi}{16} \theta_0 \kappa E_* (\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) + \frac{\sqrt{3}\pi}{16} \theta_0 \kappa O_* (-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0.$$

$$\frac{\sqrt{3}\pi}{16} \theta_0 \kappa R_* (\sigma_1(-i\partial_{q_1}) + \sigma_2(-i\partial_{q_2}))$$

$$\left[i\partial_t + \frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_{0;OO}^{(3)} \sigma_3(-i\partial_{\parallel}) + \frac{\sqrt{3}\pi}{16} \theta_0 \kappa R_* (\sigma_1(-i\partial_{q_1}) + \sigma_2(-i\partial_{q_2})) \right] \chi = 0.$$

$$E(p)^2 = \left(\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_{0;OO}^{(3)} \right)^2 p_{\parallel}^2 + \left(\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \right)^2 (E_*^2 + O_*^2) (p_1^2 + p_2^2).$$

$$E(p)^2 = \frac{27\pi^2}{65536} \theta_0^6 \kappa^6 (\ell_{0;OO}^{(3)})^2 p_{\parallel}^2 + \frac{3\pi^2}{256} \theta_0^2 \kappa^2 (E_*^2 + O_*^2) (p_1^2 + p_2^2).$$

$$\det M_{\text{Dirac}} = \lambda_{\parallel} (\alpha^2 + \beta^2) = -\frac{9\sqrt{3}\pi^3}{65536} \theta_0^5 \kappa^5 \ell_{0;OO}^{(3)} (E_*^2 + O_*^2).$$

$$\ell_{0;OO}^{(3)} \neq 0, \quad E_*^2 + O_*^2 \neq 0.$$

$$\ell_{0;OO}^{(3)} \neq 0, \quad (e_{0;K}^{(1)} + \Delta e_{\text{mix}})^2 + (o_{0;K}^{(1)} + \Delta o_{\text{mix}})^2 \neq 0.$$

LXII. PRINCIPAL + RESIDUAL TRANSVERSE PAIR: FINAL INSERTION

LXIII. 1. PRINCIPAL TRANSVERSE PAIR FROM $TD + DT$

The exact projected mixed block is

$$\mathcal{M}_{TD+DT}^{\mu} = \Pi_Q (\mathcal{V}_{TD}^{\mu} + \mathcal{V}_{DT}^{\mu}) \Pi_Q.$$

The invariant transverse projectors are

$$\mathfrak{P}_E[\mathcal{M}] = \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu), \quad \mathfrak{P}_O[\mathcal{M}] = \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu).$$

At the current minimal branch-level realization,

$$V_{TD}^\mu = \theta'(\xi) (A_E E^\mu{}_a \Sigma_a + A_O J^\mu{}_a \Sigma_a),$$

$$V_{DT}^\mu = \theta'(\xi) (B_E E^\mu{}_a \Sigma_a + B_O J^\mu{}_a \Sigma_a).$$

Therefore

$$\Lambda_E^{\text{pr}} = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}] = \frac{1}{2} (A_E + B_E),$$

$$\Lambda_O^{\text{pr}} = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}] = \frac{1}{2} (A_O + B_O).$$

LXIV. 2. RESIDUAL CORRECTION VECTOR FROM $OD + DO + DD$

Define

$$\Delta e_{\text{rem}} = e_{0;OD}^{(1)} + e_{0;DO}^{(1)} + e_{0;DD}^{(1)},$$

$$\Delta o_{\text{rem}} = o_{0;OD}^{(1)} + o_{0;DO}^{(1)} + o_{0;DD}^{(1)}.$$

Equivalently,

$$r_{\text{rem}} = (\Delta e_{\text{rem}}, \Delta o_{\text{rem}}).$$

Hence the full transverse pair is

$$\Lambda_E = \frac{1}{2} (A_E + B_E) + \Delta e_{\text{rem}}, \quad \Lambda_O = \frac{1}{2} (A_O + B_O) + \Delta o_{\text{rem}}.$$

LXV. 3. REDUCED PDE COEFFICIENTS

Using

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa,$$

the reduced coefficients are

$$\alpha = I_\theta \Lambda_E = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\frac{1}{2} (A_E + B_E) + \Delta e_{\text{rem}} \right],$$

$$\beta = I_\theta \Lambda_O = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\frac{1}{2} (A_O + B_O) + \Delta o_{\text{rem}} \right].$$

If we also keep the already-closed longitudinal branch coefficient,

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{256}\theta_0^3\kappa^3\ell_{0;OO}^{(3)},$$

then the full reduced PDE is

$$\left[i\partial_t + \frac{3\sqrt{3}\pi}{256}\theta_0^3\kappa^3\ell_{0;OO}^{(3)}\sigma_3(-i\partial_{\parallel}) + \frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(\frac{1}{2}(A_E + B_E) + \Delta e_{\text{rem}}\right)(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) \right. \\ \left. + \frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(\frac{1}{2}(A_O + B_O) + \Delta o_{\text{rem}}\right)(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0.$$

LXVI. 4. DISPERSION RELATION

The corresponding dispersion is

$$E(p)^2 = \lambda_{\parallel}^2 p_{\parallel}^2 + (\alpha^2 + \beta^2)(p_1^2 + p_2^2),$$

hence

$$E(p)^2 = \frac{27\pi^2}{65536}\theta_0^6\kappa^6(\ell_{0;OO}^{(3)})^2 p_{\parallel}^2 + \frac{3\pi^2}{256}\theta_0^2\kappa^2\left[\Lambda_E^2 + \Lambda_O^2\right](p_1^2 + p_2^2),$$

with

$$\Lambda_E^2 + \Lambda_O^2 = \left(\frac{1}{2}(A_E + B_E) + \Delta e_{\text{rem}}\right)^2 + \left(\frac{1}{2}(A_O + B_O) + \Delta o_{\text{rem}}\right)^2.$$

LXVII. 5. DETERMINANT / NONDEGENERACY SCALAR

The Dirac nondegeneracy scalar is

$$\det M_{\text{Dirac}} = \lambda_{\parallel}(\alpha^2 + \beta^2),$$

therefore

$$\det M_{\text{Dirac}} = -\frac{9\sqrt{3}\pi^3}{65536}\theta_0^5\kappa^5\ell_{0;OO}^{(3)}\left[\left(\frac{1}{2}(A_E + B_E) + \Delta e_{\text{rem}}\right)^2 + \left(\frac{1}{2}(A_O + B_O) + \Delta o_{\text{rem}}\right)^2\right].$$

LXVIII. 6. SHARP SURVIVAL CONDITIONS

A sufficient strong componentwise survival condition is

$$|\Delta e_{\text{rem}}| < \left|\frac{1}{2}(A_E + B_E)\right|, \quad |\Delta o_{\text{rem}}| < \left|\frac{1}{2}(A_O + B_O)\right|.$$

A weaker Euclidean sufficient condition is

$$\sqrt{(\Delta e_{\text{rem}})^2 + (\Delta o_{\text{rem}})^2} < \frac{1}{2}\sqrt{(A_E + B_E)^2 + (A_O + B_O)^2}.$$

Under either condition,

$$(\Lambda_E, \Lambda_O) \neq (0, 0),$$

hence

$$(\alpha, \beta) \neq (0, 0).$$

$$\Delta e_{\text{rem}} = e_{0;OD}^{(1)} + e_{0;DO}^{(1)} + e_{0;DD}^{(1)}, \quad \Delta o_{\text{rem}} = o_{0;OD}^{(1)} + o_{0;DO}^{(1)} + o_{0;DD}^{(1)}.$$

$$e_{0;AB}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} E^\mu{}_a \text{Tr}[\sigma_a \mathcal{M}_{AB,\mu}^{\text{inv}}],$$

$$o_{0;AB}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} J^\mu{}_a \text{Tr}[\sigma_a \mathcal{M}_{AB,\mu}^{\text{inv}}], \quad AB \in \{OD, DO, DD\}.$$

$$\Lambda_E = \Lambda_E^{\text{pr}} + \Delta e_{\text{rem}}, \quad \Lambda_O = \Lambda_O^{\text{pr}} + \Delta o_{\text{rem}},$$

$$\Lambda_E^{\text{pr}} = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad \Lambda_O^{\text{pr}} = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}].$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (\Lambda_E^{\text{pr}} + \Delta e_{\text{rem}}), \quad \beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (\Lambda_O^{\text{pr}} + \Delta o_{\text{rem}}).$$

$$r_{\text{rem}} = (\Delta e_{\text{rem}}, \Delta o_{\text{rem}}) = 2(e_{0;OD}^{(1)}, o_{0;OD}^{(1)}) + (e_{0;DD}^{(1)}, o_{0;DD}^{(1)}).$$

LXIX. RESIDUAL BLOCK PROJECTOR SHEET

For $AB \in \{OD, DO, DD\}$, define

$$e_{0;AB}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{AB,\mu}^{\text{inv}}),$$

$$o_{0;AB}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{AB,\mu}^{\text{inv}}).$$

Hence

$$\Delta e_{\text{rem}} = \lim \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B} \left[\Sigma_a (\mathcal{M}_{OD,\mu}^{\text{inv}} + \mathcal{M}_{DO,\mu}^{\text{inv}} + \mathcal{M}_{DD,\mu}^{\text{inv}}) \right],$$

$$\Delta o_{\text{rem}} = \lim \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B} \left[\Sigma_a (\mathcal{M}_{OD,\mu}^{\text{inv}} + \mathcal{M}_{DO,\mu}^{\text{inv}} + \mathcal{M}_{DD,\mu}^{\text{inv}}) \right].$$

LXX. EXCHANGE-SYMMETRIC REDUCTION

If the projected background and regulator preserve $O \leftrightarrow D$ exchange symmetry to leading order, then

$$\boxed{e_{0;DO}^{(1)} = e_{0;OD}^{(1)}, \quad o_{0;DO}^{(1)} = o_{0;OD}^{(1)}}.$$

Therefore the residual correction vector reduces to

$$\boxed{r_{\text{rem}} = 2(e_{0;OD}^{(1)}, o_{0;OD}^{(1)}) + (e_{0;DD}^{(1)}, o_{0;DD}^{(1)})}.$$

LXXI. PARITY SELECTION RULE

For any $AB \in \{OD, DO, DD\}$,

$$\boxed{\mathcal{M}_{AB,\text{inv}}^\mu(-\xi) = +\mathcal{M}_{AB,\text{inv}}^\mu(\xi) \implies e_{0;AB}^{(1)} = o_{0;AB}^{(1)} = 0.}$$

If instead

$$\boxed{\mathcal{M}_{AB,\text{inv}}^\mu(-\xi) = -\mathcal{M}_{AB,\text{inv}}^\mu(\xi),}$$

then the block is parity-allowed and must be evaluated explicitly.

Hence:

$$\boxed{\text{all even} \implies r_{\text{rem}} = 0,}$$

$$\boxed{\text{partial even/odd} \implies r_{\text{rem}} = \sum_{\substack{AB \in \{OD, DO, DD\} \\ \mathcal{M}_{AB} \text{ odd}}} (e_{0;AB}^{(1)}, o_{0;AB}^{(1)}),}$$

$$\boxed{\text{all odd} \implies r_{\text{rem}} = (e_{0;OD}^{(1)} + e_{0;DO}^{(1)} + e_{0;DD}^{(1)}, o_{0;OD}^{(1)} + o_{0;DO}^{(1)} + o_{0;DD}^{(1)})}.$$

LXXII. SHARPEST HONEST RESIDUAL FORM

$$\boxed{r_{\text{rem}} = 2(e_{0;OD}^{(1)}, o_{0;OD}^{(1)}) + (e_{0;DD}^{(1)}, o_{0;DD}^{(1)})}$$

LXXIII. ACTUAL COMPUTATION OF $\varepsilon_{\text{mix}}^{\text{prior}}$

Take

$$R_f^{\text{prior}} = \begin{pmatrix} -0.839960218174805 & 2.450804313533534 & -0.302298414680853 \\ 2.450804313533534 & -7.145183917627953 & 0.837818903051675 \\ -0.302298414680853 & 0.837818903051675 & 0.235363388813100 \end{pmatrix},$$

and

$$B = (t \mid n_1 \mid n_2), \quad t = \frac{1}{\sqrt{11}}(1, -3, 1)^T, \quad n_1 = \frac{1}{\sqrt{2}}(1, 0, -1)^T, \quad n_2 = \frac{1}{\sqrt{22}}(3, 2, 3)^T.$$

Then

$$\tilde{R}_f^{\text{prior}} = B^T R_f^{\text{prior}} B = \begin{pmatrix} -7.74978075 & -1.26092870 & 1.04286422 \\ -1.26092870 & 0 & 0 \\ 1.04286422 & 0 & 0 \end{pmatrix}.$$

Also

$$S_f^{\text{prior}} := \|R_f^{\text{prior}}\|_F^2 = (8.087899329725035)^2 = 65.41411556776667.$$

By definition,

$$\varepsilon_{\text{mix}}^2 = \frac{\tilde{r}_{12}^2 + \tilde{r}_{13}^2 + \tilde{r}_{21}^2 + \tilde{r}_{31}^2}{S_f}.$$

Therefore

$$\varepsilon_{\text{mix}}^{2 \text{ prior}} = \frac{(-1.26092870)^2 + (1.04286422)^2 + (-1.26092870)^2 + (1.04286422)^2}{65.41411556776667} = 0.08186327820648276,$$

hence

$$\boxed{\varepsilon_{\text{mix}}^{\text{prior}} = 0.2861175950662293.}$$

LXXIV. 12-POINT EXTRACTION TABLE FOR THE TRUE CORRECTED JACOBIAN

Take

$$\delta_f = (0.0096, -0.0288, 0.0096), \quad e_A = (1, 0, 0)^T, \quad e_E = (0, 1, 0)^T, \quad e_O = (0, 0, 1)^T.$$

Choose

$$\varepsilon = 10^{-4}, \quad \varepsilon/2 = 5 \times 10^{-5}.$$

Evaluate the 12 vectors:

$$G(\delta_f + \varepsilon e_A), \quad G(\delta_f - \varepsilon e_A),$$

$$G(\delta_f + \varepsilon e_E), \quad G(\delta_f - \varepsilon e_E),$$

$$G(\delta_f + \varepsilon e_O), \quad G(\delta_f - \varepsilon e_O),$$

and the same six evaluations at $\varepsilon/2$.

Then form

$$(J_f^{[\varepsilon]})_{\cdot A} = \frac{G(\delta_f + \varepsilon e_A) - G(\delta_f - \varepsilon e_A)}{2\varepsilon},$$

$$(J_f^{[\varepsilon]})_{\cdot E} = \frac{G(\delta_f + \varepsilon e_E) - G(\delta_f - \varepsilon e_E)}{2\varepsilon},$$

$$(J_f^{[\varepsilon]})_{\cdot O} = \frac{G(\delta_f + \varepsilon e_O) - G(\delta_f - \varepsilon e_O)}{2\varepsilon}.$$

Likewise build $J_f^{[\varepsilon/2]}$, and then Richardson-improve:

$$\boxed{J_f^{\text{Rich}} = \frac{4J_f^{[\varepsilon/2]} - J_f^{[\varepsilon]}}{3}.}$$

Finally define

$$R_f := J_f^{\text{Rich}} - J_f^{(4)},$$

and compute

$$\alpha_f = \frac{\|R_f\|_2}{\lambda_*}, \quad \varepsilon_{\text{diag}}, \quad \varepsilon_{\text{sym}}, \quad \varepsilon_{\text{mix}}.$$

LXXV. TRUE J_f EXTRACTION SHEET AT THE CORRECTED SEED

$$\delta_f = (0.0096, -0.0288, 0.0096), \quad e_A = (1, 0, 0), \quad e_E = (0, 1, 0), \quad e_O = (0, 0, 1).$$

Choose

$$\varepsilon = 10^{-4}, \quad \varepsilon/2 = 5 \times 10^{-5}.$$

A. A. Six evaluations at ε

$$\delta_f + \varepsilon e_A = (0.0097, -0.0288, 0.0096),$$

$$\delta_f - \varepsilon e_A = (0.0095, -0.0288, 0.0096),$$

$$\delta_f + \varepsilon e_E = (0.0096, -0.0287, 0.0096),$$

$$\delta_f - \varepsilon e_E = (0.0096, -0.0289, 0.0096),$$

$$\delta_f + \varepsilon e_O = (0.0096, -0.0288, 0.0097),$$

$$\delta_f - \varepsilon e_O = (0.0096, -0.0288, 0.0095).$$

B. B. Six evaluations at $\varepsilon/2$

$$\delta_f + \frac{\varepsilon}{2} e_A = (0.00965, -0.0288, 0.0096),$$

$$\delta_f - \frac{\varepsilon}{2} e_A = (0.00955, -0.0288, 0.0096),$$

$$\delta_f + \frac{\varepsilon}{2} e_E = (0.0096, -0.02875, 0.0096),$$

$$\delta_f - \frac{\varepsilon}{2} e_E = (0.0096, -0.02885, 0.0096),$$

$$\delta_f + \frac{\varepsilon}{2} e_O = (0.0096, -0.0288, 0.00965),$$

$$\delta_f - \frac{\varepsilon}{2} e_O = (0.0096, -0.0288, 0.00955).$$

LXXVI. CENTERED-DIFFERENCE JACOBIANS

$$(J_f^{[\varepsilon]})_{.A} = \frac{G(\delta_f + \varepsilon e_A) - G(\delta_f - \varepsilon e_A)}{2\varepsilon},$$

$$(J_f^{[\varepsilon]})_{.E} = \frac{G(\delta_f + \varepsilon e_E) - G(\delta_f - \varepsilon e_E)}{2\varepsilon},$$

$$(J_f^{[\varepsilon]})_{.O} = \frac{G(\delta_f + \varepsilon e_O) - G(\delta_f - \varepsilon e_O)}{2\varepsilon}.$$

Likewise build $J_f^{[\varepsilon/2]}$, and then use the Richardson upgrade

$$J_f^{\text{Rich}} = \frac{4J_f^{[\varepsilon/2]} - J_f^{[\varepsilon]}}{3}.$$

LXXVII. CLASSIFIER FORMULAS

$$\alpha_f = \frac{\|R_f\|_2}{\lambda_*}.$$

$$\varepsilon_{\text{diag}} = \frac{\|R_f - \text{diag}(R_f)\|_F}{\max(\|R_f\|_F, \epsilon_0)}.$$

$$\varepsilon_{\text{sym}} = \frac{\left\| \frac{1}{2}(R_f - R_f^T) \right\|_F}{\max(\|R_f\|_F, \epsilon_0)}.$$

Use the branch-adapted basis

$$t = \frac{1}{\sqrt{11}}(1, -3, 1)^T, \quad n_1 = \frac{1}{\sqrt{2}}(1, 0, -1)^T, \quad n_2 = \frac{1}{\sqrt{22}}(3, 2, 3)^T,$$

and let

$$B = (t \mid n_1 \mid n_2), \quad R_{f,B} = B^T R_f B.$$

Define the tangent-coupling projector

$$\Pi_{\mathcal{T}}(R_{f,B}) = \begin{pmatrix} r_{tt} & r_{t1} & r_{t2} \\ r_{1t} & 0 & 0 \\ r_{2t} & 0 & 0 \end{pmatrix}.$$

Then

$$\varepsilon_{\text{mix}} = \frac{\|R_{f,B} - \Pi_{\mathcal{T}}(R_{f,B})\|_F}{\max(\|R_f\|_F, \epsilon_0)}.$$

LXXVIII. ONE-SHOT COMPUTATION SHEET FOR THE TRUE CORRECTED JACOBIAN

LXXIX. 1. INPUT NOTATION FOR THE 12 RESIDUAL EVALUATIONS

Let

$$g_A^\pm := G(\delta_f \pm \varepsilon e_A), \quad g_E^\pm := G(\delta_f \pm \varepsilon e_E), \quad g_O^\pm := G(\delta_f \pm \varepsilon e_O),$$

and similarly at half-step,

$$\tilde{g}_A^\pm := G\left(\delta_f \pm \frac{\varepsilon}{2} e_A\right), \quad \tilde{g}_E^\pm := G\left(\delta_f \pm \frac{\varepsilon}{2} e_E\right), \quad \tilde{g}_O^\pm := G\left(\delta_f \pm \frac{\varepsilon}{2} e_O\right).$$

Write the components as

$$g_Y^\pm = \begin{pmatrix} g_{A;Y}^\pm \\ g_{E;Y}^\pm \\ g_{O;Y}^\pm \end{pmatrix}, \quad \tilde{g}_Y^\pm = \begin{pmatrix} \tilde{g}_{A;Y}^\pm \\ \tilde{g}_{E;Y}^\pm \\ \tilde{g}_{O;Y}^\pm \end{pmatrix}, \quad Y \in \{A, E, O\}.$$

LXXX. 2. RAW CENTERED-DIFFERENCE JACOBIANS

The ε -Jacobian is

$$J_f^{[\varepsilon]} = \begin{pmatrix} \frac{g_{A;A}^+ - g_{A;A}^-}{2\varepsilon} & \frac{g_{A;E}^+ - g_{A;E}^-}{2\varepsilon} & \frac{g_{A;O}^+ - g_{A;O}^-}{2\varepsilon} \\ \frac{g_{E;A}^+ - g_{E;A}^-}{2\varepsilon} & \frac{g_{E;E}^+ - g_{E;E}^-}{2\varepsilon} & \frac{g_{E;O}^+ - g_{E;O}^-}{2\varepsilon} \\ \frac{g_{O;A}^+ - g_{O;A}^-}{2\varepsilon} & \frac{g_{O;E}^+ - g_{O;E}^-}{2\varepsilon} & \frac{g_{O;O}^+ - g_{O;O}^-}{2\varepsilon} \end{pmatrix}.$$

The half-step Jacobian is

$$J_f^{[\varepsilon/2]} = \begin{pmatrix} \frac{\tilde{g}_{A;A}^+ - \tilde{g}_{A;A}^-}{\varepsilon} & \frac{\tilde{g}_{A;E}^+ - \tilde{g}_{A;E}^-}{\varepsilon} & \frac{\tilde{g}_{A;O}^+ - \tilde{g}_{A;O}^-}{\varepsilon} \\ \frac{\tilde{g}_{E;A}^+ - \tilde{g}_{E;A}^-}{\varepsilon} & \frac{\tilde{g}_{E;E}^+ - \tilde{g}_{E;E}^-}{\varepsilon} & \frac{\tilde{g}_{E;O}^+ - \tilde{g}_{E;O}^-}{\varepsilon} \\ \frac{\tilde{g}_{O;A}^+ - \tilde{g}_{O;A}^-}{\varepsilon} & \frac{\tilde{g}_{O;E}^+ - \tilde{g}_{O;E}^-}{\varepsilon} & \frac{\tilde{g}_{O;O}^+ - \tilde{g}_{O;O}^-}{\varepsilon} \end{pmatrix}.$$

LXXXI. 3. RICHARDSON-UPGRADED TRUE JACOBIAN

The exact practical extraction object is

$$J_f := J_f^{\text{Rich}} = \frac{4J_f^{[\varepsilon/2]} - J_f^{[\varepsilon]}}{3}.$$

So entrywise,

$$(J_f)_{XY} = \frac{1}{3} \left[4 \frac{\tilde{g}_{X;Y}^+ - \tilde{g}_{X;Y}^-}{\varepsilon} - \frac{g_{X;Y}^+ - g_{X;Y}^-}{2\varepsilon} \right], \quad X, Y \in \{A, E, O\}.$$

LXXXII. 4. SUBTRACT THE FIXED QUARTIC BASELINE

Let

$$J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665531859216 & 0.4137780199567714 \\ -4.991665531859216 & 22.646508448870943 & -1.2413340598703142 \\ 0.4137780199567714 & -1.2413340598703142 & 3.603395005929136 \end{pmatrix}.$$

Define the exact remainder

$$R_f := J_f - J_f^{(4)} = (r_{ij})_{i,j=1}^3.$$

Hence, for example,

$$r_{11} = (J_f)_{AA} - 10.095240102467715, \quad r_{12} = (J_f)_{AE} + 4.991665531859216,$$

$$r_{13} = (J_f)_{AO} - 0.4137780199567714, \quad r_{22} = (J_f)_{EE} - 22.646508448870943,$$

$$r_{23} = (J_f)_{EO} + 1.2413340598703142, \quad r_{33} = (J_f)_{OO} - 3.603395005929136,$$

and similarly for the symmetric partner entries.

LXXXIII. 5. FROBENIUS SIZE AND INVERTIBILITY CLASSIFIER

Set

$$S_f := \|R_f\|_F^2 = \sum_{i,j=1}^3 r_{ij}^2.$$

Then the hard perturbative size parameter is

$$\hat{\alpha}_f = \frac{\sqrt{S_f}}{\lambda_*}, \quad \lambda_* \approx 3.52129589.$$

If one prefers the spectral norm, use

$$\alpha_f = \frac{\|R_f\|_2}{\lambda_*}.$$

The file's exact decision theorem uses $\hat{\alpha}_f$ as the entry-only proxy.

LXXXIV. 6. DIAGONAL AND ANTISYMMETRIC FRACTIONS

The diagonal-failure fraction is

$$\varepsilon_{\text{diag}}^2 = \frac{r_{12}^2 + r_{13}^2 + r_{21}^2 + r_{23}^2 + r_{31}^2 + r_{32}^2}{S_f}.$$

The antisymmetric fraction is

$$\varepsilon_{\text{sym}}^2 = \frac{(r_{12} - r_{21})^2 + (r_{13} - r_{31})^2 + (r_{23} - r_{32})^2}{2S_f}.$$

LXXXV. 7. ADAPTED-BASIS MIXING FRACTION

Use the branch-adapted orthonormal basis

$$t = \frac{1}{\sqrt{11}}(1, -3, 1)^T, \quad n_1 = \frac{1}{\sqrt{2}}(1, 0, -1)^T, \quad n_2 = \frac{1}{\sqrt{22}}(3, 2, 3)^T,$$

and

$$B = (t | n_1 | n_2) \in O(3).$$

Then

$$\tilde{R}_f := B^T R_f B = (\tilde{r}_{ij})_{i,j=1}^3.$$

The tangent-coupling projector is

$$\Pi_{\mathcal{T}}(\tilde{R}_f) = \begin{pmatrix} \tilde{r}_{11} & \tilde{r}_{12} & \tilde{r}_{13} \\ \tilde{r}_{21} & 0 & 0 \\ \tilde{r}_{31} & 0 & 0 \end{pmatrix}.$$

Hence the adapted-basis mixing fraction is

$$\varepsilon_{\text{mix}}^2 = \frac{\tilde{r}_{12}^2 + \tilde{r}_{13}^2 + \tilde{r}_{21}^2 + \tilde{r}_{31}^2}{S_f}.$$

LXXXVI. 8. EXACT BRANCH DECISION RULE

With $\tau_{\text{cert}} = 0.2$, the exact next move is:

$$\hat{\alpha}_f \geq 1 \implies \text{quartic baseline invertibility control fails;}$$

$$\hat{\alpha}_f < 1, \varepsilon_{\text{sym}} > 0.2 \implies \text{the local residual map is genuinely non-gradient;}$$

$$\hat{\alpha}_f < 1, \varepsilon_{\text{sym}} \leq 0.2, \varepsilon_{\text{mix}} > 0.2 \implies \text{the coarse } (A, E, O) \text{ basis fails locally;}$$

$$\hat{\alpha}_f < 1, \varepsilon_{\text{sym}} \leq 0.2, \varepsilon_{\text{mix}} \leq 0.2, \varepsilon_{\text{diag}} \leq 0.2 \implies \text{a diagonal coarse quadratic layer is sufficient;}$$

$$\hat{\alpha}_f < 1, \varepsilon_{\text{sym}} \leq 0.2, \varepsilon_{\text{mix}} \leq 0.2, \varepsilon_{\text{diag}} > 0.2 \implies \text{a full symmetric coarse } 3 \times 3 \text{ layer is required.}$$

LXXXVII. PROJECT: TECT — MASTER CONTENTS FOR CONTINUATION (REFERENCE PDE FIXED, MIXED-BRANCH MATCHING, AND FINITE AUDIT STAGE)

LXXXVIII. CURRENT STAGE

The current research stage is the following:

$$\text{the explicit odd extra-carrier route is fixed as the current reference low-order PDE route,}$$

(71)

while

$$\boxed{\text{the canonical mixed odd-active branch is the structural target branch,}} \quad (72)$$

and the remaining work is only:

$$\boxed{\text{finite coefficient insertion, matching, and two finite microscopic audits.}} \quad (73)$$

The exact current separation is

$$\boxed{\text{Reference route: explicit odd extra-carrier route (forward-computable now),}} \quad (74)$$

$$\boxed{\text{Target route: canonical mixed odd-active branch (structurally closed, two finite audits remain).}} \quad (75)$$

LXXXIX. WHAT IS ALREADY FIXED

A. 1. Canonical mixed odd-active target branch

The canonical mixed odd-active target branch is fixed by

$$\boxed{T_E = T_O = 1, \quad R_\perp = \sqrt{2}, \quad \Delta_{00} = \frac{1}{\sqrt{2}}.} \quad (76)$$

Its exact branch-level coefficient sheet is

$$\boxed{\lambda_\parallel = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3,} \quad (77)$$

$$\boxed{\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa.} \quad (78)$$

Thus the target mixed branch is no longer waiting on branch-choice logic. What remains there is only microscopic coefficient population.

B. 2. Current reference branch choice

The current reference branch is fixed by

$$\boxed{\gamma_E = \gamma_O =: \gamma_\perp \neq 0.} \quad (79)$$

The reduced coefficient dictionary is

$$\boxed{\lambda_\parallel = \frac{\lambda_c}{2} I_3, \quad \alpha = I_c \tilde{\Lambda}_E, \quad \beta = I_c \tilde{\Lambda}_O.} \quad (80)$$

At one-mediator level in canonical explicit normalization,

$$\boxed{\tilde{\Lambda}_E^{(1)} = -\frac{\beta_X \gamma_E}{M_X^2}, \quad \tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2}.} \quad (81)$$

Therefore, under the reference choice $\gamma_E = \gamma_O = \gamma_\perp$,

$$\boxed{\alpha = \beta = -\frac{\beta_X \gamma_\perp}{M_X^2} I_c.} \quad (82)$$

C. 3. Current reference PDE

The current reference PDE is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3(-i\partial_{\parallel}) + \frac{\beta_X \gamma_{\perp}}{M_X^2} I_c (\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) + \frac{\beta_X \gamma_{\perp}}{M_X^2} I_c (-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0. \quad (83)$$

Introducing rotated transverse coordinates

$$q_1 := \frac{p_1 - p_2}{\sqrt{2}}, \quad q_2 := \frac{p_1 + p_2}{\sqrt{2}}, \quad (84)$$

the same reference PDE becomes

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3(-i\partial_{\parallel}) - v_{\perp}^{\text{ref}} (\sigma_1(-i\partial_{q_1}) + \sigma_2(-i\partial_{q_2})) \right] \chi = 0, \quad (85)$$

with

$$v_{\perp}^{\text{ref}} = -\sqrt{2} \frac{\beta_X \gamma_{\perp}}{M_X^2} I_c. \quad (86)$$

The corresponding dispersion is

$$E(p)^2 = \frac{\lambda_c^2 I_3^2}{4} p_{\parallel}^2 + 2 \left(\frac{\beta_X \gamma_{\perp}}{M_X^2} \right)^2 I_c^2 (p_1^2 + p_2^2). \quad (87)$$

D. 4. Reference-route nonzero conditions

The reference route is nondegenerate as soon as

$$\lambda_c \neq 0, \quad I_3 \neq 0, \quad I_c \neq 0, \quad \beta_X \neq 0, \quad \gamma_{\perp} \neq 0. \quad (88)$$

E. 5. Matching conditions to the canonical mixed target

The reference route matches the canonical mixed odd-active target iff

$$\frac{\lambda_c}{2} I_3 = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad (89)$$

and

$$\frac{\gamma_{\perp}}{M_X^2} I_c = \frac{\sqrt{3}\pi}{16} \theta_0 \kappa. \quad (90)$$

Under (90),

$$\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad (91)$$

so the transverse reference sheet reproduces the canonical mixed odd-active target exactly.

XC. WHAT REMAINS OPEN ON THE TARGET MIXED BRANCH

The canonical mixed odd-active branch is structurally closed, but its fully populated reduced microscopic coefficient point still awaits exactly two finite microscopic audits.

A. (A) Mixed same-shell support insertion

The branch-consistent sufficient support condition is

$$|E_{z_0}^{\text{dir}}(0) - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}. \quad (92)$$

The only uninserted same-shell quantities are

$$\mathcal{S}_{\text{diag}}^{TT}, \quad L_+^{TT}, L_-^{TT}, L_0^{TT}, \quad C_+^{TT}, C_-^{TT}, C_m^{TT}. \quad (93)$$

So Audit A is now reduced to a seven-number finite insertion problem.

B. (B) Mixed-block transverse cancellation audit

At the current conservative stage,

$$\Lambda_E = e_{0;K}^{(1)} + \Delta e_{\text{mix}}, \quad \Lambda_O = o_{0;K}^{(1)} + \Delta o_{\text{mix}}. \quad (94)$$

A strong sufficient survival condition is

$$|\Delta e_{\text{mix}}| < |e_{0;K}^{(1)}|, \quad |\Delta o_{\text{mix}}| < |o_{0;K}^{(1)}|. \quad (95)$$

Once the two audits are done, the final mixed-branch coefficient point becomes

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_{0;OO}^{(3)}, \quad (96)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (e_{0;K}^{(1)} + \Delta e_{\text{mix}}), \quad \beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (o_{0;K}^{(1)} + \Delta o_{\text{mix}}). \quad (97)$$

XCI. RESIDUAL JACOBIAN SIDE PROBLEM (FROZEN BUT IMPORTANT)

At the corrected residual stage, the exact four-class quartic baseline is already fixed, and the corrected seed is

$$\delta_f = (0.0096, -0.0288, 0.0096). \quad (98)$$

The exact quartic baseline Jacobian is

$$J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665531859216 & 0.4137780199567714 \\ -4.991665531859216 & 22.646508448870943 & -1.2413340598703142 \\ 0.4137780199567714 & -1.2413340598703142 & 3.603395005929136 \end{pmatrix}, \quad (99)$$

with

$$\lambda_* := \lambda_{\min}(J_f^{(4)}) \approx 3.52129589. \quad (100)$$

The actual centered-difference Jacobian

$$J_f := DG(\delta_f)$$

is not yet supplied in the uploaded files. Therefore the exact true local certificate is not yet available.

However, the admissible prior local completion

$$R_f^{\text{prior}} := S_{\min}$$

has already been classified numerically as

$$\alpha_f^{\text{prior}} = 2.294924967594988, \quad \varepsilon_{\text{diag}}^{\text{prior}} = 0.455959495840998, \quad \varepsilon_{\text{sym}}^{\text{prior}} = 0, \quad \varepsilon_{\text{mix}}^{\text{prior}} = 9.547253466277641 \times 10^{-17}. \quad (101)$$

Thus the currently computable local classification is:

$$\boxed{\text{not diagonal, not skew, but essentially exact tangent-coupling symmetric remainder.}} \quad (102)$$

So the honest next local finite task is the true corrected Jacobian extraction.

A. Exact next Jacobian task

Take

$$e_A = (1, 0, 0)^T, \quad e_E = (0, 1, 0)^T, \quad e_O = (0, 0, 1)^T,$$

and choose

$$\varepsilon = 10^{-4}, \quad \varepsilon/2 = 5 \times 10^{-5}.$$

Evaluate the 12 points:

$$\delta_f \pm \varepsilon e_A, \quad \delta_f \pm \varepsilon e_E, \quad \delta_f \pm \varepsilon e_O,$$

and the same six again at $\varepsilon/2$.

Then form

$$J_f^{[\varepsilon]}, \quad J_f^{[\varepsilon/2]}, \quad J_f^{\text{Rich}} = \frac{4J_f^{[\varepsilon/2]} - J_f^{[\varepsilon]}}{3},$$

and define

$$R_f := J_f^{\text{Rich}} - J_f^{(4)}.$$

Then compute the classifier quadruple

$$\boxed{(\hat{\alpha}_f, \varepsilon_{\text{sym}}, \varepsilon_{\text{mix}}, \varepsilon_{\text{diag}})}.$$

The next local refinement is then fixed by one line of arithmetic:

$$\hat{\alpha}_f \geq 1 \implies \text{quartic Jacobian no longer controls invertibility,}$$

$$\hat{\alpha}_f < 1, \varepsilon_{\text{sym}} \text{ large} \implies \text{locally non-gradient residual map,}$$

$$\hat{\alpha}_f < 1, \varepsilon_{\text{mix}} \text{ large} \implies \text{coarse basis failure / suborbit refinement,}$$

$$\hat{\alpha}_f < 1, \varepsilon_{\text{sym}}, \varepsilon_{\text{mix}} \text{ small} \implies \text{symmetric coarse tangent-coupling layer.}$$

XCII. WHAT IS ALREADY CLOSED

The following are already closed at theorem / exact-reduced-sheet level:

1. the explicit odd extra-carrier reference route;
2. the current reference reduced PDE;
3. the canonical mixed odd-active branch decision;
4. the exact target mixed branch coefficient sheet;
5. the matching conditions between reference and target branch;
6. the reduction of the remaining target-side work to exactly Audit A and Audit B;
7. the quartic corrected residual baseline and the admissible prior symmetric completion classification.

XCIII. WHAT MUST NOT BE REOPENED

The following are not to be reopened unless new microscopic data are explicitly inserted:

1. branch-choice logic for the mixed odd-active target;
2. the old JK -route frozen-harmonic sign test;
3. repeated re-derivation of the reduced coefficient dictionary

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = I_c \tilde{\Lambda}_E, \quad \beta = I_c \tilde{\Lambda}_O;$$

4. repeated re-derivation of the canonical mixed target sheet.

XCIV. EXACT NEXT TASK FOR THE NEXT CONVERSATION

The next conversation should do exactly one of the following, and should not branch away:

Path A (main PDE path): continue Audit A and Audit B only.

(103)

More precisely:

$$\text{insert } \mathcal{S}_{\text{diag}}^{TT}, L_{\pm}^{TT}, L_0^{TT}, C_{\pm}^{TT}, C_m^{TT}$$

for Audit A, or evaluate

$$\Delta e_{\text{mix}}, \quad \Delta o_{\text{mix}}$$

for Audit B.

Path B (local corrected residual path): extract the true J_f by the 12-point centered-difference audit.

(104)

In either case, do not reopen earlier symbolic reductions. Only continue finite insertion / finite audit work.

XCV. SHORTEST HONEST STATUS SENTENCE

Current reference route is fixed, target mixed branch is structurally closed, and what remains is only finite coefficient insertion,

(105)

XCVI. TRUE J_f 12-POINT AUDIT AT THE CORRECTED SEED

XCVII. 1. FROZEN CORRECTED SEED AND QUARTIC BASELINE

Freeze

$$\delta_f = (0.0096, -0.0288, 0.0096)^T, \quad e_A = (1, 0, 0)^T, \quad e_E = (0, 1, 0)^T, \quad e_O = (0, 0, 1)^T.$$

Use

$$\varepsilon = 10^{-4}, \quad \varepsilon/2 = 5 \times 10^{-5}.$$

The exact quartic baseline is

$$J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665531859216 & 0.4137780199567714 \\ -4.991665531859216 & 22.646508448870943 & -1.2413340598703142 \\ 0.4137780199567714 & -1.2413340598703142 & 3.603395005929136 \end{pmatrix},$$

with

$$\lambda_* := \lambda_{\min}(J_f^{(4)}) \approx 3.52129589.$$

XCVIII. 2. THE 12 EVALUATION POINTS

The six coordinate-basis points at step ε are

$$\delta_f + \varepsilon e_A = (0.0097, -0.0288, 0.0096),$$

$$\delta_f - \varepsilon e_A = (0.0095, -0.0288, 0.0096),$$

$$\delta_f + \varepsilon e_E = (0.0096, -0.0287, 0.0096),$$

$$\delta_f - \varepsilon e_E = (0.0096, -0.0289, 0.0096),$$

$$\delta_f + \varepsilon e_O = (0.0096, -0.0288, 0.0097),$$

$$\delta_f - \varepsilon e_O = (0.0096, -0.0288, 0.0095).$$

The six half-step points are

$$\delta_f + \frac{\varepsilon}{2} e_A = (0.00965, -0.0288, 0.0096),$$

$$\delta_f - \frac{\varepsilon}{2} e_A = (0.00955, -0.0288, 0.0096),$$

$$\delta_f + \frac{\varepsilon}{2} e_E = (0.0096, -0.02875, 0.0096),$$

$$\delta_f - \frac{\varepsilon}{2} e_E = (0.0096, -0.02885, 0.0096),$$

$$\delta_f + \frac{\varepsilon}{2} e_O = (0.0096, -0.0288, 0.00965),$$

$$\delta_f - \frac{\varepsilon}{2} e_O = (0.0096, -0.0288, 0.00955).$$

For each point evaluate

$$G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix}.$$

XCIX. 3. RAW CENTERED-DIFFERENCE JACOBIANS

Define

$$G_Y^{(+)} := G(\delta_f + \varepsilon e_Y), \quad G_Y^{(-)} := G(\delta_f - \varepsilon e_Y), \quad Y \in \{A, E, O\},$$

and similarly at half-step,

$$\tilde{G}_Y^{(+)} := G\left(\delta_f + \frac{\varepsilon}{2} e_Y\right), \quad \tilde{G}_Y^{(-)} := G\left(\delta_f - \frac{\varepsilon}{2} e_Y\right).$$

Then the six-point Jacobian is

$$(J_f^{[\varepsilon]})_{\cdot Y} = \frac{G_Y^{(+)} - G_Y^{(-)}}{2\varepsilon}, \quad Y \in \{A, E, O\}.$$

Equivalently,

$$J_f^{[\varepsilon]} = \begin{pmatrix} \frac{G_A(\delta_f + \varepsilon e_A) - G_A(\delta_f - \varepsilon e_A)}{2\varepsilon} & \frac{G_A(\delta_f + \varepsilon e_E) - G_A(\delta_f - \varepsilon e_E)}{2\varepsilon} & \frac{G_A(\delta_f + \varepsilon e_O) - G_A(\delta_f - \varepsilon e_O)}{2\varepsilon} \\ \frac{G_E(\delta_f + \varepsilon e_A) - G_E(\delta_f - \varepsilon e_A)}{2\varepsilon} & \frac{G_E(\delta_f + \varepsilon e_E) - G_E(\delta_f - \varepsilon e_E)}{2\varepsilon} & \frac{G_E(\delta_f + \varepsilon e_O) - G_E(\delta_f - \varepsilon e_O)}{2\varepsilon} \\ \frac{G_O(\delta_f + \varepsilon e_A) - G_O(\delta_f - \varepsilon e_A)}{2\varepsilon} & \frac{G_O(\delta_f + \varepsilon e_E) - G_O(\delta_f - \varepsilon e_E)}{2\varepsilon} & \frac{G_O(\delta_f + \varepsilon e_O) - G_O(\delta_f - \varepsilon e_O)}{2\varepsilon} \end{pmatrix}.$$

At half-step,

$$(J_f^{[\varepsilon/2]})_{\cdot Y} = \frac{\tilde{G}_Y^{(+)} - \tilde{G}_Y^{(-)}}{\varepsilon}, \quad Y \in \{A, E, O\}.$$

C. 4. RICHARDSON-UPGRADED JACOBIAN

The 12-point improved estimate is

$$J_f^{\text{Rich}} = \frac{4J_f^{[\varepsilon/2]} - J_f^{[\varepsilon]}}{3} = J_f + O(\varepsilon^4).$$

Set

$$J_f := J_f^{\text{Rich}}.$$

CI. 5. REMAINDER MATRIX

Define the exact local remainder

$$R_f := J_f - J_f^{(4)} = (r_{ij})_{i,j=1}^3.$$

Its Frobenius size is

$$S_f := \|R_f\|_F^2 = \sum_{i,j=1}^3 r_{ij}^2.$$

CII. 6. ENTRY-ONLY CLASSIFIER

The no-eigensolver invertibility parameter is

$$\hat{\alpha}_f = \frac{\sqrt{S_f}}{\lambda_*}.$$

The antisymmetric fraction is

$$\varepsilon_{\text{sym}}^2 = \frac{(r_{12} - r_{21})^2 + (r_{13} - r_{31})^2 + (r_{23} - r_{32})^2}{2S_f}.$$

The off-diagonal fraction is

$$\varepsilon_{\text{diag}}^2 = \frac{r_{12}^2 + r_{13}^2 + r_{21}^2 + r_{23}^2 + r_{31}^2 + r_{32}^2}{S_f}.$$

CIII. 7. ADAPTED-BASIS MIXING FRACTION

Use the corrected-branch adapted orthonormal basis

$$t = \frac{1}{\sqrt{11}}(1, -3, 1)^T, \quad n_1 = \frac{1}{\sqrt{2}}(1, 0, -1)^T, \quad n_2 = \frac{1}{\sqrt{22}}(3, 2, 3)^T,$$

and let

$$B = (t \mid n_1 \mid n_2) \in O(3).$$

Then

$$\tilde{R}_f := B^T R_f B = (\tilde{r}_{ij})_{i,j=1}^3.$$

Define the tangent-coupling projector

$$\Pi_{\mathcal{T}}(\tilde{R}_f) = \begin{pmatrix} \tilde{r}_{11} & \tilde{r}_{12} & \tilde{r}_{13} \\ \tilde{r}_{21} & 0 & 0 \\ \tilde{r}_{31} & 0 & 0 \end{pmatrix}.$$

Then

$$\varepsilon_{\text{mix}}^2 = \frac{\tilde{r}_{12}^2 + \tilde{r}_{13}^2 + \tilde{r}_{21}^2 + \tilde{r}_{31}^2}{S_f}.$$

CIV. 8. EXACT BRANCH DECISION RULE

Choose

$$\tau_{\text{cert}} = 0.2.$$

Then the next correct local refinement is determined by:

$$\hat{\alpha}_f \geq 1 \implies \text{quartic Jacobian no longer controls invertibility.}$$

$$\hat{\alpha}_f < 1, \quad \varepsilon_{\text{sym}} > 0.2 \implies \text{locally non-gradient residual map.}$$

$$\hat{\alpha}_f < 1, \quad \varepsilon_{\text{sym}} \leq 0.2, \quad \varepsilon_{\text{mix}} > 0.2 \implies \text{coarse } (A, E, O) \text{ basis fails locally; move to sign-resolved suborbits.}$$

$$\hat{\alpha}_f < 1, \quad \varepsilon_{\text{sym}} \leq 0.2, \quad \varepsilon_{\text{mix}} \leq 0.2, \quad \varepsilon_{\text{diag}} \leq 0.2 \implies \text{diagonal coarse quadratic layer is sufficient.}$$

$$\hat{\alpha}_f < 1, \quad \varepsilon_{\text{sym}} \leq 0.2, \quad \varepsilon_{\text{mix}} \leq 0.2, \quad \varepsilon_{\text{diag}} > 0.2 \implies \text{full symmetric coarse } 3 \times 3 \text{ layer is required.}$$

CV. EXACT STATUS

The “12-point” task belongs to the local corrected residual path, not to the main mixed-branch PDE path. Its role is:

$$\delta_f \pm \varepsilon e_A, \quad \delta_f \pm \varepsilon e_E, \quad \delta_f \pm \varepsilon e_O,$$

plus the same six evaluations at $\varepsilon/2$, in order to extract

$$J_f, \quad R_f := J_f - J_f^{(4)},$$

and then classify

$$(\alpha_f, \varepsilon_{\text{diag}}, \varepsilon_{\text{sym}}, \varepsilon_{\text{mix}}).$$

Therefore it is mandatory only if one wants to close the local corrected residual branch.

By contrast, on the current main PDE path for the canonical mixed odd-active branch, the only genuinely remaining microscopic tasks are

(A) mixed same-shell support insertion, (B) mixed-block transverse audit for TD, DT, DD .

Hence the 12-point Jacobian audit is not a prerequisite for continuing the main PDE completion route.

CVI. PATH A CONTINUATION: FINAL EXECUTABLE SHEET FOR AUDIT A AND AUDIT B

CVII. STATUS

We do *not* reopen branch-choice logic, the reduced coefficient dictionary, or the canonical mixed-sheet derivation. We continue only the finite insertion tasks on the canonical mixed odd-active branch.

The branch data are fixed:

$$T_E = T_O = 1, \quad R_\perp = \sqrt{2}, \quad \varphi = \frac{\pi}{4}, \quad (106)$$

and

$$a_T = \frac{2\beta_X^2}{M_X^2}. \quad (107)$$

Thus the remaining work is exactly:

1. Audit A: same-shell support insertion;
2. Audit B: mixed-block transverse cancellation audit.

CVIII. AUDIT A: EXACT ASSEMBLED SEVEN-CONSTANT PACKAGE

For the canonical mixed branch,

$$\mathcal{S}_{ab}^{(\pi/4)}(p) = \frac{1}{2}\mathcal{S}_{ab}^{EE}(p) + \frac{1}{2}\mathcal{S}_{ab}^{OO}(p) + \mathcal{S}_{ab}^{EO}(p). \quad (108)$$

Hence the mixed same-shell block is

$$d_+^{(T)}(p) = E_{z_0}^{\text{dir}}(p) - a_T \mathcal{S}_{++}^{TT}(p), \quad (109)$$

$$d_-^{(T)}(p) = E_{z_0}^{\text{dir}}(-p) - a_T \mathcal{S}_{--}^{TT}(p), \quad (110)$$

$$m^{(T)}(p) = B_{z_0}(p) - a_T \mathcal{S}_{+-}^{TT}(p). \quad (111)$$

The crucial point is that the remaining seven TT -constants are not independent unknowns. They are assembled from the already separated $EE/OO/EO$ data:

$$\boxed{\mathcal{S}_{\text{diag}}^{TT} = \frac{1}{2}\mathcal{S}_{\text{diag}}^{EE} + \frac{1}{2}\mathcal{S}_{\text{diag}}^{OO} + \mathcal{S}_{\text{diag}}^{EO},} \quad (112)$$

and

$$\boxed{L_{\bullet}^{TT} \leq \frac{1}{2}L_{\bullet}^{EE} + \frac{1}{2}L_{\bullet}^{OO} + L_{\bullet}^{EO}, \quad \bullet \in \{+, -, 0\},} \quad (113)$$

$$\boxed{C_{\bullet}^{TT} \leq \frac{1}{2}C_{\bullet}^{EE} + \frac{1}{2}C_{\bullet}^{OO} + C_{\bullet}^{EO}, \quad \bullet \in \{+, -, m\}.} \quad (114)$$

Define the derivative budgets

$$L_+^{TT} = \left(\sum_i |\ell_{+,i}^{TT}|^2 \right)^{1/2}, \quad L_-^{TT} = \left(\sum_i |\ell_{-,i}^{TT}|^2 \right)^{1/2}, \quad L_0^{TT} = \left(\sum_i |\ell_{0,i}^{TT}|^2 \right)^{1/2}. \quad (115)$$

Then the mixed-branch drift budgets satisfy

$$A_+^{(T)} \leq \left| q_* \left(Z - \frac{Y q_*^2}{2} \right) \right| + a_T L_+^{TT}, \quad (116)$$

$$A_-^{(T)} \leq \left| q_* \left(Z - \frac{Y q_*^2}{2} \right) \right| + a_T L_-^{TT}, \quad (117)$$

$$B^{(T)} \leq a_T L_0^{TT}. \quad (118)$$

Therefore

$$\varepsilon_{\text{small-}p,\rho}^{(1,T)} \leq \left[(\rho A_+^{(T)} + C_+^{TT} \rho^2)^2 + (\rho A_-^{(T)} + C_-^{TT} \rho^2)^2 + 2(\rho B^{(T)} + C_m^{TT} \rho^2)^2 \right]^{1/2}. \quad (119)$$

Hence the exact branch-consistent same-shell sufficient support condition is

$$\boxed{|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > 2a_T \mathcal{S}_{\text{diag}}^{TT} + \varepsilon_{\text{small-}p,\rho}^{(1,T)}} \quad (120)$$

Using $a_T = 2\beta_X^2/M_X^2$, this is equivalently

$$\boxed{|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{TT} + \varepsilon_{\text{small-}p,\rho}^{(1,T)}} \quad (121)$$

CIX. AUDIT A COMPRESSED LOWER-BOUND CRITERION

Introduce the directly executable assembled upper bounds

$$\bar{L}_{\bullet}^{TT} := \frac{1}{2}L_{\bullet}^{EE} + \frac{1}{2}L_{\bullet}^{OO} + L_{\bullet}^{EO}, \quad \bullet \in \{+, -, 0\}, \quad (122)$$

$$\bar{C}_{\bullet}^{TT} := \frac{1}{2}C_{\bullet}^{EE} + \frac{1}{2}C_{\bullet}^{OO} + C_{\bullet}^{EO}, \quad \bullet \in \{+, -, m\}, \quad (123)$$

$$\bar{\mathcal{S}}_{\text{diag}}^{TT} := \frac{1}{2}\mathcal{S}_{\text{diag}}^{EE} + \frac{1}{2}\mathcal{S}_{\text{diag}}^{OO} + \mathcal{S}_{\text{diag}}^{EO}. \quad (124)$$

Then

$$\bar{A}_+ = \left| q_* \left(Z - \frac{Yq_*^2}{2} \right) \right| + a_T \bar{L}_+^{TT}, \quad \bar{A}_- = \left| q_* \left(Z - \frac{Yq_*^2}{2} \right) \right| + a_T \bar{L}_-^{TT}, \quad (125)$$

$$\bar{B} = a_T \bar{L}_0^{TT}, \quad (126)$$

and therefore

$$\bar{\varepsilon}_A(\rho) := \left[(\rho \bar{A}_+ + \bar{C}_+^{TT} \rho^2)^2 + (\rho \bar{A}_- + \bar{C}_-^{TT} \rho^2)^2 + 2(\rho \bar{B} + \bar{C}_m^{TT} \rho^2)^2 \right]^{1/2}. \quad (127)$$

Define the Audit A margin

$$\mathfrak{G}_A(\rho) := |E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| - \frac{4\beta_X^2}{M_X^2} \bar{\mathcal{S}}_{\text{diag}}^{TT} - \bar{\varepsilon}_A(\rho). \quad (128)$$

Then a sufficient condition for Audit A to pass is simply

$$\mathfrak{G}_A(\rho) > 0. \quad (129)$$

This is the final nonredundant support-side test.

CX. AUDIT B: EXACT DECOMPOSITION INTO PRINCIPAL PAIR PLUS RESIDUAL TAIL

The already explicit K -sector pair is

$$e_{0;K}^{(1)} = \frac{1}{2}I_{\psi\theta}\Lambda_E^{(0)}, \quad o_{0;K}^{(1)} = \frac{1}{2}I_{\psi\theta}\Lambda_O^{(0)}. \quad (130)$$

Equivalently, for the tanh defect,

$$e_{0;K}^{(1)} = \frac{1}{2}C_{\psi}\theta_0\kappa\Lambda_E^{(0)}, \quad o_{0;K}^{(1)} = \frac{1}{2}C_{\psi}\theta_0\kappa\Lambda_O^{(0)}. \quad (131)$$

Now define the full mixed correction as

$$\Delta e_{\text{mix}} = \Delta e_{\text{pr}} + \Delta e_{\text{tail}}, \quad \Delta o_{\text{mix}} = \Delta o_{\text{pr}} + \Delta o_{\text{tail}}, \quad (132)$$

with principal pair

$$\Delta e_{\text{pr}} := e_{0;TD}^{(1)} + e_{0;DT}^{(1)}, \quad \Delta o_{\text{pr}} := o_{0;TD}^{(1)} + o_{0;DT}^{(1)}, \quad (133)$$

and residual tail

$$\Delta e_{\text{tail}} := e_{0;OD}^{(1)} + e_{0;DO}^{(1)} + e_{0;DD}^{(1)}, \quad \Delta o_{\text{tail}} := o_{0;OD}^{(1)} + o_{0;DO}^{(1)} + o_{0;DD}^{(1)}. \quad (134)$$

Hence

$$\Lambda_E = e_{0;K}^{(1)} + \Delta e_{\text{pr}} + \Delta e_{\text{tail}}, \quad \Lambda_O = o_{0;K}^{(1)} + \Delta o_{\text{pr}} + \Delta o_{\text{tail}}. \quad (135)$$

Therefore the final reduced coefficients are

$$\alpha = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(e_{0;K}^{(1)} + \Delta e_{\text{pr}} + \Delta e_{\text{tail}}\right), \quad (136)$$

$$\beta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(o_{0;K}^{(1)} + \Delta o_{\text{pr}} + \Delta o_{\text{tail}}\right). \quad (137)$$

CXI. AUDIT B SHARP SUFFICIENT SURVIVAL TESTS

A strong sufficient survival condition is

$$\boxed{|\Delta e_{\text{pr}}| + |\Delta e_{\text{tail}}| < |e_{0;K}^{(1)}|, \quad |\Delta o_{\text{pr}}| + |\Delta o_{\text{tail}}| < |o_{0;K}^{(1)}|.} \quad (138)$$

Indeed, by the triangle inequality,

$$|\Lambda_E| \geq |e_{0;K}^{(1)}| - |\Delta e_{\text{pr}}| - |\Delta e_{\text{tail}}|, \quad (139)$$

$$|\Lambda_O| \geq |o_{0;K}^{(1)}| - |\Delta o_{\text{pr}}| - |\Delta o_{\text{tail}}|. \quad (140)$$

Thus the above inequalities imply

$$\Lambda_E \neq 0, \quad \Lambda_O \neq 0.$$

A weaker but still sufficient sign-coherent version is the following: if

$$\text{sgn}(\Delta e_{\text{pr}}) = \text{sgn}(e_{0;K}^{(1)}), \quad \text{sgn}(\Delta o_{\text{pr}}) = \text{sgn}(o_{0;K}^{(1)}), \quad (141)$$

then only the residual tail must be controlled by

$$|\Delta e_{\text{tail}}| < |e_{0;K}^{(1)}| + \Delta e_{\text{pr}}, \quad |\Delta o_{\text{tail}}| < |o_{0;K}^{(1)}| + \Delta o_{\text{pr}}. \quad (142)$$

CXII. AUDIT B EXACT EXECUTION ORDER

The nonredundant order is:

1. extract the invariant projected kernel

$$\mathcal{M}_{TD+DT}^\mu = \Pi_Q(\mathcal{V}_{TD}^\mu + \mathcal{V}_{DT}^\mu)\Pi_Q;$$

2. compute

$$\Delta e_{\text{pr}} = \mathfrak{P}_E[\mathcal{M}_{TD+DT}], \quad \Delta o_{\text{pr}} = \mathfrak{P}_O[\mathcal{M}_{TD+DT}];$$

3. only after that, add the tail from $OD + DO + DD$.

This order is forced because $TD + DT$ is the principal transverse mixed source, while DD is structurally more likely to renormalize an existing transverse carrier than to create it from nothing.

CXIII. WHAT IS ACTUALLY CLOSED NOW

At this point, the exact new closure is:

1. Audit A is reduced to one assembled scalar margin $\mathfrak{G}_A(\rho)$;
2. Audit B is reduced to one principal-pair extraction $(\Delta e_{\text{pr}}, \Delta o_{\text{pr}})$ plus one residual-tail bound;
3. no independent new symbolic reduction is needed.

Therefore the true next microscopic computation is no longer “derive another theorem” but simply:

$$\boxed{\text{compute } (e_{0;TD}^{(1)}, o_{0;TD}^{(1)}, e_{0;DT}^{(1)}, o_{0;DT}^{(1)}) \text{ first.}} \quad (143)$$

CXIV. CONCLUSION

The mixed odd-active branch is structurally closed.

Audit A is now a finite insertion inequality:

$$\mathfrak{G}_A(\rho) > 0.$$

Audit B is now a finite cancellation audit around the already explicit K -sector pair:

$$\Lambda_E = e_{0;K}^{(1)} + \Delta e_{\text{pr}} + \Delta e_{\text{tail}}, \quad \Lambda_O = o_{0;K}^{(1)} + \Delta o_{\text{pr}} + \Delta o_{\text{tail}}.$$

Thus the mainline is fixed:

$$\boxed{\text{first } TD + DT, \text{ then } OD + DO + DD, \text{ without reopening any earlier symbolic layer.}} \quad (144)$$

CXV. FINAL SUMMARY

The current reference route remains fixed, and the canonical mixed odd-active branch remains structurally closed. The new nonredundant closure is:

$$\boxed{\text{Audit A reduces to one assembled scalar support margin } \mathfrak{G}_A(\rho).} \quad (145)$$

$$\boxed{\text{Audit B reduces to a principal transverse pair from } TD + DT \text{ plus a residual tail from } OD + DO + DD.} \quad (146)$$

Explicitly,

$$\mathfrak{G}_A(\rho) = |E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| - \frac{4\beta_X^2}{M_X^2} \bar{\mathcal{S}}_{\text{diag}}^{TT} - \bar{\varepsilon}_A(\rho),$$

and Audit A passes if

$$\mathfrak{G}_A(\rho) > 0.$$

Also,

$$\Lambda_E = e_{0;K}^{(1)} + \Delta e_{\text{pr}} + \Delta e_{\text{tail}}, \quad \Lambda_O = o_{0;K}^{(1)} + \Delta o_{\text{pr}} + \Delta o_{\text{tail}},$$

with

$$\Delta e_{\text{pr}} = e_{0;TD}^{(1)} + e_{0;DT}^{(1)}, \quad \Delta o_{\text{pr}} = o_{0;TD}^{(1)} + o_{0;DT}^{(1)}.$$

Therefore the exact next mainline computation is:

$$\boxed{\text{compute the principal } TD + DT \text{ transverse pair first,}}$$

$$\boxed{\text{and only then add the } OD + DO + DD \text{ tail correction.}}$$

No earlier symbolic reduction should be reopened.

CXVI. AUDIT B PRINCIPAL PAIR FROM THE $TD + DT$ BLOCK

CXVII. GOAL

On the current mixed-branch PDE path, the principal transverse correction is carried by the mixed $TD + DT$ block. The task is to identify the leading pair

$$\Delta e_{\text{pr}}, \quad \Delta o_{\text{pr}},$$

and connect it directly to the reduced PDE coefficients (α, β) .

CXVIII. 1. PROJECTED MIXED TRANSVERSE CARRIER

Define the projected mixed first-order kernel

$$\boxed{\mathcal{M}_{TD+DT}^\mu := \Pi_Q (\mathcal{V}_{TD}^\mu + \mathcal{V}_{DT}^\mu) \Pi_Q.} \quad (147)$$

The canonical invariant projector functionals are

$$\boxed{\mathfrak{P}_E[\mathcal{M}] = \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu), \quad \mathfrak{P}_O[\mathcal{M}] = \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu).} \quad (148)$$

Therefore the principal transverse coefficient point is

$$\boxed{\Lambda_E^{\text{pr}} = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad \Lambda_O^{\text{pr}} = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}].} \quad (149)$$

In the present Audit-B notation we simply set

$$\boxed{\Delta e_{\text{pr}} := \Lambda_E^{\text{pr}}, \quad \Delta o_{\text{pr}} := \Lambda_O^{\text{pr}}.} \quad (150)$$

CXIX. 2. GENERIC INVARIANT FORM OF THE TD AND DT BLOCKS

At the current explicit branch-level realization, the two mixed blocks have the generic form

$$\boxed{\mathcal{M}_{TD,\text{inv}}^\mu = a_{\parallel;TD}(\theta) n^\mu \Sigma_3 + a_{E;TD}(\theta) E^\mu{}_a \Sigma_a + a_{O;TD}(\theta) J^\mu{}_a \Sigma_a,} \quad (151)$$

$$\boxed{\mathcal{M}_{DT,\text{inv}}^\mu = a_{\parallel;DT}(\theta) n^\mu \Sigma_3 + a_{E;DT}(\theta) E^\mu{}_a \Sigma_a + a_{O;DT}(\theta) J^\mu{}_a \Sigma_a.} \quad (152)$$

Because TD and DT mix one shell-gradient/current side with one defect/tangent side, they are the natural carrier of the linear transverse monomials

$$\theta'(\xi) E^\mu{}_a \Sigma_a, \quad \theta'(\xi) J^\mu{}_a \Sigma_a.$$

Thus, at leading order, we may write

$$\boxed{V_{TD}^\mu = \theta'(\xi) (A_E E^\mu{}_a \Sigma_a + A_O J^\mu{}_a \Sigma_a),} \quad (153)$$

$$\boxed{V_{DT}^\mu = \theta'(\xi) (B_E E^\mu{}_a \Sigma_a + B_O J^\mu{}_a \Sigma_a).} \quad (154)$$

Hence

$$\boxed{\mathcal{M}_{TD+DT}^\mu = \theta'(\xi) [(A_E + B_E) E^\mu{}_a \Sigma_a + (A_O + B_O) J^\mu{}_a \Sigma_a] + \mathcal{M}_{\text{corr}}^\mu,} \quad (155)$$

where $\mathcal{M}_{\text{corr}}^\mu$ starts at higher order and/or receives residual contributions from

$$OD, \quad DO, \quad DD.$$

CXX. 3. EXPLICIT EXTRACTION OF THE PRINCIPAL PAIR

Applying the invariant projectors and using the canonical projector normalization already built into $\mathfrak{P}_E, \mathfrak{P}_O$, one obtains

$$e_{0;TD}^{(1)} = \frac{1}{2}A_E, \quad e_{0;DT}^{(1)} = \frac{1}{2}B_E, \quad (156)$$

$$o_{0;TD}^{(1)} = \frac{1}{2}A_O, \quad o_{0;DT}^{(1)} = \frac{1}{2}B_O. \quad (157)$$

Therefore the principal correction pair is

$$\Delta e_{\text{pr}} = e_{0;TD}^{(1)} + e_{0;DT}^{(1)} = \frac{1}{2}(A_E + B_E), \quad (158)$$

$$\Delta o_{\text{pr}} = o_{0;TD}^{(1)} + o_{0;DT}^{(1)} = \frac{1}{2}(A_O + B_O). \quad (159)$$

Equivalently,

$$\Lambda_E^{\text{pr}} = \Delta e_{\text{pr}}, \quad \Lambda_O^{\text{pr}} = \Delta o_{\text{pr}}. \quad (160)$$

CXXI. 4. CONNECTION TO THE REDUCED PDE COEFFICIENTS

The exact reduced coefficient dictionary on the current branch is

$$\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O, \quad I_\theta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa. \quad (161)$$

Keeping only the principal $TD + DT$ part gives

$$\alpha_{\text{pr}} = I_\theta \Delta e_{\text{pr}} = -\frac{\sqrt{3}\pi}{32}\theta_0\kappa(A_E + B_E), \quad (162)$$

$$\beta_{\text{pr}} = I_\theta \Delta o_{\text{pr}} = -\frac{\sqrt{3}\pi}{32}\theta_0\kappa(A_O + B_O). \quad (163)$$

So the principal transverse PDE point is

$$(\alpha_{\text{pr}}, \beta_{\text{pr}}) = -\frac{\sqrt{3}\pi}{32}\theta_0\kappa(A_E + B_E, A_O + B_O). \quad (164)$$

CXXII. 5. FULL AUDIT-B SPLIT

After this principal extraction, the full mixed transverse point is

$$\Lambda_E = e_{0;K}^{(1)} + \Delta e_{\text{pr}} + \Delta e_{\text{tail}}, \quad (165)$$

$$\Lambda_O = o_{0;K}^{(1)} + \Delta o_{\text{pr}} + \Delta o_{\text{tail}}, \quad (166)$$

with

$$\Delta e_{\text{tail}} = e_{0;OD}^{(1)} + e_{0;DO}^{(1)} + e_{0;DD}^{(1)}, \quad \Delta o_{\text{tail}} = o_{0;OD}^{(1)} + o_{0;DO}^{(1)} + o_{0;DD}^{(1)}. \quad (167)$$

Hence the sharp execution order is

$$TD + DT \text{ first,} \quad OD + DO + DD \text{ second.} \quad (168)$$

CXXIII. 6. HONEST STATUS

At the current explicit Class-II working level, the relation

$$\boxed{\Delta e_{\text{pr}} = \frac{1}{2}(A_E + B_E), \quad \Delta o_{\text{pr}} = \frac{1}{2}(A_O + B_O)} \quad (169)$$

is a correct branch-level reduced summary.

However, it should still be stated honestly that this is not yet the unconditional final microscopic closure of the complete TECT PDE. The remaining frontier is the literal blockwise differentiation of the full one-defect Hessian and the explicit evaluation of

$$A_E, \quad B_E, \quad A_O, \quad B_O$$

on the intended branch.

CXXIV. CONCLUSION

The principal transverse audit is now closed algebraically:

$$\boxed{\Delta e_{\text{pr}} = \frac{1}{2}(A_E + B_E), \quad \Delta o_{\text{pr}} = \frac{1}{2}(A_O + B_O).} \quad (170)$$

Therefore the principal reduced PDE coefficients are

$$\boxed{\alpha_{\text{pr}} = -\frac{\sqrt{3}\pi}{32}\theta_0\kappa(A_E + B_E), \quad \beta_{\text{pr}} = -\frac{\sqrt{3}\pi}{32}\theta_0\kappa(A_O + B_O).} \quad (171)$$

So the next nonredundant microscopic task is no longer structural. It is the explicit Hessian differentiation and projection that determines

$$A_E, \quad B_E, \quad A_O, \quad B_O.$$

CXXV. SOURCE-LEVEL ELIMINATION OF A_E, B_E, A_O, B_O

CXXVI. GOAL

The previous branch-level transverse summary

$$e_0^{(1)} = \frac{1}{2}(A_E + B_E), \quad o_0^{(1)} = \frac{1}{2}(A_O + B_O)$$

is algebraically correct, but it is not yet the sharpest source-level form.

The next nonredundant step is to eliminate the bookkeeping constants

$$A_E, \quad B_E, \quad A_O, \quad B_O$$

and rewrite the principal transverse pair directly in terms of the microscopic $JK + KJ$ source on the frozen corrected branch.

CXXVII. 1. EXACT SOURCE SPLIT

After integrating out the heavy mediator X_i^a , the exact quadratic Schur correction is

$$\boxed{\Delta_X^{(2)} = -\frac{\alpha_X^2}{M_X^2} \delta J \delta J^\dagger - \frac{\alpha_X \beta_X}{M_X^2} (\delta J \delta K^\dagger + \delta K \delta J^\dagger) - \frac{\beta_X^2}{M_X^2} \delta K \delta K^\dagger.} \quad (172)$$

Hence the microscopic source sectors are

$$JJ, \quad JK + KJ, \quad KK.$$

The source/block routing is

$$JJ \leftrightarrow OO\text{-longitudinal audit}, \quad JK + KJ \leftrightarrow TD + DT\text{-transverse audit}, \quad KK \leftrightarrow \text{transverse residual sector.} \quad (173)$$

CXXVIII. 2. PRINCIPAL TRANSVERSE PAIR FROM THE $JK + KJ$ SOURCE

On the current refined locked + Class II branch, the mixed $JK + KJ$ source yields

$$c_E = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad c_O = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}. \quad (174)$$

Equivalently, the principal source-level pair is

$$\Lambda_E^{JK} = (j \cdot k) \Lambda_E^{(0)}, \quad \Lambda_O^{JK} = (j \cdot k) \Lambda_O^{(0)}. \quad (175)$$

Therefore the principal correction pair in the Audit-B notation is

$$\Delta e_{\text{pr}} = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad \Delta o_{\text{pr}} = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}. \quad (176)$$

CXXIX. 3. ELIMINATION OF THE BOOKKEEPING CONSTANTS

Since

$$e_0^{(1)} = \frac{1}{2} (A_E + B_E), \quad o_0^{(1)} = \frac{1}{2} (A_O + B_O),$$

comparison with (176) gives

$$\frac{1}{2} (A_E + B_E) = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad (177)$$

$$\frac{1}{2} (A_O + B_O) = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}. \quad (178)$$

Hence

$$A_E + B_E = 2 \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad (179)$$

$$A_O + B_O = 2 \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}. \quad (180)$$

So the A/B -notation can be eliminated completely in favor of the source-level JK contraction.

CXXX. 4. PRINCIPAL-PLUS-RESIDUAL REDUCED PDE COEFFICIENTS

The exact reduced coefficient dictionary is

$$\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O, \quad I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa.$$

Separating the principal JK part and the residual KK part, one gets

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)} + \frac{\beta_X^2}{M_X^2} \Lambda_E^{KK} \right], \quad (181)$$

$$\beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)} + \frac{\beta_X^2}{M_X^2} \Lambda_O^{KK} \right]. \quad (182)$$

Thus the transverse audit is now an explicit competition:

$$\text{principal } JK \quad \text{vs.} \quad \text{residual } KK.$$

CXXXI. 5. SHARP DOMINANCE CRITERION

A sufficient condition that the KK -residual does not kill the principal pair is

$$|\beta_X| |\Lambda_E^{KK}| < |\alpha_X| |j \cdot k| |\Lambda_E^{(0)}|, \quad (183)$$

$$|\beta_X| |\Lambda_O^{KK}| < |\alpha_X| |j \cdot k| |\Lambda_O^{(0)}|. \quad (184)$$

Equivalently,

$$|\beta_X| \sqrt{(\Lambda_E^{KK})^2 + (\Lambda_O^{KK})^2} < |\alpha_X| |j \cdot k| \sqrt{(\Lambda_E^{(0)})^2 + (\Lambda_O^{(0)})^2}. \quad (185)$$

CXXXII. 6. HONEST ODD-CHANNEL BOTTLENECK

At the minimal rigid-internal first-shell truncation, the corrected notes show that orbit-amplitude corrections alone do not activate the odd transverse witness:

$$Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0.$$

Therefore, if

$$\Lambda_O^{(0)} = 0,$$

then the principal JK -pair does not by itself open the odd channel. In that case the first activation of β must come from a genuinely microscopic sign-sensitive insertion and/or the residual completion sector.

This is the sharp current bottleneck for the odd transverse coefficient.

CXXXIII. CONCLUSION

The principal transverse audit is now closed at the source level.

The bookkeeping constants A_E, B_E, A_O, B_O are no longer fundamental. They combine into the explicit JK -driven pair

$$\Delta e_{\text{pr}} = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad \Delta o_{\text{pr}} = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}. \quad (186)$$

Therefore the reduced transverse coefficients are

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)} + \frac{\beta_X^2}{M_X^2} \Lambda_E^{KK} \right], \quad (187)$$

$$\beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)} + \frac{\beta_X^2}{M_X^2} \Lambda_O^{KK} \right]. \quad (188)$$

So the next nonredundant microscopic problem is reduced to the literal evaluation of the three finite quantities

$$j \cdot k, \quad \Lambda_E^{(0)}, \quad \Lambda_O^{(0)},$$

followed by the residual KK -dominance check.

CXXXIV. HONEST ODD-CHANNEL CLOSURE ON THE CURRENT WORKING BRANCH

CXXXV. STATUS

The odd transverse slot is not proven to vanish universally.

What is actually established is the sharper statement:

$$\Lambda_O^{(0)} \neq 0 \text{ is not automatic, and it requires genuinely derivative-resolved and/or phase-resolved microscopic structure.} \quad (189)$$

Thus the correct logical status is:

- not “the odd channel is impossible”;
- but rather “the odd channel is not activated by the minimal rigid first-shell orbit-amplitude mechanism alone”.

CXXXVI. 1. NO LINEAR ODD ACTIVATION ON THE FROZEN CORRECTED SEED

On the sign-resolved first-shell lift of the frozen corrected seed,

$$r_{\text{sub}}^{(ct_1)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0, \quad r_{\text{sub}}^{(ct_2)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0. \quad (190)$$

Therefore the sign-sensitive first-shell transverse factor has no linear corrected shift. Equivalently, the odd transverse coefficient does not turn on linearly on the present orbit-symmetric corrected branch.

CXXXVII. 2. MINIMAL FIRST-SHELL TRUNCATION IS INSUFFICIENT

In the minimal rigid-internal first-shell truncation one has

$$Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0. \quad (191)$$

Hence orbit-amplitude corrections alone do not generate the odd transverse witness.

This closes the negative result at the current quartic orbit-amplitude level: the odd slot does not activate there.

CXXXVIII. 3. FIRST GENUINELY ACTIVE ODD SOURCE

Let

$$X_2(\delta) = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}(\delta)] \quad (192)$$

be the odd transverse source extracted from the mixed $TD + DT$ block.
Then the corrected odd transverse Hessian response is

$$H_{rs}^{(2)} := \partial_{\delta_r} \partial_{\delta_s} X_2(\delta)|_{\delta=0}. \quad (193)$$

On the current refined locked + Class II working branch, the leading realization is

$$H_{rs}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[\partial_{rs}(\delta J \delta K^\dagger + \delta K \delta J^\dagger)_{\delta=0}] + \text{subleading completion terms}. \quad (194)$$

This is the decisive point:

$$\boxed{\text{the first active odd transverse source is the quadratic Hessian response of the mixed } JK + KJ \text{ channel.}} \quad (195)$$

CXXXIX. 4. LEADING CORRECTED ODD COEFFICIENT

Since the base odd slot vanishes on the current orbit-symmetric corrected branch and the linear term also vanishes, the leading corrected odd coefficient is inferred to start at quadratic order:

$$X_2(\delta) = \frac{1}{2} H_{rs}^{(2)} \delta_r \delta_s + O(\delta^3) \quad (\text{current branch-level inference}). \quad (196)$$

Therefore

$$\boxed{\beta^{\text{corr}} = I_\theta \frac{1}{2} H_{rs}^{(2)} \delta_r \delta_s + O(\delta^3),} \quad (197)$$

provided the quadratic Hessian slot (194) is nonzero.

CXL. 5. HONEST FALLBACK IF THE QUADRATIC $JK + KJ$ SLOT STILL VANISHES

If the quadratic mixed $JK + KJ$ Hessian response vanishes in the intended truncation, then the next odd source must come from the subleading completion terms already isolated in the notes, namely genuinely microscopic derivative-/phase-resolved completion beyond the rigid first-shell orbit-amplitude ansatz.

So the honest branch logic is:

$$\boxed{\text{first try } TD + DT \text{ quadratic Hessian response;}} \quad (198)$$

$$\boxed{\text{if that still vanishes, the odd slot moves to subleading completion structure.}} \quad (199)$$

CXLI. CONCLUSION

The current bottleneck is now sharply fixed.
The odd channel is not activated by:

linear first-shell sign response, or minimal rigid orbit-amplitude corrections alone.

The first genuinely active odd source on the current refined locked + Class II working branch is

$$\partial_{\delta_r} \partial_{\delta_s} \mathfrak{P}_O [\mathcal{M}_{TD+DT}^{\text{inv}}(\delta)]_{\delta=0}, \quad (200)$$

namely the quadratic Hessian response of the mixed $JK + KJ$ channel.

Therefore the next nonredundant microscopic task is not another orbit-level closure, but the explicit evaluation of the finite Hessian matrix

$$H_{rs}^{(2)}.$$

CXLII. EXACT QUADRATIC HESSIAN EXTRACTION FOR THE ODD TRANSVERSE SLOT

CXLIII. GOAL

We now stop speaking schematically and write the odd transverse Hessian response in the exact operator form that is actually needed for the current branch.

To avoid index collision, let

$$R, S \in \{A, E, O\}$$

denote the orbit-amplitude coordinates, while $r, u, v, \dots$ are reserved for internal/tangent component indices.

CXLIV. 1. EXACT STARTING POINT

The exact odd transverse source is

$$X_2(\delta) = \mathfrak{P}_O [\mathcal{M}_{TD+DT}^{\text{inv}}(\delta)]. \quad (201)$$

Its Hessian coefficients are

$$H_{RS}^{(2)} := \partial_{\delta_R} \partial_{\delta_S} X_2(\delta)|_{\delta=0}. \quad (202)$$

On the current refined locked + Class II working branch,

$$H_{RS}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O [\partial_{RS} (\delta J \delta K^\dagger + \delta K \delta J^\dagger)_{\delta=0}] + \text{subleading completion terms.} \quad (203)$$

CXLV. 2. EXACT 8-TERM EXPANSION

Introduce the shorthand

$$J_0 := \delta J|_{\delta=0}, \quad K_0 := \delta K|_{\delta=0}, \quad (204)$$

$$J_R := \partial_{\delta_R} \delta J|_0, \quad K_R := \partial_{\delta_R} \delta K|_0, \quad (205)$$

$$J_{RS} := \partial_{\delta_R} \partial_{\delta_S} \delta J|_0, \quad K_{RS} := \partial_{\delta_R} \partial_{\delta_S} \delta K|_0. \quad (206)$$

Then

$$\begin{aligned} \partial_{RS} (\delta J \delta K^\dagger + \delta K \delta J^\dagger) \Big|_0 &= J_{RS} K_0^\dagger + J_R K_S^\dagger + J_S K_R^\dagger + J_0 K_{RS}^\dagger \\ &\quad + K_{RS} J_0^\dagger + K_R J_S^\dagger + K_S J_R^\dagger + K_0 J_{RS}^\dagger. \end{aligned} \quad (207)$$

Therefore the exact operator formula is

$$H_{RS}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O [J_{RS} K_0^\dagger + J_R K_S^\dagger + J_S K_R^\dagger + J_0 K_{RS}^\dagger + K_{RS} J_0^\dagger + K_R J_S^\dagger + K_S J_R^\dagger + K_0 J_{RS}^\dagger] + \dots \quad (208)$$

This is the exact formula that was still implicit before.

CXLVI. 3. NATURAL LINEAR-CURRENT REDUCTION

Now make the *natural linear-current approximation* around the base branch:

$$\delta J(\delta) = \sum_R \delta_R J_R + O(\delta^2), \quad \delta K(\delta) = \sum_R \delta_R K_R + O(\delta^2). \quad (209)$$

Then

$$J_0 = K_0 = 0, \quad J_{RS} = K_{RS} = 0$$

at the strict leading linear-current level, so (208) collapses to

$$H_{RS}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_R K_S^\dagger + J_S K_R^\dagger + K_R J_S^\dagger + K_S J_R^\dagger \right] + \cdots. \quad (210)$$

Thus the odd-channel frontier is reduced from a full nonlinear second-derivative problem to a finite symmetric 3×3 bilinear matrix built from the six pairs

$$(A, A), (E, E), (O, O), (A, E), (A, O), (E, O).$$

CXLVII. 4. LOAD-BEARING LOCAL KERNELS

The notes already fix the load-bearing linearized kernels as

$$(\delta J_R)_i^a = m^a(P_0) (\partial_i \phi_0) \mathcal{U}_{jR}^b (-i \partial_j) s_b, \quad (211)$$

$$(\delta K_R)_i^a = (\partial_i \xi^a) \mathcal{V}_{jR}^b (-i \partial_j) s_b. \quad (212)$$

Substituting (211) and (212) into (210), one sees that every Hessian entry is a projected mixed bilinear of the schematic form

$$H_{RS}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathcal{C}_{RS} + \cdots, \quad (213)$$

where

$$\mathcal{C}_{RS} := \mathfrak{P}_O \left[J_R K_S^\dagger + J_S K_R^\dagger + K_R J_S^\dagger + K_S J_R^\dagger \right]. \quad (214)$$

Equivalently, using

$$\mathfrak{P}_O[\mathcal{M}] = \frac{1}{4} J^\mu{}_c \text{Tr}_{Q_B}(\Sigma_c \mathcal{M}_\mu),$$

we may write

$$\mathcal{C}_{RS} = \frac{1}{4} J^\mu{}_c \text{Tr}_{Q_B} \left[\Sigma_c \Pi_Q (J_R K_S^\dagger + J_S K_R^\dagger + K_R J_S^\dagger + K_S J_R^\dagger)_\mu \Pi_Q \right]. \quad (215)$$

This is the finite projected tensor that must be computed on the frozen corrected branch.

CXLVIII. 5. SYMMETRY AND THE SIX INDEPENDENT COEFFICIENTS

Because $H_{RS}^{(2)}$ is a Hessian,

$$H_{RS}^{(2)} = H_{SR}^{(2)}. \quad (216)$$

Hence only the six entries

$$H_{AA}^{(2)}, H_{EE}^{(2)}, H_{OO}^{(2)}, H_{AE}^{(2)}, H_{AO}^{(2)}, H_{EO}^{(2)}$$

are independent.

So the odd-channel activation problem is now a literal six-number finite audit.

CXLI. 6. EXPLICIT CORRECTED-SEED QUADRATIC FORM

The corrected frozen branch is

$$\delta_{\text{corr}} = (a, e, o) = (0.0096, -0.0288, 0.0096).$$

The quadratic odd source is

$$Q_2(\delta) = \frac{1}{2} \sum_{R,S \in \{A,E,O\}} H_{RS}^{(2)} \delta_R \delta_S. \quad (217)$$

Therefore, at the corrected seed,

$$\begin{aligned} Q_2(\delta_{\text{corr}}) &= \frac{1}{2} \left[a^2 H_{AA}^{(2)} + e^2 H_{EE}^{(2)} + o^2 H_{OO}^{(2)} + 2ae H_{AE}^{(2)} + 2ao H_{AO}^{(2)} + 2eo H_{EO}^{(2)} \right] \\ &= 4.608 \times 10^{-5} H_{AA}^{(2)} + 4.1472 \times 10^{-4} H_{EE}^{(2)} + 4.608 \times 10^{-5} H_{OO}^{(2)} \\ &\quad - 2.7648 \times 10^{-4} H_{AE}^{(2)} + 9.216 \times 10^{-5} H_{AO}^{(2)} - 2.7648 \times 10^{-4} H_{EO}^{(2)}. \end{aligned} \quad (218)$$

Thus the odd coefficient becomes

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 Q_2(\delta_{\text{corr}}) + O(\delta^3). \quad (219)$$

CL. 7. SHARP PRACTICAL MEANING

Equation (218) shows immediately that the most heavily weighted entries are

$$H_{EE}^{(2)}, \quad H_{AE}^{(2)}, \quad H_{EO}^{(2)},$$

while $H_{AA}^{(2)}$ and $H_{OO}^{(2)}$ are relatively weakly weighted.

So the odd-channel sign decision on the corrected seed is dominated by a competition between:

$$+ 4.1472 \times 10^{-4} H_{EE}^{(2)} \quad (220)$$

and

$$- 2.7648 \times 10^{-4} H_{AE}^{(2)} - 2.7648 \times 10^{-4} H_{EO}^{(2)}, \quad (221)$$

with a smaller correction from

$$+ 9.216 \times 10^{-5} H_{AO}^{(2)}.$$

CLI. CONCLUSION

The odd transverse frontier is now reduced to a strict finite problem.

What remains is not a new theorem and not a new branch choice.

It is exactly:

$$\text{compute the six projected mixed bilinears } \mathcal{C}_{AA}, \mathcal{C}_{EE}, \mathcal{C}_{OO}, \mathcal{C}_{AE}, \mathcal{C}_{AO}, \mathcal{C}_{EO}, \quad (222)$$

then use

$$H_{RS}^{(2)} = -(\alpha_X \beta_X / M_X^2) \mathcal{C}_{RS} + \dots$$

inside (218) to decide the sign and nonvanishing of β^{corr} .

CLII. FINAL SUMMARY

The current odd-channel frontier is now reduced to a strict finite audit.
The exact first active odd source is

$$H_{RS}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\partial_{RS} (\delta J \delta K^\dagger + \delta K \delta J^\dagger)_{\delta=0} \right] + \cdots,$$

and its exact operator expansion is the 8-term formula.

Under the natural linear-current approximation around the base branch, this reduces to

$$H_{RS}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_R K_S^\dagger + J_S K_R^\dagger + K_R J_S^\dagger + K_S J_R^\dagger \right] + \cdots.$$

Hence only six independent coefficients remain:

$$H_{AA}^{(2)}, H_{EE}^{(2)}, H_{OO}^{(2)}, H_{AE}^{(2)}, H_{AO}^{(2)}, H_{EO}^{(2)}.$$

At the corrected seed

$$\delta_{\text{corr}} = (0.0096, -0.0288, 0.0096),$$

the odd quadratic source is exactly

$$Q_2(\delta_{\text{corr}}) = 4.608 \times 10^{-5} H_{AA}^{(2)} + 4.1472 \times 10^{-4} H_{EE}^{(2)} + 4.608 \times 10^{-5} H_{OO}^{(2)} - 2.7648 \times 10^{-4} H_{AE}^{(2)} + 9.216 \times 10^{-5} H_{AO}^{(2)} - 2.7648 \times 10^{-4} H_{EO}^{(2)}.$$

Therefore the next nonredundant microscopic task is:

compute the six projected mixed bilinears $\mathcal{C}_{RS}$, hence $H_{RS}^{(2)}$, and decide whether $Q_2(\delta_{\text{corr}}) \neq 0$.

CLIII. STRUCTURAL ZERO/NONZERO CLASSIFICATION OF THE ODD HESSIAN ROWS

CLIV. GOAL

We now classify the six symmetric odd-Hessian rows

$$H_{AA}^{(2)}, H_{EE}^{(2)}, H_{OO}^{(2)}, H_{AE}^{(2)}, H_{AO}^{(2)}, H_{EO}^{(2)}$$

into:

structural zeros in the rigid truncation

(223)

and

the first surviving non-rigid microscopic slots.

(224)

CLV. 1. FAST PRACTICAL FORMULAS

Under the fast working approximation,

$$H_{rs}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O [J_r K_s^\dagger + J_s K_r^\dagger + K_r J_s^\dagger + K_s J_r^\dagger].$$

(225)

Hence explicitly,

$$H_{EE}^{(2),\text{fast}} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O [J_E K_E^\dagger + K_E J_E^\dagger],$$

(226)

$$H_{AE}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_A K_E^\dagger + J_E K_A^\dagger + K_A J_E^\dagger + K_E J_A^\dagger], \quad (227)$$

$$H_{EO}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_E K_O^\dagger + J_O K_E^\dagger + K_E J_O^\dagger + K_O J_E^\dagger], \quad (228)$$

and similarly

$$H_{AA}^{(2),\text{fast}} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_A K_A^\dagger + K_A J_A^\dagger], \quad (229)$$

$$H_{OO}^{(2),\text{fast}} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_O K_O^\dagger + K_O J_O^\dagger], \quad (230)$$

$$H_{AO}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_A K_O^\dagger + J_O K_A^\dagger + K_A J_O^\dagger + K_O J_A^\dagger]. \quad (231)$$

CLVI. 2. RIGID-TRUNCATION NO-GO

In the minimal rigid-internal first-shell truncation,

$$K_A = K_E = K_O = 0. \quad (232)$$

Therefore every fast row vanishes identically:

$$H_{AA}^{(2),\text{fast}} = H_{EE}^{(2),\text{fast}} = H_{OO}^{(2),\text{fast}} = H_{AE}^{(2),\text{fast}} = H_{AO}^{(2),\text{fast}} = H_{EO}^{(2),\text{fast}} = 0. \quad (233)$$

Hence

$$Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0. \quad (234)$$

So the rigid first-shell orbit-amplitude ansatz cannot activate the odd witness at all.

CLVII. 3. FIRST SURVIVING NON-RIGID SLOT

Once the rigid truncation is lifted, the corrected-seed contraction weights satisfy

$$|EE| : |AE| : |EO| : |AO| : |AA| : |OO| = 9 : 6 : 6 : 2 : 1 : 1. \quad (235)$$

Therefore the first decisive row is

$$H_{EE}^{(2)}. \quad (236)$$

The second-priority rows are

$$H_{AE}^{(2)}, \quad H_{EO}^{(2)}. \quad (237)$$

The remaining rows

$$H_{AO}^{(2)}, \quad H_{AA}^{(2)}, \quad H_{OO}^{(2)}$$

are subleading in the corrected-seed contraction.

Thus the first non-rigid microscopic priority order is

$$(J_E, K_E) \text{ first,} \quad (J_A, K_E) \text{ and } (J_E, K_O) \text{ second.} \quad (238)$$

CLVIII. 4. WHY THE E -SECTOR MUST BE THE FIRST TEST

At the corrected residual level, a coarse A/E -symmetric completion is impossible: it would force

$$a^2 = e^2,$$

but the corrected seed satisfies

$$e^2 = 9a^2.$$

Therefore the missing microscopic completion must already contain

$$\boxed{\text{non-diagonal } A/E\text{-breaking mixing.}} \quad (239)$$

This is consistent with the Hessian-weight hierarchy above: the dominant odd-slot sensitivity sits precisely in the E -sector and in A/E -mixed rows.

CLIX. 5. PRACTICAL SUFFICIENT CRITERION

A convenient sufficient criterion for odd-slot activation is the EE -dominance inequality

$$\boxed{|H_{EE}^{(2)}| > \frac{1}{9}(|H_{AA}^{(2)}| + |H_{OO}^{(2)}|) + \frac{2}{3}(|H_{AE}^{(2)}| + |H_{EO}^{(2)}|) + \frac{2}{9}|H_{AO}^{(2)}|.} \quad (240)$$

If, in addition, the sign of the EE -term is constructive in the corrected contraction, then

$$Q_2(\delta_{\text{corr}}) \neq 0$$

and therefore

$$\beta^{\text{corr}} \neq 0.$$

CLX. 6. HONEST FINAL CLASSIFICATION

The six rows now split sharply into two categories.

a. Category I: structural zeros inside the rigid first-shell truncation. All six rows vanish identically because $K_A = K_E = K_O = 0$.

b. Category II: first survivors once non-rigid microscopic structure is allowed. The first actual nonzero test is

$$H_{EE}^{(2)},$$

followed by

$$H_{AE}^{(2)}, \quad H_{EO}^{(2)}.$$

So the odd-channel frontier is no longer vague. It is reduced to the explicit non-rigid E -sector current pair computation.

CLXI. CONCLUSION

The rigid first-shell route is dead for the odd channel:

$$\boxed{\beta^{\text{corr}} = 0 \quad \text{identically there.}}$$

The first meaningful microscopic test beyond that truncation is

$$\boxed{H_{EE}^{(2)} = -\frac{2\alpha_X\beta_X}{M_X^2}\Re\mathfrak{O}[J_E K_E^\dagger + K_E J_E^\dagger].}$$

Hence the next nonredundant computation is to extract the non-rigid E -sector pair

$$(J_E, K_E),$$

and only then move to the mixed rows

$$H_{AE}^{(2)}, \quad H_{EO}^{(2)}.$$

CLXII. EXACT REDUCTION OF THE FIRST NON-RIGID ODD TEST TO FOUR BACKGROUND TENSOR INPUTS

CLXIII. GOAL

We now reduce the first non-rigid odd test

$$H_{EE}^{(2)}$$

to the explicit background tensors that still remain unevaluated on the frozen corrected branch.

The correct point is that the naive undifferentiated local JK bilinear vanishes, so the first meaningful odd source must be derivative-resolved at the shell point G_* .

CLXIV. 1. EXACT STARTING POINT

The first active odd transverse Hessian response is

$$H_{EE}^{(2)} = -\frac{2\alpha_X\beta_X}{M_X^2} \mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger] + \cdots, \quad (241)$$

where the ellipsis denotes subleading completion terms beyond the current fast reduction.

However, the honest microscopic frontier is not the undifferentiated local product itself, because

$$J \sim j \Sigma_3, \quad K \sim k_E \Sigma_1 + k_O \Sigma_2$$

would imply

$$\Sigma_3(k_E \Sigma_1 + k_O \Sigma_2) + (k_E \Sigma_1 + k_O \Sigma_2) \Sigma_3 = 0$$

by Pauli anticommutation.

Therefore the true object is the shell-derivative audit

$$\left. \partial_{p_\nu} \widehat{\Sigma}_{JK}(G_*; p) \right|_{p=0}. \quad (242)$$

CLXV. 2. EXACT DERIVATIVE-RESOLVED JK BLOCK

The exact mixed derivative block is

$$\begin{aligned} \left. \partial_{p_\nu} \widehat{\Sigma}_{JK}(G_*; p) \right|_{p=0} = & -\frac{\alpha_X\beta_X}{M_X^2} \left[(\partial_{p_\nu} \widehat{\mathcal{J}})(G_*; 0) \widehat{\mathcal{K}}(G_*; 0)^\dagger + \widehat{\mathcal{J}}(G_*; 0) (\partial_{p_\nu} \widehat{\mathcal{K}})(G_*; 0)^\dagger \right. \\ & \left. + (\partial_{p_\nu} \widehat{\mathcal{K}})(G_*; 0) \widehat{\mathcal{J}}(G_*; 0)^\dagger + \widehat{\mathcal{K}}(G_*; 0) (\partial_{p_\nu} \widehat{\mathcal{J}})(G_*; 0)^\dagger \right]. \end{aligned} \quad (243)$$

Decompose this into the shell-adapted frame

$$\left. \partial_{p_\nu} \widehat{\Sigma}_{JK}(G_*; p) \right|_{p=0} = \hat{n}_\nu \mathcal{S}_\parallel^{(JK)} + e_{1\nu} \mathcal{S}_1^{(JK)} + e_{2\nu} \mathcal{S}_2^{(JK)}, \quad (244)$$

where each $\mathcal{S}_\bullet^{(JK)}$ is a Hermitian 2×2 matrix on the tangent doublet.

The odd witness projection is then

$$c_O^{(JK)} \propto \frac{1}{2} \left[\text{Tr}(\Sigma_2 \mathcal{S}_1^{(JK)}) - \text{Tr}(\Sigma_1 \mathcal{S}_2^{(JK)}) \right]. \quad (245)$$

Accordingly, the EE -row Hessian coefficient is the second δ_E -variation of this odd trace:

$$H_{EE}^{(2)} \propto \partial_{\delta_E}^2 \left[\text{Tr}(\Sigma_2 \mathcal{S}_1^{(JK)}) - \text{Tr}(\Sigma_1 \mathcal{S}_2^{(JK)}) \right]_{\delta=0}. \quad (246)$$

This is already the exact projector-trace reduction of the first non-rigid odd test.

CLXVI. 3. LOWEST-DERIVATIVE LOCAL CURRENT ANSATZ

The lowest-derivative local ansatz for the background-linearized Class II currents is

$$\delta J_i^a[\Psi_0; \delta\Psi]_{\text{lb}} = m^a(P_0) (\partial_i \phi_0) \mathcal{U}_{jr}{}^b(0) (-i\partial_j) s_b \psi_r, \quad (247)$$

$$\delta K_i^a[P_0; \delta\Psi]_{\text{lb}} = (\partial_i \xi^a) \mathcal{V}_{jr}{}^b(0) (-i\partial_j) s_b \psi_r. \quad (248)$$

Their Bloch-harmonic coefficients at the intervalley shell point are

$$J_{i,r}^a(G_*; 0) = m^a(P_0) i(G_*)_i A_{G_*} \mathcal{U}_{jr}{}^b(0) (G_*)_j s_b, \quad (249)$$

$$K_{i,r}^a(G_*; 0) = \Xi_i^a(G_*) \mathcal{V}_{jr}{}^b(0) (G_*)_j s_b, \quad (250)$$

where $\Xi_i^a(G_*)$ is the $Q = G_*$ Fourier coefficient of $\partial_i \xi^a$.

Thus the shell-derivative matrices $\mathcal{S}_1^{(JK)}, \mathcal{S}_2^{(JK)}$ are explicit bilinears in the four background tensor inputs

$$m^a(P_0), \quad \mathcal{U}_{jr}{}^b(0), \quad \Xi_i^a(G_*), \quad \mathcal{V}_{jr}{}^b(0). \quad (251)$$

CLXVII. 4. FINAL EXACT REDUCTION OF THE EE -ROW

Combining (246) with (249)–(250), the first non-rigid odd test is reduced to

$$H_{EE}^{(2)} = -\frac{2\alpha_X \beta_X}{M_X^2} \partial_{\delta_E}^2 \left\{ \frac{1}{2} \left[\text{Tr}(\Sigma_2 \mathcal{S}_1^{(JK)}) - \text{Tr}(\Sigma_1 \mathcal{S}_2^{(JK)}) \right] \right\}_{\delta=0} + \dots, \quad (252)$$

where

$$\mathcal{S}_1^{(JK)}, \mathcal{S}_2^{(JK)}$$

are built explicitly from the four background inputs in (251).

Therefore the EE -row is no longer a vague operator question. It is a finite tensor-contraction computation on the frozen corrected branch.

CLXVIII. 5. FOLDED-BRAGG COMPRESSION

Inside the explicit folded-Bragg Class II bookkeeping, the derivative audit compresses to

$$\hat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \sum_{Q+Q'=G_*} \hat{J}_i^A(Q) \hat{K}_{i,\alpha}^A(Q'), \quad \alpha = 1, 2. \quad (253)$$

Hence the same EE -row may be read equivalently as

$$H_{EE}^{(2)} \text{ is controlled by } \partial_{\delta_E}^2 \hat{T}_{(II)}^1(G_*), \quad \partial_{\delta_E}^2 \hat{T}_{(II)}^2(G_*), \quad (254)$$

followed by the odd projector combination.

CLXIX. CONCLUSION

The first non-rigid odd microscopic test has now been reduced as far as is honestly possible without inserting the final frozen-branch background tensors.

The exact remaining data are only

$$m^a(P_0), \quad \mathcal{U}_{jr}{}^b(0), \quad \Xi_i^a(G_*), \quad \mathcal{V}_{jr}{}^b(0).$$

Once these are inserted, $H_{EE}^{(2)}$ is a direct finite projector-trace evaluation.

CLXX. FIRST K -SECTOR SKEW INVARIANT FOR THE ODD CHANNEL

The shell-relevant K -sector is determined by

$$\Xi_i^a(G_*) = \xi_{\parallel} \hat{n}_i \delta^{a3} + \xi_E S^{(E)}{}_i{}^a + \xi_O S^{(O)}{}_i{}^a + \Xi_i^{a,\text{irr}},$$

$$\mathcal{V}_{jr}{}^b(0) = v_{\parallel} \hat{n}_j (\Sigma_3)_r{}^b + v_E T^{(E)}{}_{jr}{}^b + v_O T^{(O)}{}_{jr}{}^b + \mathcal{V}_{jr}^{b,\text{irr}},$$

where

$$S^{(E)}{}_i{}^a := e_{1i} \delta^{a1} + e_{2i} \delta^{a2}, \quad S^{(O)}{}_i{}^a := -e_{2i} \delta^{a1} + e_{1i} \delta^{a2},$$

$$T^{(E)}{}_{jr}{}^b := (e_{1j} \Sigma_1 + e_{2j} \Sigma_2)_r{}^b, \quad T^{(O)}{}_{jr}{}^b := (-e_{2j} \Sigma_1 + e_{1j} \Sigma_2)_r{}^b.$$

In one convenient canonical orientation convention, the shell-relevant transverse coefficients induced by the K -sector obey

$$\begin{pmatrix} c_E^{(K)} \\ c_O^{(K)} \end{pmatrix} \propto \begin{pmatrix} \xi_E & -\xi_O \\ \xi_O & \xi_E \end{pmatrix} \begin{pmatrix} v_E \\ v_O \end{pmatrix},$$

so that

$$c_E^{(K)} \propto \xi_E v_E - \xi_O v_O, \quad c_O^{(K)} \propto \xi_E v_O + \xi_O v_E,$$

or, after an equivalent orientation redefinition,

$$c_O^{(K)} \propto \xi_E v_O - \xi_O v_E.$$

Therefore the invariant structural statement is:

the odd channel is controlled by a transverse E/O -mismatch skew invariant

of the schematic form

$\Delta_K \sim \xi_E v_O - \xi_O v_E.$

Hence the first EE -row odd test is reduced to

$H_{EE}^{(2)} \propto -\frac{2\alpha_X \beta_X}{M_X^2} \partial_{\delta_E}^2 \Delta_K \Big|_{\delta=0} + \cdots,$

up to a fixed convention-dependent normalization and possible subleading completion terms.

CLXXI. FINAL SUMMARY

The K -sector has now been reduced to the canonical transverse pair

$$(\xi_E, \xi_O), \quad (v_E, v_O),$$

through the exact decompositions of

$$\Xi_i^a(G_*), \quad \mathcal{V}_{jr}{}^b(0).$$

The structural content is:

$$\boxed{\text{same-parity pairings feed the even channel,} \quad \text{cross-parity pairings feed the odd channel.}}$$

Therefore the first shell-relevant odd invariant carried by the K -sector is a skew transverse mismatch of the schematic form

$$\boxed{\Delta_K \sim \xi_E v_O - \xi_O v_E.}$$

Accordingly, the first non-rigid odd test is reduced to

$$\boxed{H_{EE}^{(2)} \propto -\frac{2\alpha_X \beta_X}{M_X^2} \partial_{\delta_E}^2 \Delta_K|_{\delta=0} + \dots}$$

So the next nonredundant microscopic task is no longer to inspect the full K -tensor, but to evaluate only the four transverse scalar responses

$$\boxed{\xi_E, \xi_O, v_E, v_O}$$

on the frozen corrected branch.

CLXXII. EXACT TWO-SOURCE SPLIT OF THE FIRST NON-RIGID ODD E -ROW

CLXXIII. GOAL

We no longer treat the first odd E -sector test in terms of abstract scalar responses

$$\partial_{\delta_E} \xi_E, \quad \partial_{\delta_E} \xi_O, \quad \partial_{\delta_E} v_E, \quad \partial_{\delta_E} v_O.$$

The uploaded notes already permit a sharper and more physical reduction.

The first non-rigid odd test is controlled by the exact E -sector current pair

$$J_E, \quad K_E,$$

and K_E itself splits into two distinct microscopic sources.

CLXXIV. 1. EXACT E -SECTOR CURRENT PAIR

The exact completed E -sector pair is

$$\boxed{J_{E,i}^a = \partial_i M_E^a}, \tag{255}$$

$$\boxed{K_{E,i}^a = \partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi} + (\partial_i \bar{q}^a) \chi_E}. \tag{256}$$

Therefore the fast odd Hessian row is

$$\boxed{H_{EE}^{(2),\text{fast}} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger]}. \tag{257}$$

CLXXV. 2. EXACT TWO-SOURCE DECOMPOSITION

Substituting (256) into (257) yields

$$H_{EE}^{(2),\text{fast}} = H_{EE,\text{proj}}^{(2)} + H_{EE,\text{qgrad}}^{(2)}, \quad (258)$$

where

$$H_{EE,\text{proj}}^{(2)} := -\frac{2\alpha_X\beta_X}{M_X^2} \mathfrak{P}_O \left[J_E \left(\partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi} \right)^\dagger + \left(\partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi} \right) J_E^\dagger \right], \quad (259)$$

and

$$H_{EE,\text{qgrad}}^{(2)} := -\frac{2\alpha_X\beta_X}{M_X^2} \mathfrak{P}_O \left[J_E ((\partial_i \bar{q}^a) \chi_E)^\dagger + (\partial_i \bar{q}^a) \chi_E J_E^\dagger \right]. \quad (260)$$

Thus the first non-rigid odd activation can come from exactly two places:

$$\partial_i F_E^a \neq 0 \quad (\text{projector/internal texture source}), \quad (261)$$

or

$$\partial_i \bar{q}^a \neq 0 \quad (\text{background charge-gradient source}). \quad (262)$$

CLXXVI. 3. RIGID-TRUNCATION NO-GO REVISITED

In the minimal rigid first-shell truncation,

$$F_E^a = 0, \quad \partial_i F_E^a = 0, \quad \partial_i \bar{q}^a = 0. \quad (263)$$

Hence both sources vanish simultaneously:

$$H_{EE,\text{proj}}^{(2)} = 0, \quad H_{EE,\text{qgrad}}^{(2)} = 0,$$

and therefore

$$H_{EE}^{(2),\text{fast}} = 0, \quad Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0. \quad (264)$$

So the rigid truncation fails for the odd channel for a completely explicit reason: it kills both microscopic E -sector odd sources.

CLXXVII. 4. HONEST STATUS OF WHAT IS AND IS NOT FIXED

What is now fixed exactly is the two-source structure

$$H_{EE}^{(2),\text{fast}} = H_{EE,\text{proj}}^{(2)} + H_{EE,\text{qgrad}}^{(2)}.$$

What is *not* yet fixed separately from the current files is a fully explicit independent formula for each of

$$\partial_{\delta_E} \xi_E, \quad \partial_{\delta_E} \xi_O, \quad \partial_{\delta_E} v_E, \quad \partial_{\delta_E} v_O.$$

So the correct microscopic decision problem is no longer phrased in terms of those four abstract scalar derivatives. It is the sharper finite test:

$$\text{which of } H_{EE,\text{proj}}^{(2)}, H_{EE,\text{qgrad}}^{(2)} \text{ is the first nonzero odd source on the frozen corrected branch?} \quad (265)$$

CLXXVIII. CONCLUSION

The first non-rigid odd E -sector test is now fully reduced to two explicit microscopic sources:

$$\boxed{H_{EE,\text{proj}}^{(2)} \quad \text{and} \quad H_{EE,\text{qgrad}}^{(2)}}.$$

Therefore the next nonredundant task is:

$$\boxed{\text{evaluate first whether } \partial_i F_E^a \neq 0 \text{ or } \partial_i \bar{q}^a \neq 0 \text{ on the intended branch.}} \quad (266)$$

Whichever of these survives first determines the first actual activation of the odd transverse coefficient.

CLXXIX. WHY THE FIRST ACTUAL ODD-SOURCE TEST SHOULD BE THE q -GRADIENT ROUTE

CLXXX. EXACT SPLIT ALREADY FIXED

The first non-rigid odd E -sector source is exactly

$$\boxed{H_{EE}^{(2),\text{fast}} = H_{EE,\text{proj}}^{(2)} + H_{EE,\text{qgrad}}^{(2)}}. \quad (267)$$

Equivalently, the two literal finite tests are

$$\boxed{\mathfrak{P}_O(\mathcal{O}_{E,\text{proj}}) \neq 0 \text{ ?}, \quad \mathfrak{P}_O(\mathcal{O}_{E,\text{qgrad}}) \neq 0 \text{ ?}.} \quad (268)$$

CLXXXI. BACKGROUND AVAILABILITY OF THE q -GRADIENT SOURCE

For the locked one-defect background,

$$\boxed{\bar{q}^a = \frac{\bar{\Psi}^\dagger T^a \bar{\Psi}}{\bar{\rho}} = z_B^\dagger T^a z_B =: m_a(\theta),} \quad (269)$$

hence

$$\boxed{\partial_i \bar{q}^a = \delta_{i\xi} \theta'(\xi) \partial_\theta m_a(\theta).} \quad (270)$$

Therefore the q -gradient source is already present at the level of the locked one-defect background, provided

$$\partial_\theta m_a(\theta) \neq 0.$$

So the q -gradient route does not require any extra non-rigid internal completion beyond the background itself.

CLXXXII. WHY THE PROJECTOR ROUTE IS SECONDARY

By contrast, the projector/internal route requires

$$\boxed{\partial_i F_E^a \neq 0,} \quad (271)$$

namely a genuinely non-rigid internal/projector texture in the E -sector.

Thus the projector route is not the minimal first candidate. It is a more refined completion channel.

CLXXXIII. FIRST-PRIORITY DECISION

Therefore the correct first-priority odd audit is:

$$\boxed{\text{compute } \mathfrak{P}_O(\mathcal{O}_{E,\text{qgrad}}) \text{ first.}} \quad (272)$$

Only if this vanishes or is symmetry-forbidden should one move next to

$$\mathfrak{P}_O(\mathcal{O}_{E,\text{proj}}).$$

CLXXXIV. EXACT FORM OF THE FIRST-PRIORITY OBJECT

The q -gradient piece is

$$\boxed{H_{EE,\text{qgrad}}^{(2)} = -\frac{2\alpha_X\beta_X}{M_X^2}\mathfrak{P}_O\left[(\partial_i M_E^a)((\partial_i \bar{q}^a)\chi_E)^\dagger + ((\partial_i \bar{q}^a)\chi_E)(\partial_i M_E^a)^\dagger\right].} \quad (273)$$

Substituting

$$\partial_i \bar{q}^a = \delta_{i\xi} \theta'(\xi) \partial_\theta m_a(\theta),$$

this becomes

$$\boxed{H_{EE,\text{qgrad}}^{(2)} = -\frac{2\alpha_X\beta_X}{M_X^2}\mathfrak{P}_O\left[(\partial_\xi M_E^a)(\theta'(\xi)\partial_\theta m_a(\theta)\chi_E)^\dagger + (\theta'(\xi)\partial_\theta m_a(\theta)\chi_E)(\partial_\xi M_E^a)^\dagger\right].} \quad (274)$$

This is the first actual finite object that should now be evaluated.

CLXXXV. HONEST STATUS

The statement is not that the q -gradient piece is already proven nonzero. The honest statement is only:

$$\boxed{\text{among the two exact odd-source candidates, the } q\text{-gradient route is the first one that is already background-available.}} \quad (275)$$

Hence it is the correct first computation target.

CLXXXVI. FINAL LITERAL OVERLAP TEST FOR THE q -GRADIENT JK -ROUTE

CLXXXVII. STATUS

The q -gradient odd source is no longer structurally open. It has already been reduced to one literal overlap decision. The exact starting formula is

$$\boxed{H_{EE,\text{qgrad}}^{(2)} = -\frac{2\alpha_X\beta_X}{M_X^2}\mathfrak{P}_O\left[(\partial_\xi M_E^a)(\theta'(\xi)k_a(\theta)\chi_E)^\dagger + (\theta'(\xi)k_a(\theta)\chi_E)(\partial_\xi M_E^a)^\dagger\right].} \quad (276)$$

Since $k_a(\theta)$ depends only on θ , and

$$\partial_i \bar{q}^a = \delta_{i\xi} \theta'(\xi) k_a(\theta),$$

the only surviving derivative is the normal one.

CLXXXVIII. 1. FIRST DECISIVE MICROSCOPIC SIGN TEST

The same expression reduces to the decisive sign test

$$\boxed{\mathfrak{P}_O \left[(\partial_\xi M_E^a) \chi_E^\dagger + \chi_E (\partial_\xi M_E^a)^\dagger \right] \stackrel{?}{\neq} 0.} \quad (277)$$

Thus the first unavoidable post-rigid odd source is controlled entirely by the overlap between the normal derivative of the E -sector current amplitude and the E -sector tangent fluctuation.

CLXXXIX. 2. FOURIER REDUCTION

Expand

$$M_E^a(\xi) = \sum_Q \widehat{M}_E^a(Q) e^{iQ\xi}, \quad \chi_E(\xi) = \sum_Q \widehat{\chi}_E(Q) e^{iQ\xi}.$$

Then

$$\partial_\xi M_E^a(\xi) = \sum_Q iQ_\xi \widehat{M}_E^a(Q) e^{iQ\xi}.$$

Substituting into (276), using the Hermitian symmetrization and the odd projector, the q -gradient route reduces to the single shell-overlap decision

$$\boxed{\mathfrak{P}_O \left[\sum_Q Q_\xi \theta'(\xi) k_a(\theta) \Im \left(\widehat{M}_E^a(Q) \widehat{\chi}_E(-Q) \right) \right] \stackrel{?}{\neq} 0.} \quad (278)$$

This is the final literal overlap test for the JK -route.

CXC. 3. HONEST STATUS

At this point the JK -route is not conceptually open any more. It is data-blocked.

The literal evaluation of (278) still requires the actual frozen-branch harmonics, schematically denoted in the notes by

$$\bar{A}_G, \quad B_G^{(E)}.$$

Therefore the correct honest statement is:

$$\boxed{\text{the } JK\text{-route odd audit has been reduced to one finite shell-overlap test,}} \quad (279)$$

$$\boxed{\text{but its numerical sign decision still needs the frozen branch harmonic data.}} \quad (280)$$

CXCI. CONCLUSION

Inside the current frozen corrected branch, the first generic odd source is

$$H_{EE, \text{qgrad}}^{(2)},$$

and its entire microscopic decision problem is exactly the single overlap test

$$\boxed{\mathfrak{P}_O \left[\sum_Q Q_\xi \theta'(\xi) k_a(\theta) \Im \left(\widehat{M}_E^a(Q) \widehat{\chi}_E(-Q) \right) \right] \stackrel{?}{\neq} 0.}$$

So the JK -route has been reduced as far as possible without inserting the frozen harmonics.

CXCII. FINAL SUMMARY

The JK -route odd audit is now fully compressed.
Its first generic odd source is

$$H_{EE,\text{qgrad}}^{(2)},$$

and its entire sign/nonzero problem is equivalent to the single shell-overlap test

$$\mathfrak{P}_O \left[\sum_Q Q_\xi \theta'(\xi) k_a(\theta) \Im \left(\widehat{M}_E^a(Q) \widehat{\chi}_E(-Q) \right) \right] \stackrel{?}{\neq} 0.$$

Thus the JK -route is no longer structurally open. It is only data-blocked by the unavailable frozen harmonics.

If one wants an immediately forward-computable odd channel with the present data, the explicit odd extra-carrier route is the correct next move.

CXCIII. ADOPTING THE EXPLICIT ODD EXTRA-CARRIER ROUTE AS THE ACTIVE FORWARD CLOSURE

CXCIV. CURRENT STAGE

The blocked JK -route is no longer the active forward path. Its sharpest sign test is already known, but its literal shell-overlap evaluation still requires unavailable frozen harmonics.

Therefore the correct active forward closure route is the explicit odd extra-carrier branch.

CXCV. 1. MINIMAL MICROSCOPIC COMPLETION

The minimal one-mediator truncation that explicitly opens the odd channel is

$$\mathcal{L}_{\text{trunc}} = \mathcal{L}_\Psi^{\text{core}} + \mathcal{L}_P^{\text{geom}} + \mathcal{L}_\sigma^{\text{vol}} + \frac{1}{2} M_X^2 X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi] + \gamma_E X_i^a \mathcal{E}_i^a[P, \psi] + \gamma_O X_i^a \mathcal{O}_i^a[P, \psi]. \quad (281)$$

The new ingredient that opens the odd extra-carrier route is exactly

$$\gamma_O X_i^a \mathcal{O}_i^a[P, \psi]. \quad (282)$$

CXCVI. 2. EXACT EXPLICIT ODD CARRIER

On the embedded defect branch,

$$q_O(\theta) = \text{Tr}(S_O M_O) = \cos \theta. \quad (283)$$

Therefore the explicit odd carrier is

$$\mathcal{I}_O = X_i^{(O)} \cos \theta(\xi) J_j D_j \psi. \quad (284)$$

Its induced tangent transport block is

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right]. \quad (285)$$

Hence the local transport densities are

$$a_3^{(EO)} = 0, \quad b_E^{(EO)} = \tilde{\Lambda}_E \theta'(\xi) \cos \theta(\xi), \quad b_O^{(EO)} = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi). \quad (286)$$

So the EO-slot is purely transverse and directly feeds the odd channel.

CXCVII. 3. EXACT REDUCED COEFFICIENT DICTIONARY

Projecting to the Bloch zero modes gives the exact reduced sheet

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = I_c \tilde{\Lambda}_E, \quad \beta = I_c \tilde{\Lambda}_O, \quad (287)$$

with

$$I_3 = \int d\xi f_+ f_- \theta'(\xi)^3, \quad I_c = \int d\xi f_+ f_- \theta'(\xi) \cos \theta(\xi). \quad (288)$$

Equivalently, the reduced first-order PDE is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3 (-i\partial_{\parallel}) - I_c \tilde{\Lambda}_E (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - I_c \tilde{\Lambda}_O (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0. \quad (289)$$

CXCVIII. 4. ONE-MEDIATOR COEFFICIENT POINT

At one-mediator level, the coefficient point is

$$\tilde{\Lambda}_E = -\frac{\beta_X \gamma_E}{M_X^2} C_E J_1, \quad \tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1. \quad (290)$$

If one uses the transport-side canonical explicit normalization where the shell-basis factors are absorbed, then the lowest-order coefficient point is simply

$$\tilde{\Lambda}_E^{(1)} = -\frac{\beta_X \gamma_E}{M_X^2}, \quad \tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2}. \quad (291)$$

Accordingly,

$$\alpha^{(1)} = -\frac{\beta_X \gamma_E}{M_X^2} I_c, \quad \beta^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2} I_c. \quad (292)$$

CXCIX. 5. EXACT NONZERO CONDITIONS

The odd channel is explicitly nonzero exactly when

$$\tilde{\Lambda}_O \neq 0 \quad \text{and} \quad I_c \neq 0. \quad (293)$$

At one-mediator level, a sufficient concrete condition is

$$\beta_X \neq 0, \quad \gamma_O \neq 0, \quad I_c \neq 0, \quad (294)$$

and in the unabsorbed normalization also

$$C_O J_1 \neq 0. \quad (295)$$

CC. 6. EXACT PROJECT STATUS AFTER SWITCHING ROUTES

What is now closed:

1. the explicit odd carrier

$$\mathcal{I}_O = X_i^{(O)} \cos \theta(\xi) J_j D_j \psi;$$

2. the exact EO-slot transport block

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) [\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a];$$

3. the reduced coefficient reading

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = I_c \tilde{\Lambda}_E, \quad \beta = I_c \tilde{\Lambda}_O;$$

4. the one-mediator odd coefficient point

$$\tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2} \quad (\text{in absorbed canonical normalization}).$$

What is still open:

the actual microscopic value/sign of $\tilde{\Lambda}_O$ beyond the lowest-order explicit one-mediator reading.

 (296)

CCI. CONCLUSION

The explicit odd extra-carrier route is now fully inserted into the reduced coefficient sheet. The odd channel is no longer waiting on unavailable frozen harmonics. Its exact branch-level formula is

$\beta = \tilde{\Lambda}_O I_c, \quad I_c = \int d\xi f_+ f_- \theta'(\xi) \cos \theta(\xi),$

 (297)

and at the explicit one-mediator level this becomes

$\beta^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2} I_c$

 (298)

in canonical absorbed normalization.

Therefore the active forward microscopic frontier is now reduced to the actual value/sign extraction of

$$\tilde{\Lambda}_O.$$

CCII. FINAL SUMMARY

The blocked JK -route is no longer the active forward path.

The explicit odd extra-carrier route has now been fully inserted into the reduced coefficient sheet.

Its exact transport block is

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) [\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a],$$

so the reduced coefficients are

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = I_c \tilde{\Lambda}_E, \quad \beta = I_c \tilde{\Lambda}_O.$$

At explicit one-mediator level,

$$\tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2}$$

in canonical absorbed normalization, hence

$$\beta^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2} I_c.$$

So the odd channel is already explicitly forward-computable. The only remaining microscopic problem is the actual value/sign extraction of $\tilde{\Lambda}_O$ beyond the lowest-order explicit reading.

CCIII. EXACT MATCHING BETWEEN THE EXPLICIT ODD ROUTE AND THE CANONICAL MIXED ODD-ACTIVE TARGET

CCIV. REFERENCE SHEET

The explicit odd extra-carrier route has the reduced coefficient sheet

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = -\frac{\beta_X \gamma_E}{M_X^2} I_c, \quad \beta = -\frac{\beta_X \gamma_O}{M_X^2} I_c. \quad (299)$$

CCV. TARGET SHEET

The canonical mixed odd-active target branch is

$$T_E = T_O = 1, \quad R_{\perp} = \sqrt{2}, \quad \Delta_{00} = \frac{1}{\sqrt{2}}, \quad (300)$$

with coefficient sheet

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad (301)$$

$$\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa. \quad (302)$$

CCVI. EXACT MATCHING CONDITIONS

Therefore the explicit odd route matches the canonical mixed odd-active target iff

$$\frac{\lambda_c}{2} I_3 = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad (303)$$

and

$$\frac{\gamma_E}{M_X^2} I_c = \frac{\sqrt{3}\pi}{16} T_E \theta_0 \kappa, \quad \frac{\gamma_O}{M_X^2} I_c = \frac{\sqrt{3}\pi}{16} T_O \theta_0 \kappa. \quad (304)$$

CCVII. CANONICAL MIXED CHOICE

For the canonical mixed target

$$T_E = T_O = 1,$$

the natural matched explicit route is

$$\boxed{\gamma_E = \gamma_O =: \gamma_\perp \neq 0,} \quad (305)$$

together with

$$\boxed{\frac{\gamma_\perp}{M_X^2} I_c = \frac{\sqrt{3}\pi}{16} \theta_0 \kappa.} \quad (306)$$

Then

$$\boxed{\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa,} \quad (307)$$

which reproduces exactly the canonical mixed odd-active transverse sheet.

CCVIII. EQUIVALENT COEFFICIENT READING

Equivalently, in terms of the reduced transport-side coefficients,

$$\boxed{\tilde{\Lambda}_E = -\frac{\beta_X \gamma_E}{M_X^2}, \quad \tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2},} \quad (308)$$

so the target matching reads

$$\boxed{I_c \tilde{\Lambda}_E = -\frac{\sqrt{3}\pi}{16} \beta_X T_E \theta_0 \kappa, \quad I_c \tilde{\Lambda}_O = -\frac{\sqrt{3}\pi}{16} \beta_X T_O \theta_0 \kappa.} \quad (309)$$

For $T_E = T_O = 1$,

$$\boxed{\tilde{\Lambda}_E = \tilde{\Lambda}_O = -\frac{\sqrt{3}\pi}{16} \beta_X \frac{\theta_0 \kappa}{I_c}.} \quad (310)$$

CCIX. CONCLUSION

Hence the explicit odd extra-carrier route is not merely compatible with the canonical mixed odd-active target. It can be matched onto it exactly by the finite relations (303)–(304).

In particular, the symmetric choice

$$\gamma_E = \gamma_O =: \gamma_\perp$$

is the unique minimal explicit route that reproduces

$$\alpha = \beta$$

at lowest order.

CCX. UNIFIED REDUCED PDE SHEET: REFERENCE ROUTE EQUALS CANONICAL MIXED TARGET

CCXI. GOAL

We now merge the current explicit odd extra-carrier reference route and the canonical mixed odd-active target into a single reduced PDE sheet.

The point is not merely compatibility. The point is exact coefficient matching.

CCXII. 1. REFERENCE REDUCED SHEET

On the explicit odd extra-carrier route, the exact reduced coefficient sheet is

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = -\frac{\beta_X \gamma_E}{M_X^2} I_c, \quad \beta = -\frac{\beta_X \gamma_O}{M_X^2} I_c. \quad (311)$$

Therefore the reference reduced PDE is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3(-i\partial_{\parallel}) + \frac{\beta_X \gamma_E}{M_X^2} I_c (\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) + \frac{\beta_X \gamma_O}{M_X^2} I_c (-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0. \quad (312)$$

CCXIII. 2. CANONICAL MIXED ODD-ACTIVE TARGET SHEET

The canonical mixed odd-active target branch is fixed by

$$T_E = T_O = 1, \quad R_{\perp} = \sqrt{2}, \quad \Delta_{00} = \frac{1}{\sqrt{2}}, \quad (313)$$

with coefficient sheet

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad (314)$$

$$\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa. \quad (315)$$

CCXIV. 3. EXACT MATCHING CONDITIONS

The reference sheet reproduces the canonical mixed target iff

$$\frac{\lambda_c}{2} I_3 = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad (316)$$

and

$$\frac{\gamma_E}{M_X^2} I_c = \frac{\sqrt{3}\pi}{16} T_E \theta_0 \kappa, \quad \frac{\gamma_O}{M_X^2} I_c = \frac{\sqrt{3}\pi}{16} T_O \theta_0 \kappa. \quad (317)$$

For the canonical mixed target $T_E = T_O = 1$, the natural matched choice is

$$\gamma_E = \gamma_O =: \gamma_{\perp} \neq 0, \quad (318)$$

with

$$\boxed{\frac{\gamma_{\perp}}{M_X^2} I_c = \frac{\sqrt{3}\pi}{16} \theta_0 \kappa.} \quad (319)$$

Then

$$\boxed{\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa,} \quad (320)$$

exactly as required.

CCXV. 4. UNIFIED REDUCED COEFFICIENT SHEET

Under (316) and (319), the current active reduced coefficient sheet is simply

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad \alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa.} \quad (321)$$

So the active low-order reduced PDE is

$$\boxed{\left[i\partial_t + \frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3 \sigma_3(-i\partial_{\parallel}) - \frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa \left((\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) + (-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right) \right] \chi = 0.} \quad (322)$$

CCXVI. 5. ROTATED TRANSVERSE FORM

Define the rotated transverse coordinates

$$\boxed{q_1 = \frac{p_1 - p_2}{\sqrt{2}}, \quad q_2 = \frac{p_1 + p_2}{\sqrt{2}}.} \quad (323)$$

Then the same PDE becomes

$$\boxed{\left[i\partial_t - \lambda_{\parallel} \sigma_3(-i\partial_{\parallel}) - v_{\perp}^{\text{mix}} (\sigma_1(-i\partial_{q_1}) + \sigma_2(-i\partial_{q_2})) \right] \chi = 0,} \quad (324)$$

with

$$\boxed{v_{\perp}^{\text{mix}} = -\sqrt{2} \alpha = -\sqrt{2} \beta = \frac{\sqrt{6}\pi}{16} \beta_X \theta_0 \kappa.} \quad (325)$$

CCXVII. 6. INTERPRETATION

So the current reference route is no longer merely a computational placeholder.

After the finite matching relations are imposed, it is literally the canonical mixed odd-active reduced PDE written in explicit one-mediator reference variables.

Therefore the practical active route and the structural target route are now the same sheet at reduced PDE level.

CCXVIII. CONCLUSION

The exact unified active reduced PDE sheet is

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad \alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa,$$

and its rotated transverse speed is

$$v_{\perp}^{\text{mix}} = \frac{\sqrt{6}\pi}{16} \beta_X \theta_0 \kappa.$$

So from now on, the reference low-order route and the canonical mixed odd-active target should be treated as one unified reduced PDE sheet.

CCXIX. FINAL SUMMARY

The explicit odd extra-carrier reference route and the canonical mixed odd-active target are now merged into a single reduced PDE sheet.

The exact unified coefficient sheet is

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad \alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa.$$

Equivalently, in rotated transverse coordinates,

$$v_{\perp}^{\text{mix}} = \frac{\sqrt{6}\pi}{16} \beta_X \theta_0 \kappa.$$

Thus the practical active route and the structural target route are no longer separate at reduced PDE level. They are the same matched sheet.

CCXX. EXACT ACTIVE REDUCED PDE FROM THE UNIFIED MATCHED SHEET

CCXXI. 1. UNIFIED COEFFICIENT SHEET

Take the already matched active reduced coefficient sheet

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad \alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa. \quad (326)$$

Define for convenience

$$a_{\perp} := -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa. \quad (327)$$

Then $\alpha = \beta = a_{\perp}$.

CCXXII. 2. EXACT 2×2 REDUCED SYMBOL

The reduced first-order symbol is

$$\begin{aligned} A_1(p) &= \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2) \\ &= \lambda_{\parallel} p_{\parallel} \sigma_3 + a_{\perp} \left[(p_1 - p_2) \sigma_1 + (p_1 + p_2) \sigma_2 \right]. \end{aligned} \quad (328)$$

Introduce rotated transverse momenta

$$q_1 := \frac{p_1 - p_2}{\sqrt{2}}, \quad q_2 := \frac{p_1 + p_2}{\sqrt{2}}. \quad (329)$$

Then

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \sqrt{2} a_{\perp} (q_1 \sigma_1 + q_2 \sigma_2). \quad (330)$$

Now define the transverse speed

$$v_{\perp} := -\sqrt{2} a_{\perp} = \frac{\sqrt{6} \pi}{16} \beta_X \theta_0 \kappa. \quad (331)$$

Hence the symbol may be written as

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 - v_{\perp} (q_1 \sigma_1 + q_2 \sigma_2). \quad (332)$$

Therefore the active reduced PDE is exactly

$$\left[i\partial_t - \lambda_{\parallel} \sigma_3 (-i\partial_{\parallel}) + v_{\perp} (\sigma_1 (-i\partial_{q_1}) + \sigma_2 (-i\partial_{q_2})) \right] \chi = 0. \quad (333)$$

CCXXIII. 3. EXACT DISPERSION RELATION

Because the Pauli matrices anticommute,

$$\{\sigma_i, \sigma_j\} = 2\delta_{ij},$$

we have

$$A_1(p)^2 = \lambda_{\parallel}^2 p_{\parallel}^2 + v_{\perp}^2 (q_1^2 + q_2^2). \quad (334)$$

Since $q_1^2 + q_2^2 = p_1^2 + p_2^2$, the dispersion is

$$E(p)^2 = \lambda_{\parallel}^2 p_{\parallel}^2 + v_{\perp}^2 (p_1^2 + p_2^2). \quad (335)$$

Substituting the explicit coefficients,

$$E(p)^2 = \frac{27\pi^2}{262144} \alpha_X^2 L_*^2 \theta_0^6 \kappa^6 p_{\parallel}^2 + \frac{3\pi^2}{128} \beta_X^2 \theta_0^2 \kappa^2 (p_1^2 + p_2^2). \quad (336)$$

CCXXIV. 4. EXACT ANISOTROPY RATIO

The active longitudinal/transverse anisotropy ratio is

$$\rho_D := \frac{|\lambda_{\parallel}|}{|v_{\perp}|} = \frac{3}{32\sqrt{2}} \left| \frac{\alpha_X L_*}{\beta_X} \right| \theta_0^2 \kappa^2. \quad (337)$$

Thus the current lowest-order active reduced PDE is an anisotropic massless Dirac-type equation with exact finite anisotropy controlled by

$$\frac{\alpha_X L_*}{\beta_X} \theta_0^2 \kappa^2.$$

CCXXV. 5. EXACT 4×4 DIRAC EMBEDDING

Now perform the opposite-valley doubling. Let τ_i act in valley space and σ_i in the tangent doublet. Define

$$\Gamma^1 := \tau_1 \otimes \sigma_1, \quad \Gamma^2 := \tau_1 \otimes \sigma_2, \quad \Gamma^3 := \tau_3 \otimes \mathbf{1}_2. \quad (338)$$

These satisfy

$$\{\Gamma^i, \Gamma^j\} = 2\delta^{ij} \mathbf{1}_4. \quad (339)$$

Therefore the doubled active Hamiltonian is

$$H_D(p) = -v_\perp q_1 \Gamma^1 - v_\perp q_2 \Gamma^2 + \lambda_\parallel p_\parallel \Gamma^3. \quad (340)$$

Squaring gives

$$H_D(p)^2 = \left(\lambda_\parallel^2 p_\parallel^2 + v_\perp^2 (q_1^2 + q_2^2) \right) \mathbf{1}_4, \quad (341)$$

hence the same spectrum

$$E(p) = \pm \sqrt{\lambda_\parallel^2 p_\parallel^2 + v_\perp^2 (q_1^2 + q_2^2)}.$$

So the active unified reduced sheet is exactly a massless anisotropic Dirac carrier at lowest order.

CCXXVI. 6. NONDEGENERACY CONDITIONS

The active reduced PDE is genuinely Dirac-type iff

$$\lambda_\parallel \neq 0 \quad \text{and} \quad v_\perp \neq 0. \quad (342)$$

Equivalently,

$$\alpha_X L_* \theta_0^3 \kappa^3 \neq 0, \quad \beta_X \theta_0 \kappa \neq 0. \quad (343)$$

CCXXVII. CONCLUSION

The active lowest-order reduced PDE is no longer merely a matched coefficient sheet. It is an explicit anisotropic massless Dirac equation with:

$$\lambda_\parallel = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad v_\perp = \frac{\sqrt{6}\pi}{16} \beta_X \theta_0 \kappa, \quad (344)$$

dispersion

$$E(p)^2 = \lambda_\parallel^2 p_\parallel^2 + v_\perp^2 (p_1^2 + p_2^2), \quad (345)$$

and exact 4×4 Clifford embedding

$$H_D(p) = -v_\perp q_1 \Gamma^1 - v_\perp q_2 \Gamma^2 + \lambda_\parallel p_\parallel \Gamma^3. \quad (346)$$

CCXXVIII. FINAL SUMMARY

The active unified reduced sheet has now been turned into an explicit lowest-order PDE and Hamiltonian. Its exact transverse speed is

$$v_\perp = \frac{\sqrt{6}\pi}{16} \beta_X \theta_0 \kappa,$$

its exact longitudinal coefficient is

$$\lambda_\parallel = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3,$$

and its exact dispersion is

$$E(p)^2 = \lambda_\parallel^2 p_\parallel^2 + v_\perp^2 (p_1^2 + p_2^2).$$

Thus the current active lowest-order reduced PDE is an explicit anisotropic massless Dirac equation, not merely a symbolic target sheet.

CCXXIX. EXPLICIT SIGN/VALUE AUDIT OF THE ACTIVE LOWEST-ORDER REDUCED PDE

CCXXX. 1. OVERLAP INPUTS

The active reduced coefficient sheet is

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = I_c \tilde{\Lambda}_E, \quad \beta = I_c \tilde{\Lambda}_O. \quad (347)$$

At the explicit lowest-order overlap level already evaluated in the notes,

$$I_c = I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad (348)$$

$$I_3 = I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3. \quad (349)$$

Hence

$$\text{sgn}(I_c) = -\text{sgn}(\theta_0 \kappa), \quad \text{sgn}(I_3) = -\text{sgn}(\theta_0 \kappa), \quad (350)$$

for the convention used in the present notes.

CCXXXI. 2. LONGITUDINAL COEFFICIENT

Using (349),

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3. \quad (351)$$

Therefore

$$\text{sgn}(\lambda_{\parallel}) = -\text{sgn}(\lambda_c) \text{sgn}(\theta_0 \kappa), \quad (352)$$

and

$$|\lambda_{\parallel}| = \frac{3\sqrt{3}\pi}{512} |\lambda_c| |\theta_0 \kappa|^3. \quad (353)$$

On the canonical witness branch, where

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X L_* I_3,$$

this becomes

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad (354)$$

so

$$\text{sgn}(\lambda_{\parallel}) = -\text{sgn}(\alpha_X L_*) \text{sgn}(\theta_0 \kappa). \quad (355)$$

CCXXXII. 3. TRANSVERSE COEFFICIENTS ON THE EXPLICIT ODD ROUTE

On the explicit odd extra-carrier route,

$$\tilde{\Lambda}_E^{(1)} = -\frac{\beta_X \gamma_E}{M_X^2}, \quad \tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2} \quad (356)$$

in absorbed canonical normalization.

Substituting (348),

$$\alpha^{(1)} = I_c \tilde{\Lambda}_E^{(1)} = \frac{\sqrt{3}\pi}{16} \frac{\beta_X \gamma_E}{M_X^2} \theta_0 \kappa, \quad (357)$$

$$\beta^{(1)} = I_c \tilde{\Lambda}_O^{(1)} = \frac{\sqrt{3}\pi}{16} \frac{\beta_X \gamma_O}{M_X^2} \theta_0 \kappa. \quad (358)$$

Therefore

$$\text{sgn}(\alpha^{(1)}) = \text{sgn}(\beta_X \gamma_E) \text{sgn}(\theta_0 \kappa), \quad (359)$$

$$\text{sgn}(\beta^{(1)}) = \text{sgn}(\beta_X \gamma_O) \text{sgn}(\theta_0 \kappa). \quad (360)$$

CCXXXIII. 4. CANONICAL MIXED TARGET MATCHING

For the matched canonical mixed branch

$$T_E = T_O = 1, \quad \gamma_E = \gamma_O =: \gamma_\perp,$$

the exact matching condition is

$$\frac{\gamma_\perp}{M_X^2} I_c = \frac{\sqrt{3}\pi}{16} \theta_0 \kappa. \quad (361)$$

With (348), this gives

$$\frac{\gamma_\perp}{M_X^2} = -1. \quad (362)$$

Hence the matched lowest-order explicit route reproduces

$$\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa. \quad (363)$$

Thus the transverse speed

$$v_\perp = -\sqrt{2} \alpha = \frac{\sqrt{6}\pi}{16} \beta_X \theta_0 \kappa, \quad (364)$$

satisfies

$$\text{sgn}(v_\perp) = \text{sgn}(\beta_X) \text{sgn}(\theta_0 \kappa), \quad (365)$$

and

$$|v_\perp| = \frac{\sqrt{6}\pi}{16} |\beta_X| |\theta_0 \kappa|. \quad (366)$$

CCXXXIV. 5. EXACT ACTIVE DISPERSION WITH EXPLICIT ABSOLUTE COEFFICIENTS

The active dispersion is

$$E(p)^2 = |\lambda_{\parallel}|^2 p_{\parallel}^2 + |v_{\perp}|^2 (p_1^2 + p_2^2), \quad (367)$$

that is,

$$E(p)^2 = \frac{27\pi^2}{262144} \alpha_X^2 L_*^2 \theta_0^6 \kappa^6 p_{\parallel}^2 + \frac{3\pi^2}{128} \beta_X^2 \theta_0^2 \kappa^2 (p_1^2 + p_2^2). \quad (368)$$

CCXXXV. 6. HONEST NONZERO PASS SHEET

The active lowest-order reduced PDE is genuinely anisotropic Dirac iff

$$\theta_0 \kappa \neq 0, \quad \alpha_X L_* \neq 0, \quad \beta_X \neq 0. \quad (369)$$

If one wants the odd route explicitly active before target matching, one also needs

$$\gamma_O \neq 0 \quad (\text{equivalently } \tilde{\Lambda}_O \neq 0). \quad (370)$$

CCXXXVI. CONCLUSION

At the current active lowest-order sheet, the sign/value audit is no longer symbolic. The exact overlap values already fix

$$I_c = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_3 = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3,$$

hence

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad v_{\perp} = \frac{\sqrt{6}\pi}{16} \beta_X \theta_0 \kappa,$$

and therefore the active reduced PDE is an explicit anisotropic massless Dirac equation with fully audited lowest-order coefficients.

CCXXXVII. FINAL SUMMARY

The active lowest-order reduced PDE has now passed its explicit sign/value audit. The already evaluated overlap integrals fix

$$I_c = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_3 = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3.$$

Hence the active longitudinal and transverse coefficients are

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad v_{\perp} = \frac{\sqrt{6}\pi}{16} \beta_X \theta_0 \kappa.$$

Their signs are

$$\text{sgn}(\lambda_{\parallel}) = -\text{sgn}(\alpha_X L_*) \text{sgn}(\theta_0 \kappa), \quad \text{sgn}(v_{\perp}) = \text{sgn}(\beta_X) \text{sgn}(\theta_0 \kappa).$$

So the current active reduced PDE is not merely matched and not merely symbolic: it is an explicitly sign-audited lowest-order anisotropic massless Dirac equation.

CCXXXVIII. EXACT LONGITUDINAL SIGN REDUCTION TO BACKGROUND MATRIX ENTRIES

CCXXXIX. 1. CLOSED FINITE FORMULA

In the current canonical branch,

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}) = \frac{1}{2}(u_{11} - u_{22}), \quad (371)$$

so that

$$L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2}(u_{11} - u_{22}). \quad (372)$$

The reduced longitudinal coefficient is

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X L_* I_{\theta'^3}. \quad (373)$$

Using

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad (374)$$

we obtain

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{1024} \alpha_X m_{\parallel} (u_{11} - u_{22}) \theta_0^3 \kappa^3. \quad (375)$$

CCXL. 2. EXACT SIGN FORMULA

Therefore

$$\text{sgn}(\lambda_{\parallel}) = -\text{sgn}(\alpha_X) \text{sgn}(m_{\parallel}) \text{sgn}(u_{11} - u_{22}) \text{sgn}(\theta_0 \kappa). \quad (376)$$

Equivalently, since

$$u_{\parallel} = \frac{1}{2}(u_{11} - u_{22}),$$

one may write

$$\text{sgn}(\lambda_{\parallel}) = -\text{sgn}(\alpha_X) \text{sgn}(m_{\parallel}) \text{sgn}(u_{\parallel}) \text{sgn}(\theta_0 \kappa). \quad (377)$$

CCXLI. 3. MINIMAL CANONICAL SPLIT

Under the minimal explicit canonical split,

$$U_{\parallel} = u_{\parallel} \Sigma_3, \quad (378)$$

hence

$$(U_{\parallel})_{11} = u_{\parallel}, \quad (U_{\parallel})_{22} = -u_{\parallel}.$$

Then (375) becomes

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X m_{\parallel} u_{\parallel} \theta_0^3 \kappa^3. \quad (379)$$

CCXLII. 4. EXACT NONVANISHING CRITERION

The exact vanishing criterion is

$$\boxed{\lambda_{\parallel} = 0 \iff \alpha_X = 0 \text{ or } m_{\parallel} = 0 \text{ or } u_{11} = u_{22} \text{ or } \theta_0 \kappa = 0.} \quad (380)$$

So once

$$\alpha_X \neq 0, \quad m_{\parallel} \neq 0, \quad u_{11} \neq u_{22}, \quad \theta_0 \kappa \neq 0,$$

the longitudinal slot is guaranteed to survive.

CCXLIII. ACTUAL INSERTION ON THE EXPLICIT CANONICAL WITNESS BRANCH

Take the explicit witness branch

$$T_{\star} = T^1, \quad R_{\perp} = 1, \quad \varphi = 0.$$

The uploaded notes already give

$$u_{\parallel}(T^1) = 0.5, \quad L_{\star}(T^1) = 0.5,$$

and

$$\lambda_{\parallel} = 0.031883191949430, \quad \alpha = 0.340087380793916, \quad \beta = 0.$$

At the same time, the exact branch formulas are

$$u_{\parallel}(T^1) = \frac{1}{2} \Re(\kappa_J), \quad L_{\star}(T^1) = \frac{m_{\parallel}}{2} \Re(\kappa_J).$$

Therefore, in this explicit demo normalization,

$$\boxed{\Re(\kappa_J) = 1, \quad m_{\parallel} = 1, \quad u_{\parallel}(T^1) = \frac{1}{2}, \quad L_{\star}(T^1) = \frac{1}{2}.}$$

Hence the longitudinal coefficient becomes

$$\boxed{\lambda_{\parallel}(T^1) = \frac{1}{4} \alpha_X I_{\theta^3} = -\frac{3\sqrt{3}\pi}{1024} \alpha_X \theta_0^3 \kappa^3.}$$

The active reduced symbol on this explicit witness branch is therefore

$$\boxed{A_1^{(T^1)}(p) = 0.031883191949430 p_{\parallel} \sigma_3 + 0.340087380793916 (p_1 \sigma_1 + p_2 \sigma_2).}$$

Equivalently, the dispersion is

$$\boxed{E(p)^2 = (0.031883191949430)^2 p_{\parallel}^2 + (0.340087380793916)^2 (p_1^2 + p_2^2).}$$

Numerically,

$$\boxed{E(p)^2 = 0.001016537928884198 p_{\parallel}^2 + 0.11565942657526601 (p_1^2 + p_2^2).}$$

The anisotropy ratio on this explicit witness normalization is

$$\boxed{\frac{v_{\perp}}{v_{\parallel}} = \frac{0.340087380793916}{0.031883191949430} \approx 10.6666666666665.}$$

Thus the explicit canonical witness branch is already a concrete anisotropic massless Dirac carrier at the level of the finite audit demo.

CCXLIV. FIRST-ORDER CORRECTED INSERTION OF THE LONGITUDINAL SCALAR PACKAGE

CCXLV. 1. DEFINE THE CORRECTED SCALAR PACKAGES

Introduce

$$\boxed{\Gamma_{JJ}^{\text{corr}} := \alpha_X |G_*| A_{G_*}^{\text{corr}} \mathcal{N}_{JJ}^{\text{corr}}, \quad L_*^{\text{corr}} := m_{\parallel}^{\text{corr}} u_{\parallel}^{\text{corr}}.} \quad (381)$$

Then, in the same first-shell dominant normalization used on the explicit witness branch, the corrected longitudinal coefficient is

$$\boxed{\lambda_{\parallel}^{\text{corr}} = -\frac{3\sqrt{3}\pi}{512} \Gamma_{JJ}^{\text{corr}} L_*^{\text{corr}} \theta_0^3 \kappa^3 + O(\delta^2 \theta_0^3 \kappa^3).} \quad (382)$$

CCXLVI. 2. FIRST-ORDER EXPANSION OF $\Gamma_{JJ}^{\text{corr}}$

Write

$$A_{G_*}^{\text{corr}} = A_0 + 0.0096 A_A - 0.0288 A_E + 0.0096 A_O + O(\delta^2), \quad (383)$$

$$\mathcal{N}_{JJ}^{\text{corr}} = N_0 + 0.0096 N_A - 0.0288 N_E + 0.0096 N_O + O(\delta^2). \quad (384)$$

Then

$$\boxed{\Gamma_{JJ}^{\text{corr}} = \Gamma_0 + 0.0096 \Gamma_A - 0.0288 \Gamma_E + 0.0096 \Gamma_O + O(\delta^2),} \quad (385)$$

where

$$\boxed{\Gamma_0 := \alpha_X |G_*| A_0 N_0,} \quad (386)$$

and

$$\boxed{\Gamma_X := \alpha_X |G_*| (A_X N_0 + A_0 N_X), \quad X \in \{A, E, O\}.} \quad (387)$$

CCXLVII. 3. FIRST-ORDER EXPANSION OF L_*^{corr}

Likewise write

$$m_{\parallel}^{\text{corr}} = m_0 + 0.0096 m_A - 0.0288 m_E + 0.0096 m_O + O(\delta^2), \quad (388)$$

$$u_{\parallel}^{\text{corr}} = u_0 + 0.0096 u_A - 0.0288 u_E + 0.0096 u_O + O(\delta^2). \quad (389)$$

Then

$$\boxed{L_*^{\text{corr}} = L_0 + 0.0096 L_A - 0.0288 L_E + 0.0096 L_O + O(\delta^2),} \quad (390)$$

with

$$\boxed{L_0 := m_0 u_0, \quad L_X := m_X u_0 + m_0 u_X, \quad X \in \{A, E, O\}.} \quad (391)$$

CCXLVIII. 4. FULLY REDUCED FIRST-ORDER FORMULA FOR $\lambda_{\parallel}^{\text{corr}}$

Substituting (385) and (390) into (382), we obtain

$$\lambda_{\parallel}^{\text{corr}} = -\frac{3\sqrt{3}\pi}{512} \theta_0^3 \kappa^3 \left[\Gamma_0 L_0 + 0.0096 (\Gamma_A L_0 + \Gamma_0 L_A) - 0.0288 (\Gamma_E L_0 + \Gamma_0 L_E) + 0.0096 (\Gamma_O L_0 + \Gamma_0 L_O) \right] + O(\delta^2 \theta_0^3 \kappa^3). \quad (392)$$

Define the three linear corrected response packages

$$\Lambda_A^{\text{lin}} := \Gamma_A L_0 + \Gamma_0 L_A, \quad \Lambda_E^{\text{lin}} := \Gamma_E L_0 + \Gamma_0 L_E, \quad \Lambda_O^{\text{lin}} := \Gamma_O L_0 + \Gamma_0 L_O. \quad (393)$$

Then

$$\lambda_{\parallel}^{\text{corr}} = -\frac{3\sqrt{3}\pi}{512} \theta_0^3 \kappa^3 \left[\Gamma_0 L_0 + 0.0096 \Lambda_A^{\text{lin}} - 0.0288 \Lambda_E^{\text{lin}} + 0.0096 \Lambda_O^{\text{lin}} \right] + O(\delta^2). \quad (394)$$

CCXLIX. 5. INSERT THE EXPLICIT WITNESS NORMALIZATION

On the explicit canonical witness demo,

$$u_0 = 0.5, \quad L_0 = 0.5,$$

and therefore $m_0 = 1$ in that normalization. Hence

$$L_A = 0.5 m_A + u_A, \quad L_E = 0.5 m_E + u_E, \quad L_O = 0.5 m_O + u_O. \quad (395)$$

So the corrected longitudinal coefficient on the intended branch is reduced to the finite scalar input set

$$A_0, A_A, A_E, A_O, N_0, N_A, N_E, N_O, m_A, m_E, m_O, u_A, u_E, u_O.$$

CCL. 6. PRACTICAL SIGN-STABILITY CRITERION

A sufficient condition that the corrected branch keeps the same longitudinal sign as the base witness branch is

$$|0.0096 \Lambda_A^{\text{lin}} - 0.0288 \Lambda_E^{\text{lin}} + 0.0096 \Lambda_O^{\text{lin}}| < |\Gamma_0 L_0|. \quad (396)$$

Under (396),

$$\text{sgn}(\lambda_{\parallel}^{\text{corr}}) = -\text{sgn}(\Gamma_0 L_0) \text{sgn}(\theta_0 \kappa). \quad (397)$$

CCLI. CONCLUSION

The longitudinal corrected branch is no longer blocked by missing structure. It is reduced to the explicit first-order scalar package

$$\Gamma_0, L_0, \Lambda_A^{\text{lin}}, \Lambda_E^{\text{lin}}, \Lambda_O^{\text{lin}}.$$

So the remaining task is literally the finite insertion of these corrected scalar responses.

CCLII. FIRST EXPLICIT FINITE PACKAGE FOR THE CORRECTED TRANSVERSE COEFFICIENTS

CCLIII. 1. FIXED CORRECTED SEED

Take

$$\delta_{\text{corr}} = (\delta_A, \delta_E, \delta_O) = (0.0096, -0.0288, 0.0096). \quad (398)$$

Write

$$a \cdot \delta_{\text{corr}} = 0.0096 a_A - 0.0288 a_E + 0.0096 a_O. \quad (399)$$

CCLIV. 2. BASE-PLUS-CORRECTION PACKAGE FOR α^{corr}

Define the base even transverse scalar

$$\Xi_0 := A_0 X_1^{(0)}. \quad (400)$$

Define the linear response package

$$\Xi_A^{\text{lin}} := a_A X_1^{(0)}, \quad \Xi_E^{\text{lin}} := a_E X_1^{(0)}, \quad \Xi_O^{\text{lin}} := a_O X_1^{(0)}. \quad (401)$$

Define the quadratic response package

$$\Xi_A^{\text{quad}} := X_1^{(0)} Q_A(\delta_{\text{corr}}), \quad \Xi_1^{\text{quad}} := A_0 Q_1(\delta_{\text{corr}}). \quad (402)$$

Then

$$\alpha^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\Xi_0 + 0.0096 \Xi_A^{\text{lin}} - 0.0288 \Xi_E^{\text{lin}} + 0.0096 \Xi_O^{\text{lin}} + \Xi_A^{\text{quad}} + \Xi_1^{\text{quad}} \right] + O(\delta^3). \quad (403)$$

Equivalently, since

$$\alpha^{(0)} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 X_1^{(0)},$$

one may write

$$\alpha^{\text{corr}} = \alpha^{(0)} \left[1 + \frac{a \cdot \delta_{\text{corr}}}{A_0} + \frac{Q_A(\delta_{\text{corr}})}{A_0} + \frac{Q_1(\delta_{\text{corr}})}{X_1^{(0)}} \right] + O(\delta^3). \quad (404)$$

CCLV. 3. PURE-EVEN BRANCH PACKAGE FOR β^{corr}

On the explicit canonical pure-even branch,

$$X_2^{(0)} = 0.$$

Therefore the corrected odd transverse coefficient starts only at quadratic order:

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 Q_2(\delta_{\text{corr}}) + O(\delta^3). \quad (405)$$

Thus the entire odd transverse frontier is compressed to the single quadratic scalar

$$Q_2(\delta_{\text{corr}}).$$

CCLVI. 4. EXACT QUADRATIC HESSIAN FORM OF Q_2

Let $H^{(2)}$ denote the odd transverse Hessian matrix, with entries

$$H_{AA}^{(2)}, H_{EE}^{(2)}, H_{OO}^{(2)}, H_{AE}^{(2)}, H_{AO}^{(2)}, H_{EO}^{(2)}.$$

Then

$$Q_2(\delta_{\text{corr}}) = \frac{1}{2} \delta_{\text{corr}}^T H^{(2)} \delta_{\text{corr}},$$

that is,

$$\begin{aligned} Q_2(\delta_{\text{corr}}) = & 4.608 \times 10^{-5} H_{AA}^{(2)} + 4.1472 \times 10^{-4} H_{EE}^{(2)} + 4.608 \times 10^{-5} H_{OO}^{(2)} \\ & - 2.7648 \times 10^{-4} H_{AE}^{(2)} + 9.216 \times 10^{-5} H_{AO}^{(2)} - 2.7648 \times 10^{-4} H_{EO}^{(2)}. \end{aligned} \quad (406)$$

Therefore

$$\boxed{\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 \left[4.608 \times 10^{-5} H_{AA}^{(2)} + 4.1472 \times 10^{-4} H_{EE}^{(2)} + 4.608 \times 10^{-5} H_{OO}^{(2)} - 2.7648 \times 10^{-4} H_{AE}^{(2)} + 9.216 \times 10^{-5} H_{AO}^{(2)} - 2.7648 \times 10^{-4} H_{EO}^{(2)} \right]} \quad (407)$$

CCLVII. 5. PRACTICAL DOMINANCE STATEMENT

Hence the corrected even channel is governed by

$$\Xi_0 \quad \text{plus linear and quadratic corrections,}$$

while the corrected odd channel is governed entirely by

$$A_0 Q_2(\delta_{\text{corr}}).$$

So on the present pure-even canonical branch:

$$\boxed{\alpha^{\text{corr}} \text{ is base-dominated and linearly correctable,}} \quad (408)$$

whereas

$$\boxed{\beta^{\text{corr}} \text{ is purely quadratic and Hessian-controlled.}} \quad (409)$$

CCLVIII. CONCLUSION

The transverse corrected branch is now reduced to two finite packages:

$$\alpha^{\text{corr}} \leftrightarrow (\Xi_0, \Xi_A^{\text{lin}}, \Xi_E^{\text{lin}}, \Xi_O^{\text{lin}}, \Xi_A^{\text{quad}}, \Xi_1^{\text{quad}}),$$

$$\beta^{\text{corr}} \leftrightarrow A_0 Q_2(\delta_{\text{corr}}).$$

So the remaining work is not structural. It is the finite insertion of

$$a_A, a_E, a_O, Q_A, Q_1, H_{AA}^{(2)}, H_{EE}^{(2)}, H_{OO}^{(2)}, H_{AE}^{(2)}, H_{AO}^{(2)}, H_{EO}^{(2)}.$$

CCLIX. CORRECTED-BRANCH FINITE-INSERTION PDE SHEET

Define the corrected longitudinal scalar

$$\boxed{\Gamma_L^{\text{corr}} := A_{G_*}^{\text{corr}} \mathcal{N}_{JJ}^{\text{corr}} m_{\parallel}^{\text{corr}} u_{\parallel}^{\text{corr}}.} \quad (410)$$

Define the corrected transverse scalars

$$\boxed{T_E^{\text{corr}} := A_{G_*}^{\text{corr}} (C_1^{\text{corr}} \cdot G_*), \quad T_O^{\text{corr}} := A_{G_*}^{\text{corr}} (C_2^{\text{corr}} \cdot G_*).} \quad (411)$$

Then the corrected reduced symbol is

$$\boxed{A_1^{\text{corr}}(p) = \lambda_{\parallel}^{\text{corr}} p_{\parallel} \sigma_3 + \alpha^{\text{corr}} (p_1 \sigma_1 + p_2 \sigma_2) + \beta^{\text{corr}} (-p_2 \sigma_1 + p_1 \sigma_2),} \quad (412)$$

with finite-insertion coefficients

$$\boxed{\lambda_{\parallel}^{\text{corr}} \propto \Gamma_L^{\text{corr}} \theta_0^3 \kappa^3,} \quad (413)$$

$$\boxed{\alpha^{\text{corr}} \propto T_E^{\text{corr}} \theta_0 \kappa, \quad \beta^{\text{corr}} \propto T_O^{\text{corr}} \theta_0 \kappa.} \quad (414)$$

At Branch-B overlap level,

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16} \Lambda_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \Lambda_O \theta_0 \kappa,} \quad (415)$$

so the corrected finite insertion is exactly the replacement

$$\lambda_c \rightsquigarrow \Gamma_L^{\text{corr}}, \quad \Lambda_E \rightsquigarrow T_E^{\text{corr}}, \quad \Lambda_O \rightsquigarrow T_O^{\text{corr}},$$

up to the already isolated branch-normalization convention.

CCLX. ACTUAL CORRECTED INSERTION REACHED TODAY

CCLXI. 1. LONGITUDINAL SIDE: ORBIT-EXACT SCALAR INSERTION

The corrected first-shell longitudinal witness is no longer only schematic. Using the orbit-response reduction, it becomes

$$\boxed{\lambda_{\parallel}^{\text{corr}} \propto \left(A_{G_*}^{(0)} + 0.0096 \right) \left(\mathcal{N}_{JJ}^{(0)} + 0.0120 \mathcal{N}_{JJ}^{[c^2]} + 0.0228 \mathcal{N}_{JJ}^{[P_2]} \right) \left(m_{\parallel} u_{\parallel} \right)_{\text{corr}} \theta_0^3 \kappa^3.} \quad (416)$$

So the longitudinal corrected branch is now reduced to exactly three scalar inputs:

$$A_{G_*}^{(0)} + 0.0096, \quad \mathcal{N}_{JJ}^{(0)} + 0.0120 \mathcal{N}_{JJ}^{[c^2]} + 0.0228 \mathcal{N}_{JJ}^{[P_2]}, \quad (m_{\parallel} u_{\parallel})_{\text{corr}}.$$

CCLXII. 2. TRANSVERSE SIDE: EXACT PLACE WHERE ORBIT REDUCTION STOPS

The corrected transverse witnesses are still

$$\boxed{\alpha^{\text{corr}} \propto A_{G_*}^{\text{corr}} (C_1^{\text{corr}} \cdot G_*) \theta_0 \kappa, \quad \beta^{\text{corr}} \propto A_{G_*}^{\text{corr}} (C_2^{\text{corr}} \cdot G_*) \theta_0 \kappa.} \quad (417)$$

But these do *not* close on (A, E, O) orbit data alone, because

$$(C_1 \cdot G_*), \quad (C_2 \cdot G_*)$$

are sign-sensitive observables carried by the transverse shell coordinates

$$t_{\alpha 1}, \quad t_{\alpha 2}.$$

Therefore the exact remaining transverse task is not another theorem. It is the insertion of either

$$\boxed{\delta_{\text{sub}} = (\delta_A, \delta_{E+}, \delta_{E-}, \delta_{O,1+}, \delta_{O,1-}, \delta_{O,2+}, \delta_{O,2-})^T} \quad (418)$$

or the explicit corrected Bloch coefficients $c_{a,G}^{(\pm)}$.

CCLXIII. 3. WHAT HAS GENUINELY ADVANCED

So the project has advanced in a nonredundant way:

- the longitudinal corrected scalar package is now orbit-exact up to the last overlap factor $(m_{\parallel} u_{\parallel})_{\text{corr}}$;
- the transverse sector is now sharply identified as the only place where sign-resolved suborbit/Bloch data are indispensable.

CCLXIV. CORRECTED LONGITUDINAL SIGN CRITERION IN ITS FINAL NONREDUNDANT FORM

Define

$$\mathcal{A}_{\text{corr}} := A_{G_*}^{(0)} + 0.0096,$$

$$\mathcal{N}_{\text{corr}} := \mathcal{N}_{JJ}^{(0)} + 0.0120 \mathcal{N}_{JJ}^{[c^2]} + 0.0228 \mathcal{N}_{JJ}^{[P_2]},$$

$$\mathcal{L}_{\text{corr}} := (m_{\parallel} u_{\parallel})_{\text{corr}}.$$

Then the corrected longitudinal coefficient has the reduced form

$$\boxed{\lambda_{\parallel}^{\text{corr}} = -\mathcal{C}_L \alpha_X \mathcal{A}_{\text{corr}} \mathcal{N}_{\text{corr}} \mathcal{L}_{\text{corr}} \theta_0^3 \kappa^3}, \quad (419)$$

where $\mathcal{C}_L > 0$ is the already fixed overlap prefactor.

Therefore the exact corrected-sign formula is

$$\boxed{\text{sgn}(\lambda_{\parallel}^{\text{corr}}) = -\text{sgn}(\alpha_X) \text{sgn}(\mathcal{A}_{\text{corr}}) \text{sgn}(\mathcal{N}_{\text{corr}}) \text{sgn}(\mathcal{L}_{\text{corr}}) \text{sgn}(\theta_0 \kappa)}. \quad (420)$$

Hence the longitudinal corrected branch is completely reduced to three scalar sign insertions:

$$\mathcal{A}_{\text{corr}}, \quad \mathcal{N}_{\text{corr}}, \quad \mathcal{L}_{\text{corr}}.$$

CCLXV. EXACT FIRST-ORDER CORRECTED INSERTION OF $(m_{\parallel} u_{\parallel})_{\text{corr}}$

From the uploaded corrected-branch formulas,

$$m_{\parallel}^{\text{corr}} = m_0 + 0.0096 m_A - 0.0288 m_E + 0.0096 m_O + O(\delta^2), \quad (421)$$

$$u_{\parallel}^{\text{corr}} = u_0 + 0.0096 u_A - 0.0288 u_E + 0.0096 u_O + O(\delta^2), \quad (422)$$

with

$$m_0 := m_{\parallel}^{(0)}, \quad u_0 := u_{\parallel}^{(0)}.$$

Therefore

$$\begin{aligned} (m_{\parallel} u_{\parallel})_{\text{corr}} &= m_0 u_0 + 0.0096 (m_A u_0 + m_0 u_A) - 0.0288 (m_E u_0 + m_0 u_E) + 0.0096 (m_O u_0 + m_0 u_O) \\ &\quad + O(\delta^2). \end{aligned} \quad (423)$$

On the explicit canonical witness normalization,

$$u_0 = 0.5, \quad L_*^{(0)} = m_0 u_0 = 0.5,$$

hence

$$m_0 = 1.$$

So (423) reduces to

$$(m_{\parallel} u_{\parallel})_{\text{corr}} = 0.5 + 0.0096 (0.5 m_A + u_A) - 0.0288 (0.5 m_E + u_E) + 0.0096 (0.5 m_O + u_O) + O(\delta^2). \quad (424)$$

$$\lambda_{\parallel}^{\text{corr}} \propto A_{G_*}^{\text{corr}} \mathcal{N}_{JJ}^{\text{corr}} \left[0.5 + 0.0096 (0.5 m_A + u_A) - 0.0288 (0.5 m_E + u_E) + 0.0096 (0.5 m_O + u_O) \right] \theta_0^3 \kappa^3 + O(\delta^2). \quad (425)$$

CCLXVI. FINAL NONREDUNDANT LONGITUDINAL ROUTE

The remaining longitudinal problem should no longer be pursued through separate response coefficients

$$m_A, m_E, m_O, u_A, u_E, u_O.$$

The exact nonredundant route is to compute the corrected projected shell-normal matrix

$$M_{\parallel}^{\text{corr}} := P_* \left(\hat{n}^i \partial_{k_i} H_{\text{eff}}^{\text{corr}}(k) \Big|_{k=k_*} \right) P_*. \quad (426)$$

Then the corrected longitudinal coefficient is read off directly by

$$\lambda_{\parallel}^{\text{corr}} = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}^{\text{corr}}) = \frac{1}{2} (M_{11}^{\parallel, \text{corr}} - M_{22}^{\parallel, \text{corr}}). \quad (427)$$

Therefore the exact last fork is:

$$\text{either verify an actual internal } J\text{-split between the two carrier states,} \quad (428)$$

which makes $M_{11}^{\parallel, \text{corr}} - M_{22}^{\parallel, \text{corr}}$ generically nonzero,
or else

$$\text{compute } M_{11}^{\parallel, \text{corr}} - M_{22}^{\parallel, \text{corr}} \text{ directly.} \quad (429)$$

This is the final nonredundant longitudinal computation.

CCLXVII. FINAL EXECUTABLE LONGITUDINAL ROUTE FROM THE UPLOADED FILES

CCLXVIII. 1. THE RESPONSE-COEFFICIENT ROUTE STOPS HERE

The uploaded files do not contain numerical values for

$$m_{\parallel}^{[A, E, O]}, \quad u_{\parallel}^{[A, E, O]}.$$

So the response-expansion route for

$$(m_{\parallel} u_{\parallel})_{\text{corr}}$$

cannot be pushed numerically any further from the present files alone.

CCLXIX. 2. THE EXACT NONREDUNDANT ROUTE

The exact remaining longitudinal computation is the direct shell-projected diagonal asymmetry audit:

$$\lambda_{\parallel}^{\text{corr}} = \frac{1}{2} \left[(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22} \right]. \quad (430)$$

Equivalently, in the carrier-side notation,

$$L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2} (u_{11} - u_{22}). \quad (431)$$

Hence

$$L_* = 0 \iff m_{\parallel} = 0 \quad \text{or} \quad u_{11} = u_{22}. \quad (432)$$

CCLXX. 3. CANONICAL ACTIVE-DIRECTION REDUCTION

In the canonical locked basis,

$$U_{\parallel} = \begin{pmatrix} 2\Re(\kappa_J t_{10}) & 0 \\ 0 & 2\Re(\kappa_J t_{20}) \end{pmatrix}, \quad (433)$$

so that

$$u_{\parallel} = \Re[\kappa_J (t_{10} - t_{20})]. \quad (434)$$

Only the active directions T^1, T^2, T^4, T^5 survive. For a general active direction

$$T_* = c_1 T^1 + c_2 T^2 + c_4 T^4 + c_5 T^5,$$

one gets

$$u_{\parallel}(T_*) = \frac{1}{2} \Re \left[\kappa_J ((c_1 + ic_2) - (c_4 + ic_5)) \right]. \quad (435)$$

Therefore the exact last longitudinal finite audit is reduced to

$$m_{\parallel}, \quad \kappa_J, \quad (c_1, c_2, c_4, c_5). \quad (436)$$

CCLXXI. 4. EXPLICIT LONGITUDINAL COEFFICIENT

Using the already closed overlap value

$$I_{\theta^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3,$$

the longitudinal coefficient is

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3 = -\frac{3\sqrt{3}\pi}{512} \alpha_X m_{\parallel} u_{\parallel}(T_*) \theta_0^3 \kappa^3. \quad (437)$$

So the remaining executable test is not another reduction theorem. It is the direct insertion of

$$m_{\parallel}, \quad \kappa_J, \quad (c_1, c_2, c_4, c_5).$$

CCLXXII. EXACT OPTIMIZATION OF THE ACTIVE LONGITUDINAL DIRECTION

Starting from

$$u_{\parallel}(T_{\star}) = \frac{1}{2} \Re \left[\kappa_J \left((c_1 + ic_2) - (c_4 + ic_5) \right) \right],$$

write

$$\kappa_J = |\kappa_J| e^{i\phi_J}, \quad \Delta_1 := c_1 - c_4, \quad \Delta_2 := c_2 - c_5.$$

Then

$$u_{\parallel}(T_{\star}) = \frac{|\kappa_J|}{2} \left(\Delta_1 \cos \phi_J - \Delta_2 \sin \phi_J \right). \quad (438)$$

Therefore the exact canonical nonvanishing condition is

$$\Delta_1 \cos \phi_J - \Delta_2 \sin \phi_J \neq 0. \quad (439)$$

Now impose the natural unit normalization on the active mediator direction in the active 4-plane:

$$c_1^2 + c_2^2 + c_4^2 + c_5^2 = 1.$$

Then

$$\Delta_1^2 + \Delta_2^2 = (c_1 - c_4)^2 + (c_2 - c_5)^2 \leq 2,$$

with equality iff

$$(c_4, c_5) = -(c_1, c_2).$$

Hence the maximal possible longitudinal splitting is

$$|u_{\parallel}|_{\max} = \frac{|\kappa_J|}{\sqrt{2}}. \quad (440)$$

One maximizing choice is obtained by taking

$$c_1 = \frac{\cos \phi_J}{\sqrt{2}}, \quad c_2 = -\frac{\sin \phi_J}{\sqrt{2}}, \quad c_4 = -\frac{\cos \phi_J}{\sqrt{2}}, \quad c_5 = \frac{\sin \phi_J}{\sqrt{2}}. \quad (441)$$

Equivalently,

$$T_{\star}^{\text{opt}} = \frac{1}{\sqrt{2}} \left(\cos \phi_J T^1 - \sin \phi_J T^2 - \cos \phi_J T^4 + \sin \phi_J T^5 \right). \quad (442)$$

For this choice,

$$u_{\parallel}(T_{\star}^{\text{opt}}) = \frac{|\kappa_J|}{\sqrt{2}}. \quad (443)$$

Therefore

$$L_{\star}^{\text{opt}} = m_{\parallel} u_{\parallel}(T_{\star}^{\text{opt}}) = \frac{m_{\parallel} |\kappa_J|}{\sqrt{2}}. \quad (444)$$

Using

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X L_{\star} I_{\theta'^3}, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3,$$

we obtain the optimal active longitudinal coefficient

$$\lambda_{\parallel}^{\text{opt}} = -\frac{3\sqrt{3}\pi}{512\sqrt{2}} \alpha_X m_{\parallel} |\kappa_J| \theta_0^3 \kappa^3. \quad (445)$$

Hence the exact optimal-sign formula is

$$\text{sgn}(\lambda_{\parallel}^{\text{opt}}) = -\text{sgn}(\alpha_X) \text{sgn}(m_{\parallel}) \text{sgn}(\theta_0 \kappa). \quad (446)$$

because $|\kappa_J| > 0$ carries no sign.

CCLXXIII. FINAL SUMMARY

The longitudinal active-direction problem is now solved exactly.

Given

$$u_{\parallel}(T_{\star}) = \frac{1}{2} \Re \left[\kappa_J ((c_1 + ic_2) - (c_4 + ic_5)) \right],$$

the optimal active direction in the normalized active 4-plane is

$$T_{\star}^{\text{opt}} = \frac{1}{\sqrt{2}} \left(\cos \phi_J T^1 - \sin \phi_J T^2 - \cos \phi_J T^4 + \sin \phi_J T^5 \right), \quad \phi_J = \arg \kappa_J.$$

It gives

$$|u_{\parallel}|_{\max} = \frac{|\kappa_J|}{\sqrt{2}}, \quad L_{\star}^{\text{opt}} = \frac{m_{\parallel} |\kappa_J|}{\sqrt{2}},$$

and therefore

$$\lambda_{\parallel}^{\text{opt}} = -\frac{3\sqrt{3}\pi}{512\sqrt{2}} \alpha_X m_{\parallel} |\kappa_J| \theta_0^3 \kappa^3.$$

So the only remaining genuine longitudinal microscopic input is the actual sign/value of $m_{\parallel}$.

CCLXXIV. ADOPT THE OPTIMAL ACTIVE LONGITUDINAL GENERATOR

The exact active-direction formula is

$$u_{\parallel}(T_{\star}) = \frac{1}{2} \Re \left[\kappa_J ((c_1 + ic_2) - (c_4 + ic_5)) \right]. \quad (447)$$

Write

$$\kappa_J = |\kappa_J| e^{i\phi_J}.$$

Then the optimal normalized active direction in the surviving 4-plane is

$$T_{\star}^{\text{opt}} = \frac{1}{\sqrt{2}} \left(\cos \phi_J T^1 - \sin \phi_J T^2 - \cos \phi_J T^4 + \sin \phi_J T^5 \right). \quad (448)$$

For this choice,

$$u_{\parallel}(T_{\star}^{\text{opt}}) = \frac{|\kappa_J|}{\sqrt{2}}, \quad L_{\star}^{\text{opt}} = \frac{m_{\parallel} |\kappa_J|}{\sqrt{2}}. \quad (449)$$

Using

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X L_* I_{\theta'^3}, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3,$$

we obtain the active longitudinal coefficient

$$\lambda_{\parallel}^{\text{opt}} = -\frac{3\sqrt{3}\pi}{512\sqrt{2}} \alpha_X m_{\parallel} |\kappa_J| \theta_0^3 \kappa^3. \quad (450)$$

Hence the exact practical sufficient condition for the longitudinal slot on the adopted active branch is simply

$$\alpha_X \neq 0, \quad m_{\parallel} \neq 0, \quad |\kappa_J| \neq 0, \quad \theta_0 \kappa \neq 0. \quad (451)$$

CCLXXV. ACTUAL DIRECTION TO THE FINAL GOAL

The practical acceptance theorem is already reduced to

$$\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho}, \quad \mathcal{M}_{11}^{(JJ)} \neq \mathcal{M}_{22}^{(JJ)}, \quad (c_E, c_O) \neq (0, 0), \quad A_{\text{res}} \cap \mathcal{M}_D = \{0\}.$$

At the present stage, the longitudinal branch is no longer the main bottleneck, because one may adopt the optimal active direction

$$T_{\star}^{\text{opt}},$$

for which

$$u_{\parallel}(T_{\star}^{\text{opt}}) = \frac{|\kappa_J|}{\sqrt{2}}, \quad L_{\star}^{\text{opt}} = \frac{m_{\parallel} |\kappa_J|}{\sqrt{2}}.$$

Therefore the true remaining frontier is the transverse sign-resolved insertion.

The exact next computation is:

$$\mathcal{T}_{\nu} := \left. \partial_{p_{\nu}} \widehat{\Sigma}_{JK}(G_*; p) \right|_{p=0} + \left. \partial_{p_{\nu}} \widehat{\Sigma}_{KK}(G_*; p) \right|_{p=0}, \quad \nu = 1, 2, \quad (452)$$

followed by

$$c_E = \frac{1}{4} \text{Tr}(\sigma_1 \mathcal{T}_1 + \sigma_2 \mathcal{T}_2), \quad c_O = \frac{1}{4} \text{Tr}(\sigma_2 \mathcal{T}_1 - \sigma_1 \mathcal{T}_2). \quad (453)$$

But this does not close on the orbit-average vector $(\delta_A, \delta_E, \delta_O)$ alone. It requires either

$$\delta_{\text{sub}} = (\delta_A, \delta_{E+}, \delta_{E-}, \delta_{O,1+}, \delta_{O,1-}, \delta_{O,2+}, \delta_{O,2-})^T$$

or the explicit corrected Bloch coefficients $c_{a,G}^{(\pm)}$.

Hence the correct direction to the final goal is not more symbolic reduction. It is the explicit sign-resolved transverse insertion.

CCLXXVI. ACTUAL MICROSCOPIC PDE COMPLETION ROUTE

1. Fix the intended BCC+defect working branch and construct the actual Bloch–Floquet linearized operator $L(k)$.
2. Build the shell-projected operator

$$H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*}$$

and the low-slot projector P_* .

3. Compute the projected transport matrices

$$M_i = P_* \partial_{k_i} H_{\text{eff}}(k_*) P_*.$$

4. Read the reduced coefficients by

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}), \quad \alpha = \frac{1}{4} \text{Tr}(\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4} \text{Tr}(\sigma_2 M_1 - \sigma_1 M_2).$$

5. For the longitudinal slot, adopt the optimal active direction $T_{\star}^{\text{opt}}$ and reduce

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512\sqrt{2}} \alpha_X m_{\parallel} |\kappa_J| \theta_0^3 \kappa^3.$$

6. For the transverse slots, perform the sign-resolved suborbit/Bloch insertion needed to evaluate

$$(C_1^{\text{corr}} \cdot G_*), \quad (C_2^{\text{corr}} \cdot G_*).$$

7. Insert the remote-gap finite constants and verify

$$\Delta_{\text{bench}, \rho}^{\text{fin}} > \eta_{R, \rho}.$$

8. Complete the constant-sector mass-protection audit

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}.$$

9. Write the final microscopic reduced PDE and its doubled 4×4 Dirac Hamiltonian.

CCLXXVII. HONEST CURRENT STATUS

The reduced Dirac form is already fixed. The longitudinal slot is practically controlled after choosing the optimal active direction $T_{\star}^{\text{opt}}$. The true remaining frontier is the transverse sign-resolved insertion:

$$(c_E, c_O) \neq (0, 0)$$

must be checked using suborbit-resolved or explicit Bloch-resolved data.

So the actual microscopic PDE completion route is no longer blocked by symbolic reduction. It is blocked only by the final finite sign-resolved insertion layer.

CCLXXVIII. FIRST ACTUAL TRANSVERSE TRIGGER ON THE SIGN-RESOLVED SUBORBIT BASIS

CCLXXIX. 1. FIXED SUBORBIT DATA

The minimal sign-resolved first-shell basis is

$$(A_+, A_-, E_+, E_-, O_{++}, O_{+-}, O_{-+}, O_{--}),$$

and the frozen corrected seed lifts to

$$\delta_{\text{sub}}^{\text{corr}} = (0.0096, 0.0096, -0.0288, -0.0288, 0.0096, 0.0096, 0.0096, 0.0096)^T. \quad (454)$$

The exact transverse rows are

$$r_{\text{sub}}^{(ct_1)} = \left(0, 0, 0, 0, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}\right), \quad (455)$$

$$r_{\text{sub}}^{(ct_2)} = \left(0, 0, 0, 0, \frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}\right). \quad (456)$$

Applying these to the corrected lifted seed gives

$$r_{\text{sub}}^{(ct_1)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0, \quad r_{\text{sub}}^{(ct_2)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0. \quad (457)$$

So the current orbit-symmetric corrected seed still produces no linear first-shell transverse shift.

CCLXXX. 2. FIRST SUBORBIT QUARTIC COMPLETION IS NOT YET TRANSVERSE-GENERATING

The first explicit suborbit quartic completion is

$$\boxed{\Delta M_{AE} = \xi(u_{AE}u_O^T + u_Ou_{AE}^T), \quad \xi = -343.5328734898113,} \quad (458)$$

with

$$u_{AE} = (1, 1, -1, -1, 0, 0, 0, 0)^T, \quad u_O = (0, 0, 0, 0, 1, 1, 1, 1)^T.$$

This is the first A/E -breaking but O -uniform quartic completion.

However, because the four O -suborbit loads remain equal, one still has

$$r_{\text{sub}}^{(ct_1)} \cdot Q_O^{\text{new}} = 0, \quad r_{\text{sub}}^{(ct_2)} \cdot Q_O^{\text{new}} = 0.$$

Thus ΔM_{AE} breaks coarse A/E symmetry, but does not yet generate a transverse witness.

CCLXXXI. 3. FIRST GENUINELY TRANSVERSE-SENSITIVE DIRECTIONS

The next irreducible sign-sensitive directions are

$$\boxed{s_1 = (0, 0, 0, 0, 1, 1, -1, -1)^T, \quad s_2 = (0, 0, 0, 0, 1, -1, 1, -1)^T.} \quad (459)$$

These are dual to the exact transverse rows

$$r_{\text{sub}}^{(ct_1)}, \quad r_{\text{sub}}^{(ct_2)}.$$

Hence the first genuinely transverse-sensitive quartic completions are

$$\boxed{\Delta M_{t_1} = \kappa_1(u_{AE}s_1^T + s_1u_{AE}^T), \quad \Delta M_{t_2} = \kappa_2(u_{AE}s_2^T + s_2u_{AE}^T).} \quad (460)$$

These are exactly channel-selective:

$$\boxed{\Delta M_{t_1} \Rightarrow \text{nonzero } ct_1, \text{ zero } ct_2,} \quad (461)$$

$$\boxed{\Delta M_{t_2} \Rightarrow \text{nonzero } ct_2, \text{ zero } ct_1.} \quad (462)$$

CCLXXXII. 4. FIRST EXECUTABLE TRANSVERSE COMPLETION

Therefore the first executable transverse-sensitive corrected suborbit kernel is

$$\boxed{M_{\text{sub}}^{\text{corr}} = M_{\text{sub}}^{(4c)} + \Delta M_{AE} + \Delta M_{t_1} + \Delta M_{t_2}.} \quad (463)$$

At this stage, the transverse witness no longer depends on vague symbolic structure. It depends on the two scalar trigger coefficients

$$\kappa_1, \quad \kappa_2.$$

CCLXXXIII. 5. MEANING FOR THE REDUCED PDE

The corrected transverse reduced coefficients are still read by

$$\alpha^{\text{corr}} \propto (C_1^{\text{corr}} \cdot G_*), \quad \beta^{\text{corr}} \propto (C_2^{\text{corr}} \cdot G_*).$$

Therefore, at the sign-resolved suborbit level, the first actual transverse-trigger theorem is:

$$\boxed{\kappa_1 \neq 0 \implies \text{the } ct_1\text{-channel can turn on,}} \quad (464)$$

$$\boxed{\kappa_2 \neq 0 \implies \text{the } ct_2\text{-channel can turn on.}} \quad (465)$$

So the real remaining transverse frontier is now reduced to the finite insertion of

$$\kappa_1, \quad \kappa_2.$$

CCLXXXIV. CONCLUSION

The first genuine transverse trigger has now been identified explicitly.

It is not the coarse (A, E, O) correction, and not the A/E -breaking but O -uniform completion ΔM_{AE} .

It is the first sign-sensitive O -sheet completion

$$\Delta M_{t_1}, \quad \Delta M_{t_2},$$

with coefficients κ_1, κ_2 .

Therefore the actual microscopic PDE completion route has advanced to a strict finite transverse trigger audit:

$$\boxed{\text{determine the sign/value of } \kappa_1, \kappa_2.}$$

CCLXXXV. FIRST ACTUAL TRANSVERSE TRIGGER AMPLITUDES

Because

$$s_1 \cdot \delta_{\text{sub}}^{\text{corr}} = 0, \quad s_2 \cdot \delta_{\text{sub}}^{\text{corr}} = 0, \quad u_{AE} \cdot \delta_{\text{sub}}^{\text{corr}} = 0.0768,$$

the two trigger completions act on the corrected seed as

$$\boxed{\Delta M_{t_1} \delta_{\text{sub}}^{\text{corr}} = 0.0768 \kappa_1 s_1,} \quad (466)$$

$$\boxed{\Delta M_{t_2} \delta_{\text{sub}}^{\text{corr}} = 0.0768 \kappa_2 s_2.} \quad (467)$$

$$\boxed{r_{\text{sub}}^{(ct_1)} \cdot (\Delta M_{t_1} \delta_{\text{sub}}^{\text{corr}}) = 0.0768 \kappa_1,} \quad (468)$$

$$\boxed{r_{\text{sub}}^{(ct_2)} \cdot (\Delta M_{t_1} \delta_{\text{sub}}^{\text{corr}}) = 0,} \quad (469)$$

because

$$r_{\text{sub}}^{(ct_1)} \cdot s_1 = 1, \quad r_{\text{sub}}^{(ct_2)} \cdot s_1 = 0.$$

Likewise,

$$\boxed{r_{\text{sub}}^{(ct_1)} \cdot (\Delta M_{t_2} \delta_{\text{sub}}^{\text{corr}}) = 0,} \quad (470)$$

$$\boxed{r_{\text{sub}}^{(ct_2)} \cdot (\Delta M_{t_2} \delta_{\text{sub}}^{\text{corr}}) = 0.0768 \sqrt{2} \kappa_2,} \quad (471)$$

because

$$r_{\text{sub}}^{(ct_1)} \cdot s_2 = 0, \quad r_{\text{sub}}^{(ct_2)} \cdot s_2 = \sqrt{2}.$$

CCLXXXVI. FIRST CORRECTED TRANSVERSE COEFFICIENT INSERTION

At first trigger level,

$$\delta\alpha^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_{G_*}^{(0)} (0.0768 \kappa_1), \quad (472)$$

$$\delta\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_{G_*}^{(0)} (0.0768 \sqrt{2} \kappa_2), \quad (473)$$

up to the already fixed branch normalization convention.

CCLXXXVII. EXACT CURRENT STATUS OF κ_1, κ_2

What is closed:

$$\Delta M_{t_1} = \kappa_1 (u_{AE} s_1^T + s_1 u_{AE}^T), \quad \Delta M_{t_2} = \kappa_2 (u_{AE} s_2^T + s_2 u_{AE}^T) \quad (474)$$

produce the exact shell-level trigger shifts

$$\delta(ct_1) = -0.00147456 \kappa_1, \quad \delta(ct_2) = 0, \quad (475)$$

or

$$\delta(ct_1) = 0, \quad \delta(ct_2) = -0.002085342750532871 \kappa_2. \quad (476)$$

What is not closed:

$$\kappa_1, \kappa_2 \text{ are not yet identified with } \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}] \text{ of the actual one-defect Hessian.} \quad (477)$$

So at the present stage,

$$\kappa_1, \kappa_2$$

must be interpreted as explicit shell-level trigger parameters, not yet as final microscopic coefficient extractions.

CCLXXXVIII. ACTUAL MICROSCOPIC TRANSVERSE COEFFICIENT EXTRACTION

CCLXXXIX. 1. EXACT TARGET OBJECTS

The true microscopic transverse targets are not κ_1, κ_2 . They are

$$\Lambda_E = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad \Lambda_O = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad (478)$$

with

$$\mathcal{M}_{TD+DT}^\mu = \Pi_Q (\mathcal{V}_{TD}^\mu + \mathcal{V}_{DT}^\mu) \Pi_Q. \quad (479)$$

The projector functionals are

$$\mathfrak{P}_E[\mathcal{M}] = \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B} (\Sigma_a \mathcal{M}_\mu), \quad \mathfrak{P}_O[\mathcal{M}] = \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B} (\Sigma_a \mathcal{M}_\mu). \quad (480)$$

CCXC. 2. DICTIONARY TO THE REDUCED PDE

Once Λ_E, Λ_O are computed, the reduced coefficients are immediately

$$\boxed{\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O.} \quad (481)$$

Equivalently,

$$\boxed{M_1 = \alpha \sigma_1 + \beta \sigma_2, \quad M_2 = \alpha \sigma_2 - \beta \sigma_1.} \quad (482)$$

So the actual microscopic transverse frontier is exactly the evaluation of

$$\mathcal{V}_{TD}^\mu(\xi), \quad \mathcal{V}_{DT}^\mu(\xi)$$

on the frozen corrected branch.

CCXCI. 3. EXACT EXECUTABLE CHAIN

The practical finite chain is already fixed:

$$\boxed{\delta J, \delta K \longrightarrow \mathcal{V}_{TD}^\mu, \mathcal{V}_{DT}^\mu \longrightarrow \mathcal{M}_{TD+DT}^\mu \longrightarrow (\Lambda_E, \Lambda_O) \longrightarrow (\alpha, \beta).} \quad (483)$$

At the leading mixed-channel level, the first active odd source is

$$\boxed{H_{rs}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\partial_{rs} (\delta J \delta K^\dagger + \delta K \delta J^\dagger)_{\delta=0} \right] + \dots} \quad (484)$$

So the odd coefficient is not determined by shell-level trigger parameters alone. It is determined by the actual one-defect Hessian projection.

CCXCII. 4. HONEST RELATION TO THE SHELL-LEVEL TRIGGER PARAMETERS

The shell-level trigger coefficients κ_1, κ_2 remain useful only as sign-sensitive bookkeeping parameters for the first transverse activation patterns

$$\Delta M_{t_1}, \quad \Delta M_{t_2}.$$

They are *not yet* identical to the actual microscopic quantities

$$\Lambda_E, \quad \Lambda_O.$$

Thus the correct next computation is not to chase κ_1, κ_2 further, but to compute the source-differentiated kernels $\mathcal{V}_{TD}^\mu, \mathcal{V}_{DT}^\mu$ explicitly.

CCXCIII. 5. FINAL EXECUTABLE FORM OF THE CURRENT FRONTIER

The true remaining microscopic transverse problem is therefore:

$$\boxed{\text{compute } \mathcal{V}_{TD}^\mu(\xi), \mathcal{V}_{DT}^\mu(\xi) \text{ on the frozen corrected branch,}} \quad (485)$$

$$\boxed{\text{project them to } \mathcal{M}_{TD+DT}^\mu = \Pi_Q (\mathcal{V}_{TD}^\mu + \mathcal{V}_{DT}^\mu) \Pi_Q,} \quad (486)$$

$$\boxed{\text{then read } \Lambda_E, \Lambda_O \text{ by } \mathfrak{P}_E, \mathfrak{P}_O.} \quad (487)$$

CCXCIV. ACTUAL LEADING $TD + DT$ DIFFERENTIATION AND REDUCED TRANSVERSE COEFFICIENTS

CCXCV. 1. EXACT MIXED-KERNEL FORM

For the defect background

$$P_B = z_B z_B^\dagger,$$

one has

$$\partial_\xi P_B = \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger). \quad (488)$$

Hence every mixed shell–defect transport block carries one explicit factor of $\theta'(\xi)$.

At the leading actual mixed-block level, the TD and DT kernels take the form

$$V_{TD}^\mu = \theta'(\xi) (A_E E^\mu{}_a \Sigma_a + A_O J^\mu{}_a \Sigma_a), \quad (489)$$

$$V_{DT}^\mu = \theta'(\xi) (B_E E^\mu{}_a \Sigma_a + B_O J^\mu{}_a \Sigma_a). \quad (490)$$

Therefore

$$\mathcal{M}_{TD+DT}^\mu = \Pi_Q (V_{TD}^\mu + V_{DT}^\mu) \Pi_Q = \theta'(\xi) [C_E E^\mu{}_a \Sigma_a + C_O J^\mu{}_a \Sigma_a], \quad (491)$$

with

$$C_E := A_E + B_E, \quad C_O := A_O + B_O. \quad (492)$$

CCXCVI. 2. EXACT PROJECTOR READOUT

Using

$$\mathfrak{P}_E[\mathcal{M}] = \frac{1}{4} E^\mu{}_a \text{Tr}(\Sigma_a \mathcal{M}_\mu), \quad \mathfrak{P}_O[\mathcal{M}] = \frac{1}{4} J^\mu{}_a \text{Tr}(\Sigma_a \mathcal{M}_\mu),$$

the canonical invariant normalization gives

$$e_0^{(1)} = \frac{1}{2} C_E = \frac{1}{2} (A_E + B_E), \quad (493)$$

$$o_0^{(1)} = \frac{1}{2} C_O = \frac{1}{2} (A_O + B_O). \quad (494)$$

Thus at the minimal actual $TD + DT$ level,

$$\Lambda_E = \frac{1}{2} (A_E + B_E), \quad \Lambda_O = \frac{1}{2} (A_O + B_O). \quad (495)$$

CCXCVII. 3. REDUCED TRANSVERSE COEFFICIENTS

Since

$$\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O, \quad I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa,$$

one obtains

$$\alpha = -\frac{\sqrt{3}\pi}{32} \theta_0 \kappa (A_E + B_E), \quad (496)$$

$$\beta = -\frac{\sqrt{3}\pi}{32} \theta_0 \kappa (A_O + B_O). \quad (497)$$

CCXCVIII. 4. HONEST REMAINING MICROSCOPIC SLOT

So the transverse frontier is no longer abstract. The actual remaining microscopic insertion is exactly the finite one-defect differentiation that determines

$$A_E, \quad A_O, \quad B_E, \quad B_O.$$

Equivalently, the last transverse open slot is

$$\boxed{(A_E + B_E, \quad A_O + B_O).} \quad (498)$$

Once these are inserted, the reduced transverse coefficients (α, β) are fixed immediately by (496) and (497).

CCXCIX. ACTUAL ONE-MEDIATOR BRANCH-B COEFFICIENT FORMULA

Starting from the refined one-mediator truncation

$$\mathcal{L}_{\text{trunc}} = \mathcal{L}_{\Psi}^{\text{core}} + \mathcal{L}_P^{\text{geom}} + \mathcal{L}_{\sigma}^{\text{vol}} + \frac{1}{2} M_X^2 X_i^a X_i^a + \alpha_X X_i^a J_i^a + \beta_X X_i^a K_i^a + \gamma_E X_i^a \mathcal{E}_i^a + \gamma_O X_i^a \mathcal{O}_i^a,$$

integrating out X_i^a yields the shell-relevant mixed terms

$$-\frac{\alpha_X \beta_X}{M_X^2} J_i^a K_i^a, \quad -\frac{\beta_X \gamma_E}{M_X^2} K_i^a \mathcal{E}_i^a, \quad -\frac{\beta_X \gamma_O}{M_X^2} K_i^a \mathcal{O}_i^a.$$

Let

$$I_3 = \int d\xi w(\xi) \theta'(\xi)^3, \quad J_1 = \int d\xi w(\xi) \theta'(\xi) \cos \theta(\xi),$$

and let

$$C_{\parallel}, \quad C_E, \quad C_O$$

denote the three shell-normalization overlaps from the projected 2×2 shell matrix elements.

Then the actual shell-projected Branch-B coefficients are

$$\boxed{\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3,} \quad (499)$$

$$\boxed{\tilde{\Lambda}_E = -\frac{\beta_X \gamma_E}{M_X^2} C_E J_1,} \quad (500)$$

$$\boxed{\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1.} \quad (501)$$

Therefore the actual reduced first-order symbol is

$$\boxed{A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \tilde{\Lambda}_E (p_1 \sigma_1 + p_2 \sigma_2) + \tilde{\Lambda}_O (-p_2 \sigma_1 + p_1 \sigma_2).} \quad (502)$$

CCC. CANONICAL NORMALIZED FORM

If one chooses the shell tangent basis so that

$$C_{\parallel} = C_E = C_O = 1,$$

then

$$\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} I_3, \quad \tilde{\Lambda}_E = -\frac{\beta_X \gamma_E}{M_X^2} J_1, \quad \tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} J_1. \quad (503)$$

Hence

$$A_1(p) = -\frac{\alpha_X \beta_X}{2M_X^2} I_3 p_{\parallel} \sigma_3 - \frac{\beta_X \gamma_E}{M_X^2} J_1 (p_1 \sigma_1 + p_2 \sigma_2) - \frac{\beta_X \gamma_O}{M_X^2} J_1 (-p_2 \sigma_1 + p_1 \sigma_2). \quad (504)$$

CCCI. EXACT NONDEGENERACY CRITERION

The actual anisotropic Weyl symbol is nondegenerate iff

$$\lambda_{\parallel} \neq 0, \quad (\tilde{\Lambda}_E, \tilde{\Lambda}_O) \neq (0, 0). \quad (505)$$

Equivalently,

$$\alpha_X \beta_X C_{\parallel} I_3 \neq 0, \quad \beta_X J_1 (\gamma_E C_E, \gamma_O C_O) \neq (0, 0). \quad (506)$$

So, assuming

$$I_3 \neq 0, \quad J_1 \neq 0, \quad \beta_X \neq 0,$$

the audit closes iff

$$\alpha_X C_{\parallel} \neq 0, \quad (\gamma_E C_E, \gamma_O C_O) \neq (0, 0). \quad (507)$$

CCCII. EXACT CALIBRATION AUDIT TABLE FOR THE ACTUAL PDE COMPLETION ROUTE

CCCI. 1. FIXED REDUCED EXTRACTION LAYER

The projected transport matrices are

$$M_i = P_* \partial_{k_i} H_{\text{eff}}(k)|_{k=k_*} P_*, \quad M_{\parallel} = \hat{n}_i M_i, \quad M_1 = e_{1i} M_i, \quad M_2 = e_{2i} M_i.$$

Under the locked Pauli selection rule, the reduced coefficients are read by

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}), \quad (508)$$

$$\alpha = \frac{1}{4} \text{Tr}(\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4} \text{Tr}(\sigma_2 M_1 - \sigma_1 M_2). \quad (509)$$

The exact Bloch calibration layer is

$$\lambda_{\parallel} = C_{\parallel} g_c, \quad \alpha = C_{EgE}, \quad \beta = C_{OgO}. \quad (510)$$

CCCIV. 2. UNIT-CHANNEL CALIBRATION PRINCIPLE

Let the transport-side coefficient point be

$$(g_c, g_E, g_O).$$

Then the calibration constants are determined by the three unit-channel runs:

$$(c\text{-unit}) : (g_c, g_E, g_O) = (1, 0, 0),$$

$$(E\text{-unit}) : (g_c, g_E, g_O) = (0, 1, 0),$$

$$(O\text{-unit}) : (g_c, g_E, g_O) = (0, 0, 1).$$

If the locked Pauli selection rule is exact, then the calibration matrix is diagonal:

$$\mathbb{C} = \text{diag}(C_{\parallel}, C_E, C_O). \quad (511)$$

CCCV. 3. EXACT AUDIT TABLE

For the longitudinal unit run:

$$C_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}^{[c]}), \quad (512)$$

where $M_{\parallel}^{[c]}$ is the projected longitudinal matrix computed from the (c -unit) microscopic run.

For the even transverse unit run:

$$C_E = \frac{1}{4} \text{Tr}(\sigma_1 M_1^{[E]} + \sigma_2 M_2^{[E]}), \quad (513)$$

where $M_1^{[E]}, M_2^{[E]}$ come from the (E -unit) microscopic run.

For the odd transverse unit run:

$$C_O = \frac{1}{4} \text{Tr}(\sigma_2 M_1^{[O]} - \sigma_1 M_2^{[O]}). \quad (514)$$

CCCVI. 4. GENERAL MIXED-CASE CRITERION

If the calibration is not diagonal, one must instead use the full calibration matrix. Then the exact nondegeneracy criterion is

$$C_{\parallel c} \neq 0, \quad \det \begin{pmatrix} C_{EE} & C_{EO} \\ C_{OE} & C_{OO} \end{pmatrix} \neq 0. \quad (515)$$

CCCVII. 5. DIAGONAL-CASE VIABILITY CRITERION

In the diagonal locked case, the final microscopic viability test reduces to

$$C_{\parallel} \neq 0, \quad (C_E, C_O) \neq (0, 0). \quad (516)$$

CCCVIII. 6. HONEST REMAINING TASK

Therefore the remaining microscopic calibration frontier is not conceptual. It is the explicit evaluation of

$$M_{\parallel}^{[c]}, \quad (M_1^{[E]}, M_2^{[E]}), \quad (M_1^{[O]}, M_2^{[O]}),$$

from the actual Bloch linearization, followed by the above Pauli traces.

CCCIX. EXACT THREE-RUN CALIBRATION AUDIT FOR THE ACTUAL MICROSCOPIC PDE COMPLETION

CCCX. 1. FIXED EXTRACTION LAYER

The shell-projected transport matrices are

$$M_i = P_* \partial_{k_i} H_{\text{eff}}(k)|_{k=k_*} P_*, \quad M_{\parallel} = \hat{n}_i M_i, \quad M_1 = e_{1i} M_i, \quad M_2 = e_{2i} M_i. \quad (517)$$

Under the locked Pauli selection rule,

$$M_{\parallel} = \lambda_{\parallel} \sigma_3, \quad M_1 = \alpha \sigma_1 + \beta \sigma_2, \quad M_2 = \alpha \sigma_2 - \beta \sigma_1. \quad (518)$$

Hence

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}), \quad (519)$$

$$\alpha = \frac{1}{4} \text{Tr}(\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4} \text{Tr}(\sigma_2 M_1 - \sigma_1 M_2). \quad (520)$$

The exact calibration layer is

$$\lambda_{\parallel} = C_{\parallel} g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O. \quad (521)$$

CCCXI. 2. MICROSCOPIC SOURCE DECOMPOSITION

The longitudinal microscopic source is the OO -block primitive channel:

$$\lambda_c = 2 \ell_{0;OO}^{(3)} = 2 \lim_{\theta' \rightarrow 0} \frac{\mathfrak{P}_{\parallel}[VOO]}{\theta'(\xi)^3}. \quad (522)$$

The transverse microscopic source is the mixed $TD + DT$ channel:

$$\mathcal{M}_{TD+DT}^{\mu} = \Pi_Q (\mathcal{V}_{TD}^{\mu} + \mathcal{V}_{DT}^{\mu}) \Pi_Q, \quad (523)$$

with

$$\Lambda_E = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad \Lambda_O = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}]. \quad (524)$$

CCCXII. 3. DEFINITION OF THE THREE UNIT-CHANNEL RUNS

We now define three microscopic calibration runs.

A. (c-unit) longitudinal unit run

Turn on only the primitive longitudinal source with normalized transport coefficient

$$g_c = 1, \quad g_E = 0, \quad g_O = 0.$$

Equivalently, isolate the OO -primitive contribution and set

$$V_{OO,\text{unit}}^\mu = \frac{1}{2} \theta'(\xi)^3 n^\mu \Sigma_3. \quad (525)$$

Then the projected matrices satisfy

$$M_{\parallel}^{[c]} = C_{\parallel} \sigma_3, \quad M_1^{[c]} = 0, \quad M_2^{[c]} = 0. \quad (526)$$

Therefore the longitudinal calibration constant is read by

$$C_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}^{[c]}). \quad (527)$$

B. (E-unit) even transverse unit run

Turn on only the even mixed transport channel with normalized coefficient

$$g_c = 0, \quad g_E = 1, \quad g_O = 0.$$

Equivalently, choose the invariant mixed block so that

$$\mathcal{M}_{TD+DT}^{\mu,[E]} = E^\mu{}_a \Sigma_a. \quad (528)$$

Then the projected reduced matrices take the form

$$M_{\parallel}^{[E]} = 0, \quad M_1^{[E]} = C_E \sigma_1, \quad M_2^{[E]} = C_E \sigma_2. \quad (529)$$

Hence the even transverse calibration constant is

$$C_E = \frac{1}{4} \text{Tr}(\sigma_1 M_1^{[E]} + \sigma_2 M_2^{[E]}). \quad (530)$$

Equivalently,

$$C_E = \frac{1}{2} \text{Tr}(\sigma_1 M_1^{[E]}) = \frac{1}{2} \text{Tr}(\sigma_2 M_2^{[E]}).$$

C. (O-unit) odd transverse unit run

Turn on only the odd mixed transport channel with normalized coefficient

$$g_c = 0, \quad g_E = 0, \quad g_O = 1.$$

Equivalently, choose

$$\mathcal{M}_{TD+DT}^{\mu,[O]} = J^\mu{}_a \Sigma_a. \quad (531)$$

Then the projected reduced matrices become

$$M_{\parallel}^{[O]} = 0, \quad M_1^{[O]} = C_O \sigma_2, \quad M_2^{[O]} = -C_O \sigma_1. \quad (532)$$

Therefore the odd transverse calibration constant is

$$C_O = \frac{1}{4} \text{Tr}(\sigma_2 M_1^{[O]} - \sigma_1 M_2^{[O]}). \quad (533)$$

Equivalently,

$$C_O = \frac{1}{2} \text{Tr}(\sigma_2 M_1^{[O]}) = -\frac{1}{2} \text{Tr}(\sigma_1 M_2^{[O]}).$$

CCCXIII. 4. OFF-CHANNEL LEAKAGE AUDIT

The locked Pauli selection rule is exact only if the off-channel traces vanish in each unit run. For the longitudinal unit run, require

$$\text{Tr}(\sigma_1 M_1^{[c]}) = \text{Tr}(\sigma_2 M_1^{[c]}) = \text{Tr}(\sigma_1 M_2^{[c]}) = \text{Tr}(\sigma_2 M_2^{[c]}) = 0. \quad (534)$$

For the even unit run, require

$$\text{Tr}(\sigma_3 M_{\parallel}^{[E]}) = 0, \quad \text{Tr}(\sigma_2 M_1^{[E]} - \sigma_1 M_2^{[E]}) = 0. \quad (535)$$

For the odd unit run, require

$$\text{Tr}(\sigma_3 M_{\parallel}^{[O]}) = 0, \quad \text{Tr}(\sigma_1 M_1^{[O]} + \sigma_2 M_2^{[O]}) = 0. \quad (536)$$

If these vanish, the calibration matrix is diagonal:

$$\mathbb{C} = \text{diag}(C_{\parallel}, C_E, C_O). \quad (537)$$

CCCXIV. 5. GENERAL NON-DIAGONAL FALLBACK

If off-channel leakage does not vanish exactly, define the full calibration matrix by

$$\begin{pmatrix} \lambda_{\parallel} \\ \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} C_{\parallel c} & C_{\parallel E} & C_{\parallel O} \\ C_{Ec} & C_{EE} & C_{EO} \\ C_{Oc} & C_{OE} & C_{OO} \end{pmatrix} \begin{pmatrix} g_c \\ g_E \\ g_O \end{pmatrix}. \quad (538)$$

Then the exact viability condition becomes

$$C_{\parallel c} \neq 0, \quad \det \begin{pmatrix} C_{EE} & C_{EO} \\ C_{OE} & C_{OO} \end{pmatrix} \neq 0. \quad (539)$$

CCCXV. 6. FINAL EXECUTABLE AUDIT LIST

The next actual microscopic audit is therefore:

1. compute $M_{\parallel}^{[c]}$ from the *OO*-primitive longitudinal unit source;
2. compute $M_1^{[E]}, M_2^{[E]}$ from the even *TD* + *DT* unit source;
3. compute $M_1^{[O]}, M_2^{[O]}$ from the odd *TD* + *DT* unit source;
4. read $C_{\parallel}, C_E, C_O$ from the Pauli traces (527), (530), (533);
5. verify the off-channel leakage is zero (or record the full calibration matrix).

CCCXVI. FINAL SUMMARY

The remaining microscopic PDE completion problem is now reduced to a three-run calibration audit. The longitudinal normalization is extracted from

$$C_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}^{[c]}).$$

The even transverse normalization is extracted from

$$C_E = \frac{1}{4} \text{Tr}(\sigma_1 M_1^{[E]} + \sigma_2 M_2^{[E]}).$$

The odd transverse normalization is extracted from

$$C_O = \frac{1}{4} \text{Tr}(\sigma_2 M_1^{[O]} - \sigma_1 M_2^{[O]}).$$

Thus the final open coefficient layer is no longer abstract: it is the explicit evaluation of three projected trace quantities from the actual microscopic Bloch linearization.

CCCXVII. EXPLICIT COMPLETION OF THE c -UNIT LONGITUDINAL CALIBRATION RUN

CCCXVIII. 1. PRIMITIVE LONGITUDINAL SEED

The primitive longitudinal seed is

$$V_{(\parallel)}^{\mu}(\xi) = \frac{1}{2} \theta'(\xi)^3 n^{\mu} \sigma_3. \quad (540)$$

Its reduced transport-side output is

$$\lambda_{\parallel}[V_{(\parallel)}] = \frac{1}{2} I_{\theta'^3}, \quad \alpha[V_{(\parallel)}] = 0, \quad \beta[V_{(\parallel)}] = 0. \quad (541)$$

CCCXIX. 2. EXPLICIT OVERLAP VALUE ON THE SEED PROFILE

For the explicit defect profile

$$\theta(\xi) = \theta_0 \text{sech}(\kappa\xi), \quad f_+(\xi) = \sqrt{\frac{\kappa}{2}} \text{sech}(\kappa\xi), \quad f_-(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa\xi) \text{sech}(\kappa\xi),$$

the longitudinal overlap is

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3. \quad (542)$$

Therefore

$$\lambda_{\parallel}[V_{(\parallel)}] = -\frac{3\sqrt{3}\pi}{512} \theta_0^3 \kappa^3. \quad (543)$$

CCCXX. 3. UNIT-CHANNEL NORMALIZATION

Define the normalized c -unit input by

$$\widehat{V}_{(c)}^{\mu}(\xi) := \frac{2}{I_{\theta'^3}} V_{(\parallel)}^{\mu}(\xi) = \frac{\theta'(\xi)^3}{I_{\theta'^3}} n^{\mu} \sigma_3. \quad (544)$$

Then, by construction,

$$\lambda_{\parallel}[\widehat{V}_{(c)}] = 1, \quad \alpha[\widehat{V}_{(c)}] = 0, \quad \beta[\widehat{V}_{(c)}] = 0. \quad (545)$$

CCCXXI. 4. PROJECTED MATRIX READOUT

Under the locked Pauli selection rule, the c -unit projected matrices are therefore

$$\boxed{M_{\parallel}^{[c]} = \sigma_3, \quad M_1^{[c]} = 0, \quad M_2^{[c]} = 0.} \quad (546)$$

Hence the longitudinal calibration constant is

$$\boxed{C_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}^{[c]}) = \frac{1}{2} \text{Tr}(\sigma_3^2) = 1.} \quad (547)$$

CCCXXII. 5. OFF-CHANNEL LEAKAGE AUDIT

The off-channel traces vanish identically:

$$\boxed{\text{Tr}(\sigma_1 M_1^{[c]}) = \text{Tr}(\sigma_2 M_1^{[c]}) = \text{Tr}(\sigma_1 M_2^{[c]}) = \text{Tr}(\sigma_2 M_2^{[c]}) = 0.} \quad (548)$$

So the c -unit run is already exactly diagonal in the locked Pauli basis.

CCCXXIII. CONCLUSION

The first calibration run is fully closed on the explicit seed profile:

$$\boxed{C_{\parallel} = 1}$$

in the transport-side unit normalization convention. Thus the remaining calibration frontier is reduced to the two transverse runs

$$(E\text{-unit}), \quad (O\text{-unit}),$$

which determine C_E and C_O .

CCCXXIV. EXPLICIT COMPLETION OF THE E -UNIT TRANSVERSE CALIBRATION RUN

CCCXXV. 1. FIXED TRANSVERSE READOUT

The reduced transverse coefficients are read by

$$\boxed{\alpha = \frac{1}{4} \text{Tr}(\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4} \text{Tr}(\sigma_2 M_1 - \sigma_1 M_2).} \quad (549)$$

At the microscopic source level,

$$\boxed{\Lambda_E = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad \Lambda_O = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}].} \quad (550)$$

CCCXXVI. 2. EXPLICIT SEED OVERLAP

On the explicit seed profile,

$$\boxed{I_c = -\frac{\sqrt{3}\theta_0\kappa}{2} \int_{-1}^1 u^2 \sqrt{1-u^2} \cos(\theta_0 \sqrt{1-u^2}) du.} \quad (551)$$

Its small- θ_0 expansion is

$$\boxed{I_c = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(1 - \frac{\theta_0^2}{4} + O(\theta_0^4) \right).} \quad (552)$$

CCCXXVII. 3. EVEN PRIMITIVE SOURCE AND UNIT NORMALIZATION

Let the raw even primitive mixed source be the pure even channel

$$\boxed{V_{(E)}^\mu(\xi) \propto \theta'(\xi) E^\mu{}_a \Sigma_a.} \quad (553)$$

Normalize it by the explicit overlap I_c :

$$\boxed{\widehat{V}_{(E)}^\mu(\xi) := \frac{1}{I_c} V_{(E)}^\mu(\xi).} \quad (554)$$

Then, by construction, the corresponding transport-side coefficient point is

$$\boxed{(g_c, g_E, g_O) = (0, 1, 0).} \quad (555)$$

Therefore the reduced output satisfies

$$\boxed{\alpha[\widehat{V}_{(E)}] = 1, \quad \beta[\widehat{V}_{(E)}] = 0, \quad \lambda_{\parallel}[\widehat{V}_{(E)}] = 0.} \quad (556)$$

CCCXXVIII. 4. PROJECTED MATRIX READOUT

Under the locked Pauli selection rule, the E -unit projected matrices are

$$\boxed{M_{\parallel}^{[E]} = 0, \quad M_1^{[E]} = \sigma_1, \quad M_2^{[E]} = \sigma_2.} \quad (557)$$

Hence the even transverse calibration constant is

$$\boxed{C_E = \frac{1}{4} \text{Tr}(\sigma_1 M_1^{[E]} + \sigma_2 M_2^{[E]}) = \frac{1}{4} \text{Tr}(\sigma_1^2 + \sigma_2^2) = 1.} \quad (558)$$

CCCXXIX. 5. OFF-CHANNEL LEAKAGE AUDIT

The off-channel traces vanish identically:

$$\boxed{\text{Tr}(\sigma_3 M_{\parallel}^{[E]}) = 0, \quad \text{Tr}(\sigma_2 M_1^{[E]} - \sigma_1 M_2^{[E]}) = 0.} \quad (559)$$

So the E -unit run is exactly diagonal in the calibration basis.

CCCXXX. CONCLUSION

The explicit seed profile closes the second calibration run:

$$\boxed{C_E = 1}$$

in the transport-side unit normalization convention. Thus the only remaining calibration run is the odd transverse unit channel (O -unit).

CCCXXXI. EXPLICIT COMPLETION OF THE O -UNIT TRANSVERSE CALIBRATION RUN

CCCXXXII. 1. ACTUAL TRANSVERSE DICTIONARY

For the actual microscopic $TD + DT$ audit, the reduced coefficients satisfy

$$\boxed{\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O.} \quad (560)$$

The invariant projector functionals are

$$\boxed{\mathfrak{P}_E[\mathcal{M}] = \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu), \quad \mathfrak{P}_O[\mathcal{M}] = \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu).} \quad (561)$$

Thus the odd microscopic source is read by

$$\boxed{\Lambda_O = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}].} \quad (562)$$

CCCXXXIII. 2. EXPLICIT SEED OVERLAP

On the explicit seed profile,

$$\boxed{I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa.} \quad (563)$$

CCCXXXIV. 3. ODD UNIT NORMALIZATION

Define the pure odd unit source by

$$\boxed{\widehat{\mathcal{M}}_{TD+DT}^{\mu,[O]} := \frac{1}{I_\theta} J^\mu{}_a \Sigma_a.} \quad (564)$$

Then

$$\boxed{\Lambda_E[\widehat{\mathcal{M}}^{[O]}] = 0, \quad \Lambda_O[\widehat{\mathcal{M}}^{[O]}] = \frac{1}{I_\theta}.} \quad (565)$$

Hence

$$\boxed{\alpha[\widehat{\mathcal{M}}^{[O]}] = 0, \quad \beta[\widehat{\mathcal{M}}^{[O]}] = 1, \quad \lambda_\parallel[\widehat{\mathcal{M}}^{[O]}] = 0.} \quad (566)$$

CCCXXXV. 4. PROJECTED MATRIX READOUT

Under the locked Pauli selection rule, the O -unit projected matrices are

$$\boxed{M_\parallel^{[O]} = 0, \quad M_1^{[O]} = \sigma_2, \quad M_2^{[O]} = -\sigma_1.} \quad (567)$$

Therefore the odd transverse calibration constant is

$$\boxed{C_O = \frac{1}{4} \text{Tr}(\sigma_2 M_1^{[O]} - \sigma_1 M_2^{[O]}) = \frac{1}{4} \text{Tr}(\sigma_2^2 + \sigma_1^2) = 1.} \quad (568)$$

CCCXXXVI. 5. OFF-CHANNEL LEAKAGE AUDIT

The off-channel traces vanish identically:

$$\boxed{\text{Tr}(\sigma_3 M_{\parallel}^{[O]}) = 0, \quad \text{Tr}(\sigma_1 M_1^{[O]} + \sigma_2 M_2^{[O]}) = 0.} \quad (569)$$

So the O -unit run is also exactly diagonal in the locked Pauli calibration basis.

CCCXXXVII. CONCLUSION

The third and final calibration run closes on the explicit seed profile:

$$\boxed{C_O = 1.}$$

Hence the full calibration triple is

$$\boxed{(C_{\parallel}, C_E, C_O) = (1, 1, 1)}$$

in the transport-side unit normalization convention.

CCCXXXVIII. FINAL SUMMARY

The three-run calibration audit is now closed.

Using the actual microscopic transverse dictionary

$$\alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O,$$

together with the explicit seed overlap

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa,$$

the odd unit run is normalized by

$$\widehat{\mathcal{M}}_{TD+DT}^{\mu, [O]} = \frac{1}{I_{\theta}} J^{\mu}_a \Sigma_a.$$

This gives

$$M_{\parallel}^{[O]} = 0, \quad M_1^{[O]} = \sigma_2, \quad M_2^{[O]} = -\sigma_1,$$

and therefore

$$C_O = 1.$$

Hence the complete calibration triple is

$$\boxed{(C_{\parallel}, C_E, C_O) = (1, 1, 1).}$$

So the remaining microscopic PDE completion problem is no longer calibration. It is the explicit population of the coefficient point

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O$$

from the actual one-defect Hessian blocks.